Associate Cloud Engineer

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No job outcome is promised.

Question 1
An external member of your team needs list access to compute images and disks in one of your projects. You want to follow Google-recommended practices when you grant the required permissions to this user. What should you do?

Options
A. Create a custom role, and add all the required compute.disks.list and compute.images.list permissions as includedPermissions. Grant the custom role to the user at the project level.
B. Create a custom role based on the Compute Image User role. Add the compute.disks.list to the IncludedPermissions field. Grant the custom role to the user at the project level.
C. Create a custom role based on the Compute Storage Admin role. Exclude unnecessary permissions from the custom role. Grant the custom role to the user at the project level.
D. Grant the Compute Storage Admin role at the project level.

Answer
A

Detailed explanation
A. - Correct. Creating a minimal custom role with only compute.disks.list and compute.images.list follows Google's principle of least privilege.
B. - Incorrect. Compute Image User role plus extra permissions is broader than needed and not the recommended practice.
C. - Incorrect. Compute Storage Admin role is unrelated to compute disks/images and would give unnecessary permissions.
D. - Incorrect. Compute Storage Admin is overly broad and violates least privilege.

Question 2
You are running a web application on Cloud Run for a few hundred users. Some of your users complain that the initial web page of the application takes much longer to load than the following pages. You want to follow Google's recommendations to mitigate the issue. What should you do?

Options
A. Set the minimum number of instances for your Cloud Run service to 3.
B. Set the concurrency number to 1 for your Cloud Run service.
C. Set the maximum number of instances for your Cloud Run service to 100.
D. Update your web application to use the protocol HTTP/2 instead of HTTP/1.1.

Answer
A

Detailed explanation
A. - Correct. Setting a minimum number of instances ensures that at least some containers are kept warm, which reduces cold-start latency for the first request.
B. - Incorrect. Setting concurrency to 1 isolates requests but does not solve the cold-start issue.
C. - Incorrect. Setting maximum instances to 100 only affects scale-out, not initial startup latency.
D. - Incorrect. HTTP/2 improves request multiplexing but does not address cold starts.

Question 4
You are running out of primary internal IP addresses in a subnet for a custom mode VPC. The subnet has the IP range 10.0.0.0/20, and the IP addresses are primarily used by virtual machines in the project. You need to provide more IP addresses for the virtual machines. What should you do?

Options
A. Add a secondary IP range 10.1.0.0/20 to the subnet.
B. Change the subnet IP range from 10.0.0.0/20 to 10.0.0.0/18.
C. Change the subnet IP range from 10.0.0.0/20 to 10.0.0.0/22.
D. Convert the subnet IP range from IPv4 to IPv6.

Answer
B

Detailed explanation
A. - Incorrect. Adding a secondary IP range provides IPs for alias IPs (like pods or containers), not primary VM internal IP addresses.
B. - Correct. Expanding the primary subnet range from /20 to /18 increases the available IP addresses from 4096 to 16384, solving the exhaustion issue.
C. - Incorrect. /22 reduces the available IP addresses (only 1024), making the problem worse.
D. - Incorrect. IPv6 addresses cannot be used for primary internal IPs of Compute Engine VMs in this context.

Question 5
Your company requires all developers to have the same permissions, regardless of the Google Cloud project they are working on. Your company's security policy also restricts developer permissions to Compute Engine, Cloud Functions, and Cloud SQL. You want to implement the security policy with minimal effort. What should you do?

Options
A. Create a custom role with Compute Engine, Cloud Functions, and Cloud SQL permissions in one project within the Google Cloud organization. Assign the role to developers in those projects.
B. Add all developers to a Google group in Google Groups for Workspace. Assign the predefined role of Compute Admin to the Google group at the Google Cloud organization level.
C. Add all developers to a Google group in Cloud Identity. Assign predefined roles for Compute Engine, Cloud Functions, and Cloud SQL permissions to the Google group for each project.
D. Add all developers to a Google group in Cloud Identity. Create a custom role with Compute Engine, Cloud Functions, and Cloud SQL permissions at the Google Cloud organization level. Assign the custom role to the Google group.

Answer
D

Detailed explanation
A. - Incorrect. Creating roles in individual projects requires duplication across projects, which increases administrative overhead.
B. - Incorrect. Compute Admin is broader than required; it violates least privilege.
C. - Incorrect. Assigning roles project by project increases operational effort and is not minimal.
D. - Correct. A custom organization-level role with only required permissions, assigned to a Cloud Identity group, enforces least privilege and is minimal to manage.

Question 6
You are working for a hospital that stores its medical images in an on-premises data room. The hospital wants to use Cloud Storage for archival storage of these images. The hospital wants an automated process to upload any new medical images to Cloud Storage. You need to design and implement a solution. What should you do?

Options
A. Create a Pub/Sub topic, and enable a Cloud Storage trigger for the Pub/Sub topic. Create an application that sends all medical images to the Pub/Sub topic.
B. Create a script that uses the gcloud storage command to synchronize the on-premises storage with Cloud Storage. Schedule the script as a cron job.
C. Create a Pub/Sub topic, and create a Cloud Function connected to the topic that writes data to Cloud Storage. Create an application that sends all medical images to the Pub/Sub topic.
D. In the Google Cloud console, go to Cloud Storage. Upload the relevant images to the appropriate bucket.

Answer
B

Detailed explanation
A. - Incorrect. Pub/Sub is not needed; the use case is syncing files, not message streaming.
B. - Correct. gcloud storage with cron provides a simple, automated, and efficient synchronization between on-prem storage and Cloud Storage.
C. - Incorrect. Pub/Sub + Cloud Functions is more complex than needed and not recommended for simple archival uploads.
D. - Incorrect. Manual uploads are not automated and don't meet requirements.

Question 7
Your company has an internal application for managing transactional orders. The application is used exclusively by employees in a single physical location. The application requires strong consistency, fast queries, and ACID guarantees for multi-table transactional updates. The first version of the application is implemented in PostgreSQL, and you want to deploy it to the cloud with minimal code changes. Which database is most appropriate for this application?

Options
A. Bigtable
B. BigQuery
C. Cloud SQL
D. Firestore

Answer
C

Detailed explanation
A. - Incorrect. Bigtable is NoSQL and does not provide ACID multi-table transactions.
B. - Incorrect. BigQuery is for analytical queries, not transactional workloads.
C. - Correct. Cloud SQL (PostgreSQL) provides relational ACID transactions and minimal code changes since the app is already in PostgreSQL.
D. - Incorrect. Firestore provides strong consistency but is NoSQL, and does not support multi-table relational joins.

Question 8
Your company runs one batch process in an on-premises server that takes around 30 hours to complete. The task runs monthly, can be performed offline, and must be restarted if interrupted. You want to migrate this workload to the cloud while minimizing cost. What should you do?

Options
A. Create an Instance Template with Spot VMs On. Create a Managed Instance Group from the template and adjust Target CPU Utilization. Migrate the workload.
B. Migrate the workload to a Compute Engine VM. Start and stop the instance as needed.
C. Migrate the workload to a Google Kubernetes Engine cluster with Spot nodes.
D. Migrate the workload to a Compute Engine Spot VM.

Answer
B

Detailed explanation
A. - Incorrect. Spot VMs can be preempted, which is unsuitable for long 30-hour jobs.
B. - Correct. Running the batch on a Compute Engine VM that you start/stop as needed avoids interruptions and minimizes cost.
C. - Incorrect. GKE with Spot nodes risks preemption, unsuitable for workloads requiring restart if interrupted.
D. - Incorrect. Spot VMs will not guarantee completion of a 30-hour workload.

Question 9
You are planning to migrate the following on-premises data management solutions to Google Cloud:

- One MySQL cluster for your main database
- Apache Kafka for your event streaming platform
- One Cloud SQL for PostgreSQL database for your analytical and reporting needs

You want to implement Google-recommended solutions for the migration. You need to ensure that the new solutions provide global scalability and require minimal operational and infrastructure management. What should you do?

Options
A. Migrate from MySQL to Cloud SQL, from Kafka to Pub/Sub, and from Cloud SQL for PostgreSQL to BigQuery.
B. Migrate from MySQL to Cloud Spanner, from Kafka to Pub/Sub, and from Cloud SQL for PostgreSQL to BigQuery.
C. Migrate from MySQL to Cloud Spanner, from Kafka to Memorystore, and from Cloud SQL for PostgreSQL to Cloud SQL.
D. Migrate from MySQL to Cloud SQL, from Kafka to Memorystore, and from Cloud SQL for PostgreSQL to Cloud SQL.

Answer
B

Detailed explanation
A. - Incorrect. Cloud SQL does not provide global scalability like Cloud Spanner.
B. - Correct. Cloud Spanner offers global scalability for transactional DB, Pub/Sub is the recommended Kafka alternative, and BigQuery is optimal for analytics.
C. - Incorrect. Memorystore is for caching, not an alternative to Kafka.
D. - Incorrect. This option does not scale globally and misuses Memorystore.

Question 10
During a recent audit of your existing Google Cloud resources, you discovered several users with email addresses outside of your Google Workspace domain. You want to ensure that your resources are only shared with users whose email addresses match your domain. You need to remove any mismatched users, and you want to avoid having to audit your resources to identify mismatched users. What should you do?

Options
A. Create a Cloud Scheduler task to regularly scan your projects and delete mismatched users.
B. Create a Cloud Scheduler task to regularly scan your resources and delete mismatched users.
C. Set an organizational policy constraint to limit identities by domain to automatically remove mismatched users.
D. Set an organizational policy constraint to limit identities by domain, and then retroactively remove the existing mismatched users.

Answer
D

Detailed explanation
A. - Incorrect. Custom scripts increase operational overhead and are not scalable.
B. - Incorrect. Same issue as A; scheduler-based deletion is not the recommended Google practice.
C. - Incorrect. The constraint prevents new mismatched users but does not remove existing ones.
D. - Correct. The constraint plus retroactive cleanup enforces policy and removes existing mismatched accounts.

Question 11
Your application is running on Google Cloud in a managed instance group (MIG). You see errors in Cloud Logging for one VM that one of the processes is not responsive. You want to replace this VM in the MIG quickly. What should you do?

Options
A. Use the gcloud compute instances update command with a REFRESH action for the VM.
B. Use the gcloud compute instance-groups managed recreate-instances command to recreate the VM.
C. Select the MIG from the Compute Engine console and, in the menu, select Replace VMs.
D. Update and apply the instance template of the MIG.

Answer
B

Detailed explanation
A. - Incorrect. REFRESH action is not supported for this use case.
B. - Correct. The recreate-instances command is designed to quickly recreate an unhealthy VM in a managed instance group.
C. - Incorrect. Replace VMs in console is broader and replaces all, not one instance.
D. - Incorrect. Updating the instance template affects future VMs, not the current failing one.

Question 12
You want to permanently delete a Pub/Sub topic managed by Config Connector in your Google Cloud project. What should you do?

Options
A. Use kubectl to create the label deleted-by-cnrm and to change its value to true for the topic resource.
B. Use kubectl to delete the topic resource.
C. Use gcloud CLI to delete the topic.
D. Use gcloud CLI to update the topic label managed-by-cnrm to false.

Answer
B

Detailed explanation
A. - Incorrect. Label changes won't delete the topic.
B. - Correct. In Config Connector, you delete resources like Pub/Sub topics using kubectl delete.
C. - Incorrect. gcloud CLI does not manage Config Connector resources directly.
D. - Incorrect. Updating the managed label does not delete the resource.

Question 13
Your company is using Google Workspace to manage employee accounts. Anticipated growth will increase the number of personnel from 100 employees to 1,000 within 2 years. Most employees will need access to your company's Google Cloud account. The systems and processes must support 10x growth without performance degradation, unnecessary complexity, or security issues. What should you do?

Options
A. Migrate the users to Active Directory. Connect HR system to Active Directory. Turn on Google Cloud Directory Sync and Identity Federation.
B. Organize the users in Cloud Identity into groups. Enforce multi-factor authentication in Cloud Identity.
C. Turn on identity federation between Cloud Identity and Google Workspace. Enforce multi-factor authentication for domain wide delegation.
D. Use a third-party identity provider service through federation. Synchronize users from Google Workspace to the third-party provider in real time.

Answer
C

Detailed explanation
A. - Incorrect. Migrating to AD adds unnecessary complexity.
B. - Incorrect. While grouping and MFA help, they don't provide scalability and federation integration.
C. - Correct. Identity federation between Cloud Identity and Workspace with MFA provides scalability, security, and minimal complexity.
D. - Incorrect. Third-party identity provider adds complexity, not the recommended approach.

Question 14
You want to host your video encoding software on Compute Engine. Your user base is growing rapidly, and users need to be able to encode their videos at any time without interruption or CPU limitations. You must ensure the encoding solution is highly available and want to automate operations. What should you do?

Options
A. Deploy your solution on multiple standalone Compute Engine instances, and increase instances when CPU utilization exceeds a threshold.
B. Deploy your solution on multiple standalone Compute Engine instances, and replace existing instances with high-CPU instances when utilization exceeds a threshold.
C. Deploy your solution to an instance group, and increase available instances when utilization is high.
D. Deploy your solution to an instance group, and set autoscaling based on CPU utilization.

Answer
D

Detailed explanation
A. - Incorrect. Manually increasing instances is not automated.
B. - Incorrect. Replacing instances manually with bigger machines is not efficient.
C. - Incorrect. Increasing instances manually in a group is not automated.
D. - Correct. Autoscaling in instance groups based on CPU ensures high availability and automation.

Question 15
Your managed instance group raised an alert stating that new instance creation has failed. You need to solve the instance creation problem. What should you do?

Options
A. Create an instance template that contains valid syntax. Delete any persistent disks with the same name as instance names.
B. Create an instance template with valid syntax. Verify persistent disk names are not the same as instance names.
C. Verify the template being used contains valid syntax. Delete persistent disks with same names as instance names. Set disks.autoDelete property to true.
D. Delete the current template and replace with new one. Verify persistent disk and instance names are different. Set disks.autoDelete property to true.

Answer
A

Detailed explanation
A. - Correct. Invalid syntax or disk conflicts cause failures. Deleting duplicate disks and fixing syntax resolves the issue.
B. - Incorrect. Only verifying names does not solve syntax errors.
C. - Incorrect. This mixes steps and requires both verification and settings but not the root fix.
D. - Incorrect. Replacing templates alone doesn’t address syntax issues.

Question 16
You have created an application packaged into a Docker image. You want to deploy it to Google Kubernetes Engine. What should you do?

Options
A. Upload the image to Cloud Storage and create a Kubernetes Service referencing the image.
B. Upload the image to Cloud Storage and create a Kubernetes Deployment referencing the image.
C. Upload the image to Artifact Registry and create a Kubernetes Service referencing the image.
D. Upload the image to Artifact Registry and create a Kubernetes Deployment referencing the image.

Answer
D

Detailed explanation
A. - Incorrect. GKE cannot pull container images directly from Cloud Storage.
B. - Incorrect. Same as A, Cloud Storage is not a container registry.
C. - Incorrect. A Service is for networking exposure, not running containers.
D. - Correct. Artifact Registry is the correct container registry, and Deployment is the correct workload object in Kubernetes.

Question 17
You are using Looker Studio to visualize a table from BigQuery. Data is appended daily, and nightly summaries overwrite the table. Charts in Looker Studio are broken. What should you do?

Options
A. In Cloud Logging, create a filter for your Looker Studio report.
B. Use Snapshot Debugger to find out why data was not refreshed.
C. Review the Error Reporting page in the Google Cloud console.
D. Use the BigQuery interface to review the nightly job and look for errors.

Answer
D

Detailed explanation
A. - Incorrect. Cloud Logging won’t fix data overwrite issues.
B. - Incorrect. Snapshot Debugger is for debugging apps, not data pipelines.
C. - Incorrect. Error Reporting is not tied to query overwrites.
D. - Correct. Reviewing the BigQuery nightly job reveals issues with overwriting the table, which explains broken charts.

Question 18
You have a batch workload running nightly on many VMs. It is fault-tolerant and can tolerate some VM terminations. The cost is too high. What should you do?

Options
A. Run a test using simulated maintenance events. If successful, use Spot N2 Standard VMs for future jobs.
B. Run a test using simulated maintenance events. If successful, use N2 Standard VMs.
C. Run a test using a managed instance group. If successful, use N2 Standard VMs.
D. Run a test using N1 Standard VMs instead of N2. If successful, use N1 Standard VMs.

Answer
A

Detailed explanation
A. - Correct. Spot VMs reduce cost, and testing ensures tolerance for preemption.
B. - Incorrect. N2 Standard VMs are regular priced and do not reduce costs.
C. - Incorrect. Managed instance groups help orchestration but do not reduce costs.
D. - Incorrect. N1 is cheaper but not as effective as Spot for cost reduction.

Question 19
You created resources in multiple Google Cloud projects with linked billing accounts. You want a single visual representation of all costs and need new cost data as soon as possible. What should you do?

Options
A. Fill all resources in Pricing Calculator to estimate cost.
B. Use Reports in Billing Console to view detailed info.
C. Visit Cost Table page to export CSV and visualize.
D. Configure Billing Data Export to BigQuery and visualize in Looker Studio.

Answer
D

Detailed explanation
A. - Incorrect. Pricing Calculator only provides estimates, not actual incurred costs.
B. - Incorrect. Reports show data but lack flexibility and real-time analysis.
C. - Incorrect. CSV export is manual and not automated for updates.
D. - Correct. Exporting billing data to BigQuery and visualizing in Looker provides near real-time and scalable cost reporting.

Question 20
Your company has large amounts of unstructured data in different file formats. You want to make the data accessible on Google Cloud so it can be processed by a Dataflow job. What should you do?

Options
A. Upload the data to BigQuery using bq command line.
B. Upload the data to Cloud Storage using gcloud storage command.
C. Upload the data into Cloud SQL using import function.
D. Upload the data into Cloud Spanner using import function.

Answer
B

Detailed explanation
A. - Incorrect. BigQuery requires structured/tabular data, not raw unstructured files.
B. - Correct. Cloud Storage is the correct place to store unstructured files, and Dataflow can directly process them.
C. - Incorrect. Cloud SQL is relational and unsuitable for raw unstructured files.
D. - Incorrect. Cloud Spanner is for relational distributed workloads, not unstructured data.

Question 21
You have deployed an application on a single Compute Engine instance. The application writes logs to disk. Users start reporting errors with the application. You want to diagnose the problem. What should you do?

Options
A. Navigate to Cloud Logging and view the application logs.
B. Configure a health check on the instance and set a “consecutive successes” Healthy threshold value of 1.
C. Connect to the instance’s serial console and read the application logs.
D. Install and configure the Ops agent and view the logs from Cloud Logging.

Answer
D

Detailed explanation
A. - Incorrect. If Ops Agent is not installed, logs won’t be automatically exported to Cloud Logging.
B. - Incorrect. A health check is for load balancing and does not help diagnose application log errors.
C. - Incorrect. The serial console is used for boot/debugging issues, not application log monitoring.
D. - Correct. Installing and configuring the Ops Agent ensures application/system logs are sent to Cloud Logging for centralized troubleshooting.

Question 22
You recently received a new Google Cloud project with an attached billing account where you will work. You need to create instances, set firewalls, and store data in Cloud Storage. You want to follow Google-recommended practices. What should you do?

Options
A. Use the gcloud CLI services enable cloudresourcemanager.googleapis.com command to enable all resources.
B. Use the gcloud services enable compute.googleapis.com command to enable Compute Engine and the gcloud services enable storage-api.googleapis.com command to enable the Cloud Storage APIs.
C. Open the Google Cloud console and enable all Google Cloud APIs from the API dashboard.
D. Open the Google Cloud console and run gcloud init --project in a Cloud Shell.

Answer
B

Detailed explanation
A. - Incorrect. cloudresourcemanager.googleapis.com alone won't enable all necessary APIs.
B. - Correct. Enabling only the required APIs (Compute Engine and Cloud Storage) follows least-privilege and recommended practices.
C. - Incorrect. Enabling all APIs is not recommended and grants unnecessary capabilities.
D. - Incorrect. gcloud init sets project context but does not enable APIs.

Question 23
Your application development team has created Docker images for an application that will be deployed on Google Cloud. Your team does not want to manage the infrastructure associated with this application. You need to ensure that the application can scale automatically as it gains popularity. What should you do?

Options
A. Create an instance template with the container image, and deploy a Managed Instance Group with Autoscaling.
B. Upload Docker images to Artifact Registry, and deploy the application on Google Kubernetes Engine using Standard mode.
C. Upload Docker images to Cloud Storage, and deploy the application on Google Kubernetes Engine using Standard mode.
D. Upload Docker images to Artifact Registry, and deploy the application on Cloud Run.

Answer
D

Detailed explanation
A. - Incorrect. Managed instance groups require VM management, which the team wants to avoid.
B. - Incorrect. GKE requires cluster management and is not fully serverless.
C. - Incorrect. Cloud Storage is not a container registry, and using it for images is not recommended.
D. - Correct. Cloud Run is fully managed, scales automatically, and works directly with Artifact Registry for container images.

Question 24
You are migrating a business critical application from your local data center into Google Cloud. As part of your high-availability strategy, you want to ensure that any data used by the application will be immediately available if a zonal failure occurs. What should you do?

Options
A. Store the application data on a zonal persistent disk. Create a snapshot schedule for the disk. If an outage occurs, create a new disk from the most recent snapshot and attach it to a new VM in another zone.
B. Store the application data on a zonal persistent disk. If an outage occurs, create an instance in another zone with this disk attached.
C. Store the application data on a regional persistent disk. Create a snapshot schedule for the disk. If an outage occurs, create a new disk from the most recent snapshot and attach it to a new VM in another zone.
D. Store the application data on a regional persistent disk. If an outage occurs, create an instance in another zone with this disk attached.

Answer
D

Detailed explanation
A. - Incorrect. Snapshots introduce delay in recovery and are not immediate.
B. - Incorrect. A zonal persistent disk cannot be attached to VMs in another zone.
C. - Incorrect. Snapshots again add delay, not immediate availability.
D. - Correct. Regional persistent disks replicate data synchronously across two zones, ensuring immediate availability in case of zonal failure.

Question 25
The DevOps group in your organization needs full control of Compute Engine resources in your development project, but they should not have permission to create or update other resources. You want to follow Google's recommendations for setting permissions. What should you do?

Options
A. Grant the basic role roles/viewer and the predefined role roles/compute.admin to the DevOps group.
B. Create an IAM policy and grant all compute.instanceAdmin.* permissions to the policy. Attach the policy to the DevOps group.
C. Create a custom role at the folder level and grant all compute.instanceAdmin.* permissions to the role. Grant the custom role to the DevOps group.
D. Grant the basic role roles/editor to the DevOps group.

Answer
C

Detailed explanation
A. - Incorrect. Viewer role is unnecessary if compute.admin is already granted.
B. - Incorrect. Granting all instanceAdmin.* permissions directly is not recommended; policies should use roles.
C. - Correct. A custom role scoped only to compute resources follows least-privilege and best practice.
D. - Incorrect. Editor role grants permissions beyond Compute Engine, violating least-privilege.

Question 26
Your team is running an on-premises ecommerce application with microservices in Docker containers. You need to deploy your current application to a serverless Google Cloud solution. What should you do?

Options
A. Use your existing CI/CD pipeline. Use the generated Docker images and deploy them to Cloud Run. Update configurations and endpoints.
B. Use your existing CI/CD pipeline. Use the generated Docker images and deploy them to Cloud Functions. Use the same configurations as on-premises.
C. Use your existing codebase and deploy each service as a separate Cloud Function. Update configurations and endpoints.
D. Use your existing codebase and deploy each service as a separate Cloud Run. Use the same configurations as on-premises.

Answer
A

Detailed explanation
A. - Correct. Cloud Run supports Docker containers, requires minimal changes, and scales serverlessly.
B. - Incorrect. Cloud Functions do not run Docker containers directly.
C. - Incorrect. Cloud Functions require major code refactoring.
D. - Incorrect. Deploying code directly ignores Docker packaging and would need rework.

Question 27
You are running multiple microservices in a Kubernetes Engine cluster. One microservice renders images and needs much higher CPU than the others. You need to optimize cluster resources. What should you do?

Options
A. Assign the pods of the image rendering microservice a higher pod priority.
B. Create a node pool with compute-optimized machine type nodes for the image rendering microservice. Use a general-purpose node pool for the other services.
C. Use the node pool with general-purpose nodes for the rendering service. Create another pool with compute-optimized nodes for other services.
D. Configure resource requests for CPU and memory in the image rendering deployment. Leave others at default.

Answer
B

Detailed explanation
A. - Incorrect. Priority changes scheduling order, not resource optimization.
B. - Correct. Separate node pools allow the image rendering service to run on compute-optimized nodes while others use general-purpose nodes.
C. - Incorrect. Reversed — rendering service should use compute-optimized nodes.
D. - Incorrect. Requests alone don’t ensure efficiency; node type must match workload.

Question 28
You are working in a team developing a critical application to deploy on Kubernetes. It must be reliable and optimized for production. You want to follow Google's recommended practices. What should you do?

Options
A. Create a GKE Autopilot cluster and enroll it in the rapid release channel.
B. Create a GKE Autopilot cluster and enroll it in the stable release channel.
C. Create a zonal GKE standard cluster and enroll it in the stable release channel.
D. Create a regional GKE standard cluster and enroll it in the rapid release channel.

Answer
B

Detailed explanation
A. - Incorrect. Rapid release risks stability issues.
B. - Correct. Autopilot with stable release ensures reliability and minimizes ops burden.
C. - Incorrect. Zonal clusters are less reliable than regional.
D. - Incorrect. Regional rapid release risks instability.

Question 29
You are responsible for a web application on Compute Engine. You want support to be notified automatically if users experience latency for at least 5 minutes. What should you do?

Options
A. Export Cloud Monitoring metrics to BigQuery and use Looker Studio to monitor.
B. Create an alert policy to notify when HTTP response latency exceeds threshold.
C. Implement an App Engine service that calls Cloud Monitoring API and sends notifications.
D. Use Cloud Monitoring dashboard and manually take action when latency is high.

Answer
B

Detailed explanation
A. - Incorrect. Exporting to BigQuery is for analytics, not real-time alerts.
B. - Correct. Alert policies in Cloud Monitoring are the Google-recommended way to automate notifications.
C. - Incorrect. Creating custom API-based service adds unnecessary complexity.
D. - Incorrect. Manual monitoring does not ensure timely alerts.

Question 30
You have on-premises analytics jobs running binaries that process 1–16 GB data in memory for ~45 minutes nightly. You want to migrate to Google Cloud with minimal cost. What should you do?

Options
A. Create a container for the binaries. Use Cloud Scheduler to start a Cloud Run job.
B. Create a container for the binaries. Deploy to GKE and use Kubernetes scheduler to start.
C. Upload code to Cloud Functions and use Cloud Scheduler to start.
D. Lift and shift to a VM on Compute Engine. Use instance schedule to start/stop.

Answer
D

Detailed explanation
A. - Incorrect. Cloud Run jobs have a max execution time of 60 minutes but not optimal for heavy binaries of 16 GB memory.
B. - Incorrect. GKE requires more ops overhead.
C. - Incorrect. Cloud Functions execution time is max 9 minutes, not suitable.
D. - Correct. Compute Engine VMs with instance schedules give full control and minimal cost for nightly 45-min workloads.

Question 31
You are working with a user to set up an application in a new VPC behind a firewall. The user is concerned about data egress. You want to configure the fewest open egress ports. What should you do?

Options
A. Set up a low-priority (65534) rule that blocks all egress and a high-priority rule (1000) that allows only the appropriate ports.
B. Set up a high-priority (1000) rule that pairs both ingress and egress ports.
C. Set up a high-priority (1000) rule that blocks all egress and a low-priority (65534) rule that allows only the appropriate ports.
D. Set up a high-priority (1000) rule to allow the appropriate ports.

Answer
A

Detailed explanation
A. Correct - The recommended practice is to block all egress by default (low-priority rule) and then explicitly allow only the necessary ports with a high-priority rule. This minimizes exposure.
B. Incorrect - Pairing ingress and egress in one rule is not supported; ingress and egress rules are configured separately.
C. Incorrect - If you block all egress with a high-priority rule, traffic will never be allowed. The correct order is to block broadly at low priority and allow specifically at high priority.
D. Incorrect - This would allow traffic but doesn’t provide a default block rule. Without the block, unnecessary egress ports remain open.

Question 32
Your company runs its Linux workloads on Compute Engine instances. Your company will be working with a new operations partner that does not use Google Accounts. You need to grant access to the instances to your operations partner so they can maintain the installed tooling. What should you do?

Options
A. Enable Cloud IAP for the Compute Engine instances, and add the operations partner as a Cloud IAP Tunnel User.
B. Tag all the instances with the same network tag. Create a firewall rule in the VPC to grant TCP access on port 22 for traffic from the operations partner to instances with the network tag.
C. Set up Cloud VPN between your Google Cloud VPC and the internal network of the operations partner.
D. Ask the operations partner to generate SSH key pairs, and add the public keys to the VM instances.

Answer
A

Detailed explanation
A. Correct - Cloud IAP (Identity-Aware Proxy) allows secure SSH access without requiring Google accounts. Assigning the IAP Tunnel User role is the best practice.
B. Incorrect - Firewall rules only expose SSH ports; they do not solve the authentication problem for non-Google accounts.
C. Incorrect - A VPN provides connectivity but doesn't provide user-level authentication or secure access management.
D. Incorrect - Manually adding SSH keys bypasses centralized management and increases security risks.

Question 33
You have created a code snippet that should be triggered whenever a new file is uploaded to a Cloud Storage bucket. You want to deploy this code snippet. What should you do?

Options
A. Use App Engine and configure Cloud Scheduler to trigger the application using Pub/Sub.
B. Use Cloud Functions and configure the bucket as a trigger resource.
C. Use Google Kubernetes Engine and configure a CronJob to trigger the application using Pub/Sub.
D. Use Dataflow as a batch job, and configure the bucket as a data source.

Answer
B

Detailed explanation
A. Incorrect - App Engine with Cloud Scheduler is not needed for this use case. Scheduler runs jobs at fixed intervals, not event-driven triggers.
B. Correct - Cloud Functions natively support Cloud Storage event triggers. When a new file is uploaded, the function can execute the code automatically.
C. Incorrect - Kubernetes CronJob is for time-based jobs, not event-based triggers.
D. Incorrect - Dataflow is used for stream and batch processing pipelines, but it's overkill for executing a small snippet on file upload.

Question 34
You have been asked to set up Object Lifecycle Management for objects stored in storage buckets. The objects are written once and accessed frequently for 30 days. After 30 days, the objects are not read again unless there is a special need. The objects should be kept for three years, and you need to minimize cost. What should you do?

Options
A. Set up a policy that uses Nearline storage for 30 days and then moves to Archive storage for three years.
B. Set up a policy that uses Standard storage for 30 days and then moves to Archive storage for three years.
C. Set up a policy that uses Nearline storage for 30 days, then moves to Coldline for one year, and then moves to Archive storage for two years.
D. Set up a policy that uses Standard storage for 30 days, then moves to Coldline for one year, and then moves to Archive storage for two years.

Answer
B

Detailed explanation
A. Incorrect - Nearline is meant for monthly access. The data is accessed frequently during the first 30 days, so Standard storage is required, not Nearline.
B. Correct - Standard storage is appropriate for frequent access during the first 30 days. After that, Archive storage is the lowest-cost option for long-term storage when access is rare, aligning perfectly with the requirement.
C. Incorrect - Using Nearline first doesn't match the frequent access requirement. Also, Coldline is an unnecessary intermediate step, which increases cost.
D. Incorrect - While Standard storage for 30 days is correct, Coldline before Archive adds extra cost unnecessarily. Directly moving to Archive after 30 days is more cost-effective.

Question 35
You are storing sensitive information in a Cloud Storage bucket. For legal reasons, you need to be able to record all requests that read any of the stored data. You want to make sure you comply with these requirements. What should you do?

Options
A. Enable the Identity Aware Proxy API on the project.
B. Scan the bucket using the Data Loss Prevention API.
C. Allow only a single Service Account access to read the data.
D. Enable Data Access audit logs for the Cloud Storage API.

Answer
D

Detailed explanation
A. Incorrect - Identity-Aware Proxy controls access but does not log every read request.
B. Incorrect - DLP scans for sensitive data patterns but does not log access events.
C. Incorrect - Restricting access does not ensure compliance with auditing requirements, as it doesn't log reads.
D. Correct - Data Access audit logs record every read, write, or admin action on the Cloud Storage API. This is the Google-recommended solution for compliance.

Question 36
You are the team lead of a group of 10 developers. You provided each developer with an individual Google Cloud Project that they can use as their personal sandbox to experiment with different Google Cloud solutions. You want to be notified if any of the developers are spending above $500 per month on their sandbox environment. What should you do?

Options
A. Create a single budget for all projects and configure budget alerts on this budget.
B. Create a separate billing account per sandbox project and enable BigQuery billing exports. Create a Data Studio dashboard to plot the spending per billing account.
C. Create a budget per project and configure budget alerts on all of these budgets.
D. Create a single billing account for all sandbox projects and enable BigQuery billing exports. Create a Data Studio dashboard to plot the spending per project.

Answer
C

Detailed explanation
A. Incorrect - A single budget across all projects won't alert per developer, only on the total, so individual overages won't be caught.
B. Incorrect - Creating separate billing accounts is unnecessary and increases overhead.
C. Correct - A budget per project ensures each developer's sandbox spending is tracked individually. Budget alerts can be configured at $500 to notify when thresholds are exceeded.
D. Incorrect - While exporting to BigQuery helps with visualization, it doesn't create per-project budget alerts directly.

Question 37
You are deploying a production application on Compute Engine. You want to prevent anyone from accidentally destroying the instance by clicking the wrong button. What should you do?

Options
A. Disable the flag Delete boot disk when instance is deleted.
B. Enable delete protection on the instance.
C. Disable Automatic restart on the instance.
D. Enable Preemptibility on the instance.

Answer
B

Detailed explanation
A. Incorrect - Disabling the "Delete boot disk with instance" only prevents the disk from being deleted but doesn't stop the instance itself from being deleted.
B. Correct - Delete protection ensures that users cannot accidentally delete the instance from the console or API without first removing this protection explicitly.
C. Incorrect - Disabling automatic restart affects uptime, not accidental deletions.
D. Incorrect - Preemptible VMs are short-lived and meant for cost savings, not protecting against accidental deletions.

Question 38
Your company uses a large number of Google Cloud services centralized in a single project. All teams have specific projects for testing and development. The DevOps team needs access to all of the production services in order to perform their job. You want to prevent Google Cloud product changes from broadening their permissions in the future. You want to follow Google-recommended practices. What should you do?

Options
A. Grant all members of the DevOps team the role of Project Editor on the organization level.
B. Grant all members of the DevOps team the role of Project Editor on the production project.
C. Create a custom role that combines the required permissions. Grant the DevOps team the custom role on the production project.
D. Create a custom role that combines the required permissions. Grant the DevOps team the custom role on the organization level.

Answer
C

Detailed explanation
A. Incorrect - Granting Editor role at the organization level provides far more access than needed, violating least privilege.
B. Incorrect - Granting Editor role on the production project also provides broad access, including future permissions automatically, which is not safe.
C. Correct - A custom IAM role gives only the specific required permissions for the DevOps team. This avoids granting broad Editor rights and prevents privilege creep when Google updates predefined roles.
D. Incorrect - Assigning the custom role at the organization level is unnecessary and too broad. The role should only apply to the production project.

Question 39
You are building an application that processes data files uploaded from thousands of suppliers. Your primary goals for the application are data security and the expiration of aged data. You need to design the application to:

- Restrict access so that suppliers can access only their own data.
- Give suppliers write access to data only for 30 minutes.
- Delete data that is over 45 days old.
  You have a very short development cycle, and you need to make sure that the application requires minimal maintenance. Which two strategies should you use? (Choose two.)

Options
A. Build a lifecycle policy to delete Cloud Storage objects after 45 days.
B. Use signed URLs to allow suppliers limited time access to store their objects.
C. Set up an SFTP server for your application, and create a separate user for each supplier.
D. Build a Cloud function that triggers a timer of 45 days to delete objects that have expired.
E. Develop a script that loops through all Cloud Storage buckets and deletes any buckets that are older than 45 days.

Answer
A, B

Detailed explanation
A. Correct - A lifecycle policy can automatically delete objects older than 45 days without manual intervention.
B. Correct - Signed URLs provide suppliers secure, time-limited write access to their own objects. This matches the 30-minute write requirement.
C. Incorrect - Setting up an SFTP server and managing separate accounts for thousands of suppliers adds complexity and maintenance.
D. Incorrect - Cloud Functions with timers are unnecessary when lifecycle policies can manage deletion automatically.
E. Incorrect - Scripts for deletion are not scalable or recommended. Buckets may contain data for multiple suppliers.

Question 40
Your company wants to standardize the creation and management of multiple Google Cloud resources using Infrastructure as Code. You want to minimize the amount of repetitive code needed to manage the environment. What should you do?

Options
A. Develop templates for the environment using Cloud Deployment Manager.
B. Use curl in a terminal to send a REST request to the relevant Google API for each individual resource.
C. Use the Cloud Console interface to provision and manage all related resources.
D. Create a bash script that contains all requirement steps as gcloud commands.

Answer
A

Detailed explanation
A. Correct - Cloud Deployment Manager allows you to define reusable templates for infrastructure as code, reducing repetition and enabling automation.
B. Incorrect - Using curl for each resource is manual, repetitive, and error-prone.
C. Incorrect - Cloud Console is GUI-based and not efficient for repeatable infrastructure management.
D. Incorrect - Bash scripts can automate tasks but are harder to maintain and less structured than declarative templates.

Question 41
You are performing a monthly security check of your Google Cloud environment and want to know who has access to view data stored in your Google Cloud Project. What should you do?

Options
A. Enable Audit Logs for all APIs that are related to data storage.
B. Review the IAM permissions for any role that allows for data access.
C. Review the Identity-Aware Proxy settings for each resource.
D. Create a Data Loss Prevention job.

Answer
B

Detailed explanation
A. Incorrect. Audit Logs tell you what happened, but they don't directly show who currently has access.
B. Correct. Reviewing IAM permissions shows exactly who has access to data resources.
C. Incorrect. Identity-Aware Proxy settings apply only if IAP is used; it's not the general way to check access.
D. Incorrect. DLP jobs classify/sanitize data, not check who has access.

Question 42
Your company has embraced a hybrid cloud strategy where some applications are deployed on Google Cloud. A VPN tunnel connects your VPC in Google Cloud with your company's on-premises network. Multiple applications in Google Cloud need to connect to an on-premises database server, and you want to avoid having to change the IP configuration in all your applications when the IP of the database changes. What should you do?

Options
A. Configure Cloud NAT for all subnets of your VPC to be used when egressing from the VM instances.
B. Create a private zone on Cloud DNS, and configure the applications with the DNS name.
C. Configure the IP of the database as custom metadata for each instance, and query the metadata server.
D. Query the Compute Engine internal DNS from the applications to retrieve the IP of the database.

Answer
B

Detailed explanation
A. Incorrect. Cloud NAT manages outbound internet traffic, not DNS resolution.
B. Correct. Using a private zone in Cloud DNS with a DNS name avoids IP dependency.
C. Incorrect. Metadata server is per-VM, not a scalable solution for central database IP resolution.
D. Incorrect. Compute Engine internal DNS only resolves internal GCP resources, not on-prem databases.

Question 43
You have developed a containerized web application to serve internal colleagues during business hours. You want to ensure that no costs are incurred outside business hours. You just created a new Google Cloud project and want to deploy the application. What should you do?

Options
A. Deploy the container on Cloud Run for Anthos, and set the minimum number of instances to zero.
B. Deploy the container on Cloud Run (fully managed), and set the minimum number of instances to zero.
C. Deploy the container on App Engine flexible environment with autoscaling, and set the value min_instances to zero in the app.yaml.
D. Deploy the container on App Engine flexible environment with manual scaling, and set the value instances to zero in the app.yaml.

Answer
B

Detailed explanation
A. Incorrect. Cloud Run for Anthos depends on GKE and still incurs cluster costs.
B. Correct. Cloud Run (fully managed) allows min instances = 0, incurring no cost when idle.
C. Incorrect. App Engine flexible requires at least 1 instance running.
D. Incorrect. Manual scaling cannot scale to zero.

Question 44
You have experimented with Google Cloud using your own credit card. Your company wants billing streamlined and project costs charged to the company's monthly invoice. What should you do?

Options
A. Grant the financial team the IAM role of Billing Account User on the billing account linked to your credit card.
B. Set up BigQuery billing export and grant your financial department IAM access to query the data.
C. Create a ticket with Google Billing Support to ask them to send the invoice to your company.
D. Change the billing account of your projects to the billing account of your company.

Answer
D

Detailed explanation
A. Incorrect. Giving IAM role on your personal billing account won't move charges to the company.
B. Incorrect. Exporting billing data gives visibility, not billing control.
C. Incorrect. Billing support won't transfer charges; it must be configured in GCP.
D. Correct. Changing the billing account to the company's account ensures costs are charged centrally.

Question 45
You are running a BigQuery data warehouse. A partner company needs access to your dataset to run their recommendation engine in their own project. What should you do?

Options
A. Create a Service Account in your project, and grant this Service Account access to BigQuery in your project.
B. Create a Service Account in your project, and ask the partner to grant this Service Account access in their project.
C. Ask the partner to create a Service Account in their project, and have them give this Service Account access to BigQuery in their project.
D. Ask the partner to create a Service Account in their project, and grant their Service Account access to the BigQuery dataset in your project.

Answer
D

Detailed explanation
A. Incorrect. Your service account would require distribution of private keys, not secure.
B. Incorrect. Cross-project access doesn't work this way.
C. Incorrect. This grants access to their project, not your dataset.
D. Correct. Best practice: partner creates their Service Account, and you grant dataset access to that account.

Question 46
Your web application runs on Cloud Run for Anthos. You want to test a new version with a percentage of production users (canary deployment). What should you do?

Options
A. Create a new service with the new version of the application. Split traffic between this version and the version that is currently running.
B. Create a new revision with the new version of the application. Split traffic between this version and the version that is currently running.
C. Create a new service with the new version of the application. Add an HTTP Load Balancer in front of both services.
D. Create a new revision with the new version of the application. Add an HTTP Load Balancer in front of both revisions.

Answer
B

Detailed explanation
A. Incorrect. Canary deployment in Cloud Run Anthos is done at revision level, not separate service.
B. Correct. Splitting traffic between revisions is the supported method for canary releases.
C. Incorrect. HTTP LB is unnecessary for canary in Cloud Run Anthos.
D. Incorrect. Revisions do not require an external LB.

Question 47
You developed a mobile game deployed on Google Cloud. It sends UDP packets and needs to expose VMs under a single IP for multiplayer gaming. What should you do?

Options
A. Configure an SSL Proxy load balancer in front of the application servers.
B. Configure an Internal UDP load balancer in front of the application servers.
C. Configure an External HTTP(s) load balancer in front of the application servers.
D. Configure an External Network load balancer in front of the application servers.

Answer
D

Detailed explanation
A. Incorrect. SSL Proxy LB only works with TCP.
B. Incorrect. Internal UDP LB won't expose resources externally.
C. Incorrect. HTTP(S) LB only supports HTTP/S traffic, not raw UDP.
D. Correct. External Network LB supports UDP and provides single external IP.

Question 48
You are working for a hospital that stores medical images in an on-premises data room. The hospital wants an automated process to upload new medical images to Cloud Storage for archival. What should you do?

Options
A. Create a Pub/Sub topic, and enable a Cloud Storage trigger for the Pub/Sub topic. Create an application that sends all medical images to the Pub/Sub topic.
B. Deploy a Dataflow job from the batch template, Datastore to Cloud Storage. Schedule the batch job on the desired interval.
C. Create a script that uses the gsutil command-line interface to synchronize the on-premises storage with Cloud Storage. Schedule the script as a cron job.
D. In the Cloud Console, go to Cloud Storage. Upload the relevant images to the appropriate bucket.

Answer
C

Detailed explanation
A. Incorrect. Pub/Sub is not designed for direct file upload triggers.
B. Incorrect. Datastore → Cloud Storage isn't relevant here.
C. Correct. gsutil rsync with cron automates syncing on-prem files to Cloud Storage.
D. Incorrect. Manual upload does not meet the requirement for automation.

Question 49
Your auditor wants to view who accessed data in Cloud Storage buckets. What should you do?

Options
A. Turn on Data Access Logs for the buckets they want to audit, and build a query in the log viewer that filters on Cloud Storage.
B. Assign the appropriate permissions, and then create a Data Studio report on Admin Activity Audit Logs.
C. Assign the appropriate permissions, and then use Cloud Monitoring to review metrics.
D. Use the export logs API to provide the Admin Activity Audit Logs in the format they want.

Answer
A

Detailed explanation
A. Correct. Data Access Logs track who accessed Cloud Storage objects.
B. Incorrect. Admin Activity Logs show admin actions, not data access.
C. Incorrect. Monitoring shows metrics, not user access details.
D. Incorrect. Export logs API still relies on enabling Data Access Logs first.

Question 50
You received a JSON file containing a private key of a Service Account. You want to use it for authentication when performing gcloud commands. What should you do?

Options
A. Use the command gcloud auth login and point it to the private key.
B. Use the command gcloud auth activate-service-account and point it to the private key.
C. Place the private key file in the installation directory of the Cloud SDK and rename it to credentials.json.
D. Place the private key file in your home directory and rename it to GOOGLE_APPLICATION_CREDENTIALS.

Answer
B

Detailed explanation
A. Incorrect. gcloud auth login is for user accounts, not service accounts.
B. Correct. gcloud auth activate-service-account with the JSON key is the right method.
C. Incorrect. Renaming to credentials.json doesn't automatically authenticate.
D. Incorrect. GOOGLE_APPLICATION_CREDENTIALS is used by client libraries, not gcloud CLI.

Question 51
You are working with a Cloud SQL MySQL database at your company. You need to retain a month-end copy of the database for three years for audit purposes. What should you do?

Options
A. Set up an export job for the first of the month. Write the export file to an Archive class Cloud Storage bucket.
B. Save the automatic first-of-the-month backup for three years. Store the backup file in an Archive class Cloud Storage bucket.
C. Set up an on-demand backup for the first of the month. Write the backup to an Archive class Cloud Storage bucket.
D. Convert the automatic first-of-the-month backup to an export file. Write the export file to a Coldline class Cloud Storage bucket.

Answer
A

Detailed explanation
A. Correct - exporting the database monthly and storing it in an Archive class Cloud Storage bucket is the most cost-effective and compliant way to retain data for 3 years. Exports are designed for long-term storage and are easy to retrieve later.
B. Incorrect - automatic backups are only kept for a limited retention period (not 3 years), so you cannot rely on them for long-term audit purposes.
C. Incorrect - on-demand backups also have the same retention limits as automatic backups, not suitable for 3-year retention.
D. Incorrect - Coldline is more expensive for long-term storage compared to Archive, and converting backups adds unnecessary steps.

Question 52
You are monitoring an application and receive user feedback that a specific error is spiking. You notice that the error is caused by a Service Account having insufficient permissions. You are able to solve the problem but want to be notified if the problem recurs. What should you do?

Options
A. In the Log Viewer, filter the logs on severity "Error" and the name of the Service Account.
B. Create a sink to BigQuery to export all the logs. Create a Data Studio dashboard on the exported logs.
C. Create a custom log-based metric for the specific error to be used in an Alerting Policy.
D. Grant Project Owner access to the Service Account.

Answer
C

Detailed explanation
A. Incorrect - filtering logs manually does not provide automatic alerts. You won't get notified proactively.
B. Incorrect - exporting to BigQuery and building a dashboard only helps with analysis, not real-time alerts.
C. Correct - creating a log-based metric for the error and using it in an Alerting Policy ensures you get real-time notifications if the error recurs.
D. Incorrect - granting Project Owner to a service account is a security risk and not the right way to fix or monitor the issue.

Question 53
You are developing a financial trading application that will be used globally. Data is stored and queried using a relational structure, and clients from all over the world should get the exact identical state of the data. The application will be deployed in multiple regions to provide the lowest latency to end users. You need to select a storage option for the application data while minimizing latency. What should you do?

Options
A. Use Cloud Bigtable for data storage.
B. Use Cloud SQL for data storage.
C. Use Cloud Spanner for data storage.
D. Use Firestore for data storage.

Answer
C

Detailed explanation
A. Incorrect - Cloud Bigtable is NoSQL, optimized for large scale analytical/IoT workloads, not relational or transactional global consistency.
B. Incorrect - Cloud SQL is regional only and cannot provide global consistency and low latency to worldwide users.
C. Correct - Cloud Spanner provides globally distributed, strongly consistent relational database storage, ideal for financial trading systems.
D. Incorrect - Firestore is NoSQL, eventually consistent in some modes, and not suitable for strict global relational consistency.

Question 54
You are about to deploy a new Enterprise Resource Planning (ERP) system on Google Cloud. The application holds the full database in-memory for fast data access, and you need to configure the most appropriate resources on Google Cloud for this application. What should you do?

Options
A. Provision preemptible Compute Engine instances.
B. Provision Compute Engine instances with GPUs attached.
C. Provision Compute Engine instances with local SSDs attached.
D. Provision Compute Engine instances with M1 machine type.

Answer
D

Detailed explanation
A. Incorrect - preemptible instances can be terminated anytime, not suitable for ERP systems that need reliability.
B. Incorrect - GPUs are for compute-intensive tasks like ML, not needed for ERP in-memory workloads.
C. Incorrect - local SSDs help with I/O, but the question specifies in-memory workloads, so RAM capacity matters more.
D. Correct - using Compute Engine instances with M1 machine type provides high memory optimized instances, perfect for in-memory databases.

Question 55
You have developed an application that consists of multiple microservices, with each microservice packaged in its own Docker container image. You want to deploy the entire application on Google Kubernetes Engine so that each microservice can be scaled individually. What should you do?

Options
A. Create and deploy a Custom Resource Definition per microservice.
B. Create and deploy a Docker Compose File.
C. Create and deploy a Job per microservice.
D. Create and deploy a Deployment per microservice.

Answer
D

Detailed explanation
A. Incorrect - Custom Resource Definitions are for extending Kubernetes APIs, not for deploying standard workloads like microservices.
B. Incorrect - Docker Compose is not a native Kubernetes deployment method; it's meant for local container orchestration.
C. Incorrect - Jobs are for short-lived, batch tasks, not long-running microservices.
D. Correct - Deployments are the correct way to manage scalable, long-running workloads like microservices in Kubernetes.

Question 56
You will have several applications running on different Compute Engine instances in the same project. You want to specify at a more granular level the service account each instance uses when calling Google Cloud APIs. What should you do?

Options
A. When creating the instances, specify a Service Account for each instance.
B. When creating the instances, assign the name of each Service Account as instance metadata.
C. After starting the instances, use gcloud compute instances update to specify a Service Account for each instance.
D. After starting the instances, use gcloud compute instances update to assign the name of the relevant Service Account as instance metadata.

Answer
A

Detailed explanation
A. Correct - the right way is to attach the correct service account at instance creation, so the instance uses it for API calls.
B. Incorrect - simply adding service account names in metadata does not give authentication capabilities.
C. Incorrect - you cannot assign a new service account to an existing instance after creation (only at creation).
D. Incorrect - again, instance metadata does not configure authentication properly.

Question 57
You are creating an application that will run on Google Kubernetes Engine. You have identified MongoDB as the most suitable database system for your application and want to deploy a managed MongoDB environment that provides a support SLA. What should you do?

Options
A. Create a Cloud Bigtable cluster, and use the HBase API.
B. Deploy MongoDB Atlas from the Google Cloud Marketplace.
C. Download a MongoDB installation package, and run it on Compute Engine instances.
D. Download a MongoDB installation package, and run it on a Managed Instance Group.

Answer
B

Detailed explanation
A. Incorrect - Bigtable with HBase API is not MongoDB, and it doesn't meet the requirement for MongoDB support SLA.
B. Correct - MongoDB Atlas is a managed service available via Google Cloud Marketplace, offering MongoDB with SLA.
C. Incorrect - running MongoDB manually on Compute Engine instances requires you to manage everything (no SLA).
D. Incorrect - Managed Instance Groups still require self-management of MongoDB, not SLA-backed.

Question 58
You are managing a project for the Business Intelligence (BI) department in your company. A data pipeline ingests data into BigQuery via streaming. You want the users in the BI department to be able to run the custom SQL queries against the latest data in BigQuery. What should you do?

Options
A. Create a Data Studio dashboard that uses the related BigQuery tables as a source and give the BI team view access to the Data Studio dashboard.
B. Create a Service Account for the BI team and distribute a new private key to each member of the BI team.
C. Use Cloud Scheduler to schedule a batch Dataflow job to copy the data from BigQuery to the BI team's internal data warehouse.
D. Assign the IAM role of BigQuery User to a Google Group that contains the members of the BI team.

Answer
D

Detailed explanation
A. Incorrect - Data Studio dashboards give only predefined reports, not custom SQL query access.
B. Incorrect - distributing private keys is insecure and not best practice for team-wide access.
C. Incorrect - copying data to another warehouse is unnecessary and adds complexity.
D. Correct - assigning the IAM role BigQuery User to a Google Group allows BI team members to run custom SQL queries directly on the latest BigQuery data.

Question 59
Your company is moving its entire workload to Compute Engine. Some servers should be accessible through the Internet, and other servers should only be accessible over the internal network. All servers need to talk to each other over specific ports and protocols. The current on-premises network relies on a demilitarized zone (DMZ) for the public servers and a Local Area Network (LAN) for the private servers. You need to design the networking infrastructure on Google Cloud to match these requirements. What should you do?

Options
A. Create a single VPC with a subnet for the DMZ and a subnet for the LAN. Set up firewall rules to open up relevant traffic between the DMZ and the LAN subnets, and another firewall rule to allow public ingress traffic for the DMZ.
B. Create a single VPC with a subnet for the DMZ and a subnet for the LAN. Set up firewall rules to open up relevant traffic between the DMZ and the LAN subnets, and another firewall rule to allow public egress traffic for the DMZ.
C. Create a VPC with a subnet for the DMZ and another VPC with a subnet for the LAN. Set up firewall rules to open up relevant traffic between the DMZ and the LAN subnets, and another firewall rule to allow public ingress traffic for the DMZ.
D. Create a VPC with a subnet for the DMZ and another VPC with a subnet for the LAN. Set up firewall rules to open up relevant traffic between the DMZ and the LAN subnets, and another firewall rule to allow public egress traffic for the DMZ.

Answer
A

Detailed explanation
A. Correct - one VPC with two subnets (DMZ + LAN) and proper firewall rules ensures communication between internal and external servers securely.
B. Incorrect - “public egress traffic” is not what’s needed; you need ingress for the DMZ to accept outside traffic.
C. Incorrect - multiple VPCs would complicate communication between LAN and DMZ unnecessarily.
D. Incorrect - same issue as option B, egress traffic rule is wrong for DMZ.

Question 60
You have just created a new project which will be used to deploy a globally distributed application. You will use Cloud Spanner for data storage. You want to create a Cloud Spanner instance. You want to perform the first step in preparation of creating the instance. What should you do?

Options
A. Enable the Cloud Spanner API.
B. Configure your Cloud Spanner instance to be multi-regional.
C. Create a new VPC network with subnetworks in all desired regions.
D. Grant yourself the IAM role of Cloud Spanner Admin.

Answer
A

Detailed explanation
A. Correct - before creating a Cloud Spanner instance, the first step is to enable the Cloud Spanner API in the project.
B. Incorrect - configuring multi-regional is part of the instance creation process, not the first step.
C. Incorrect - a new VPC is not required specifically for Cloud Spanner, and not the initial step.
D. Incorrect - assigning IAM role might be necessary, but the very first required step is enabling the API.

Question 61
You have created a new project in Google Cloud through the gcloud command line interface (CLI) and linked a billing account. You need to create a new Compute Engine instance using the CLI. You need to perform the prerequisite steps. What should you do?

Options
A. Create a Cloud Monitoring Workspace.
B. Create a VPC network in the project.
C. Enable the compute.googleapis.com API.
D. Grant yourself the IAM role of Computer Admin.

Answer
C

Detailed explanation
A. Incorrect - Cloud Monitoring Workspace is for monitoring, not a prerequisite to creating Compute Engine instances.
B. Incorrect - a default VPC network is automatically created in new projects, so this is not a required first step.
C. Correct - enabling the Compute Engine API is the required prerequisite before you can create any instances.
D. Incorrect - while IAM roles are important, the Compute Engine API must be enabled first.

Question 62
Your company has developed a new application that consists of multiple microservices. You want to deploy the application to Google Kubernetes Engine (GKE), and you want to ensure that the cluster can scale as more applications are deployed in the future. You want to avoid manual intervention when each new application is deployed. What should you do?

Options
A. Deploy the application on GKE, and add a HorizontalPodAutoscaler to the deployment.
B. Deploy the application on GKE, and add a VerticalPodAutoscaler to the deployment.
C. Create a GKE cluster with autoscaling enabled on the node pool. Set a minimum and maximum for the size of the node pool.
D. Create a separate node pool for each application, and deploy each application to its dedicated node pool.

Answer
C

Detailed explanation
A. Incorrect - HorizontalPodAutoscaler scales pods, not the underlying cluster nodes.
B. Incorrect - VerticalPodAutoscaler adjusts pod resources but does not scale the cluster automatically.
C. Correct - enabling autoscaling on the node pool ensures the cluster scales automatically as applications grow.
D. Incorrect - creating separate node pools for each application adds unnecessary management complexity.

Question 63
You need to manage a third-party application that will run on a Compute Engine instance. Other Compute Engine instances are already running with default configuration. Application installation files are hosted on Cloud Storage. You need to access these files from the new instance without allowing other virtual machines (VMs) to access these files. What should you do?

Options
A. Create the instance with the default Compute Engine service account. Grant the service account permissions on Cloud Storage.
B. Create the instance with the default Compute Engine service account. Add metadata to the objects on Cloud Storage that matches the metadata on the new instance.
C. Create a new service account and assign this service account to the new instance. Grant the service account permissions on Cloud Storage.
D. Create a new service account and assign this service account to the new instance. Add metadata to the objects on Cloud Storage that matches the metadata on the new instance.

Answer
C

Detailed explanation
A. Incorrect - the default Compute Engine service account would allow all instances using it to access the files, which is insecure.
B. Incorrect - object metadata matching instance metadata is not a valid method for controlling access.
C. Correct - creating a dedicated service account and granting it access ensures only the new instance can access the Cloud Storage files.
D. Incorrect - metadata matching is not a proper security mechanism; IAM roles with service accounts are the right approach.

Question 64
You need to configure optimal data storage for files stored in Cloud Storage for minimal cost. The files are used in a mission-critical analytics pipeline that is used continually. The users are in Boston, MA (United States). What should you do?

Options
A. Configure regional storage for the region closest to the users. Configure a Nearline storage class.
B. Configure regional storage for the region closest to the users. Configure a Standard storage class.
C. Configure dual-regional storage for the dual region closest to the users. Configure a Nearline storage class.
D. Configure dual-regional storage for the dual region closest to the users. Configure a Standard storage class.

Answer
D

Detailed explanation
A. Incorrect - Nearline is for infrequently accessed data, not suitable for continually accessed mission-critical analytics.
B. Incorrect - regional Standard would provide performance but no redundancy across regions.
C. Incorrect - Nearline is again not suitable because the pipeline needs continuous access.
D. Correct - dual-regional Standard provides high availability, durability, and low latency for critical analytics workloads.

Question 65
You are developing a new web application that will be deployed on Google Cloud Platform. As part of your release cycle, you want to test updates to your application on a small portion of real user traffic. The majority of the users should still be directed towards a stable version of your application. What should you do?

Options
A. Deploy the application on App Engine. For each update, create a new version of the same service. Configure traffic splitting to send a small percentage of traffic to the new version.
B. Deploy the application on App Engine. For each update, create a new service. Configure traffic splitting to send a small percentage of traffic to the new service.
C. Deploy the application on Kubernetes Engine. For a new release, update the deployment to use the new version.
D. Deploy the application on Kubernetes Engine. For a new release, create a new deployment for the new version. Update the service to use the new deployment.

Answer
A

Detailed explanation
A. Correct - App Engine natively supports traffic splitting between versions, making it easy to test updates with a small user percentage.
B. Incorrect - traffic splitting works between versions of the same service, not between separate services.
C. Incorrect - updating the deployment immediately directs all traffic to the new version, no partial rollout.
D. Incorrect - same as option C, there’s no traffic splitting mechanism built-in like App Engine.

Question 66
You need to add a group of new users to Cloud Identity. Some of the users already have existing Google accounts. You want to follow one of Google's recommended practices and avoid conflicting accounts. What should you do?

Options
A. Invite the user to transfer their existing account.
B. Invite the user to use an email alias to resolve the conflict.
C. Tell the user that they must delete their existing account.
D. Tell the user to remove all personal email from the existing account.

Answer
A

Detailed explanation
A. Correct - the recommended practice is to invite users to transfer their existing Google accounts into Cloud Identity.
B. Incorrect - email aliases don't resolve account conflicts in Cloud Identity.
C. Incorrect - users do not need to delete their accounts; they can transfer.
D. Incorrect - removing personal email doesn't solve the identity conflict.

Question 67
You need to manage a Cloud Spanner instance for best query performance. Your instance in production runs in a single Google Cloud region. You need to improve performance in the shortest amount of time. You want to follow Google best practices for service configuration. What should you do?

Options
A. Create an alert in Cloud Monitoring to alert when the percentage of high priority CPU utilization reaches 45%. If you exceed this threshold, add nodes to your instance.
B. Create an alert in Cloud Monitoring to alert when the percentage of high priority CPU utilization reaches 45%. Use database query statistics to identify queries that result in high CPU usage, and then rewrite those queries to optimize their resource usage.
C. Create an alert in Cloud Monitoring to alert when the percentage of high priority CPU utilization reaches 65%. If you exceed this threshold, add nodes to your instance.
D. Create an alert in Cloud Monitoring to alert when the percentage of high priority CPU utilization reaches 65%. Use database query statistics to identify queries that result in high CPU usage, and then rewrite those queries to optimize their resource usage.

Answer
C

Detailed explanation
A. Incorrect - 45% is too low a threshold; adding nodes prematurely increases costs.
B. Incorrect - query tuning is good but may take longer than simply adding nodes for quick performance improvement.
C. Correct - best practice is to scale when high-priority CPU usage exceeds 65%, adding nodes for faster performance improvement.
D. Incorrect - while query optimization is good, the question asks for the shortest amount of time solution, which is adding nodes.

Question 68
Your company has an internal application for managing transactional orders. The application is used exclusively by employees in a single physical location. The application requires strong consistency, fast queries, and ACID guarantees for multi-table transactional updates. The first version of the application is implemented in PostgreSQL, and you want to deploy it to the cloud with minimal code changes. Which database is most appropriate for this application?

Options
A. BigQuery
B. Cloud SQL
C. Cloud Spanner
D. Cloud Datastore

Answer
B

Detailed explanation
A. Incorrect - BigQuery is for analytics, not transactional workloads.
B. Correct - Cloud SQL supports PostgreSQL, provides ACID compliance, and requires minimal code changes to migrate.
C. Incorrect - Cloud Spanner requires schema and application changes, not minimal migration.
D. Incorrect - Cloud Datastore is NoSQL and not suitable for relational ACID transactions.

Question 69
You are assigned to maintain a Google Kubernetes Engine (GKE) cluster named "dev" that was deployed on Google Cloud. You want to manage the GKE configuration using the command line interface (CLI). You have just downloaded and installed the Cloud SDK. You want to ensure that future CLI commands by default address this specific cluster. What should you do?

Options
A. Use the command gcloud config set container/cluster dev.
B. Use the command gcloud container clusters update dev.
C. Create a file called gke.default in the ~/.gcloud folder that contains the cluster name.
D. Create a file called defaults.json in the ~/.gcloud folder that contains the cluster name.

Answer
A

Detailed explanation
A. Correct - setting the cluster in gcloud config ensures future commands target that cluster by default.
B. Incorrect - update command changes configuration of the cluster, not CLI default settings.
C. Incorrect - there’s no such gke.default file supported by gcloud.
D. Incorrect - defaults.json is not a valid method for setting cluster defaults.

Question 70
The sales team has a project named Sales Data Digest that has the ID acme-data-digest. You need to set up similar Google Cloud resources for the marketing team but their resources must be organized independently of the sales team. What should you do?

Options
A. Grant the Project Editor role to the Marketing team for acme-data-digest.
B. Create a Project Lien on acme-data-digest and then grant the Project Editor role to the Marketing team.
C. Create another project with the ID acme-marketing-data-digest for the Marketing team and deploy the resources there.
D. Create a new project named Marketing Data Digest and use the ID acme-data-digest. Grant the Project Editor role to the Marketing team.

Answer
C

Detailed explanation
A. Incorrect - giving the Marketing team access to the Sales project mixes resources, not independent organization.
B. Incorrect - Project Liens prevent deletion, they do not help in organizing separate team resources.
C. Correct - the best approach is to create a new project for the Marketing team with its own resources, independent of Sales.
D. Incorrect - project IDs must be globally unique; you cannot reuse acme-data-digest.

Question 71
You have deployed multiple Linux instances on Compute Engine. You plan on adding more instances in the coming weeks. You want to be able to access all of these instances through your SSH client over the internet without having to configure specific access on the existing and new instances. You do not want the Compute Engine instances to have a public IP. What should you do?

Options
A. Configure Cloud Identity-Aware Proxy for HTTPS resources.
B. Configure Cloud Identity-Aware Proxy for SSH and TCP resources.
C. Create an SSH keypair and store the public key as a project-wide SSH Key.
D. Create an SSH keypair and store the private key as a project-wide SSH Key.

Answer
B

Detailed explanation
A. Incorrect - HTTPS IAP only works for web traffic, not SSH/TCP connections.
B. Correct - IAP for SSH/TCP allows secure connections to VMs without external IPs, perfect for this use case.
C. Incorrect - project-wide SSH keys would expose access to all instances but still require external IPs.
D. Incorrect - private keys are never stored in the project; only public keys are used.

Question 72
You have created an application that is packaged into a Docker image. You want to deploy the Docker image as a workload on Google Kubernetes Engine. What should you do?

Options
A. Upload the image to Cloud Storage and create a Kubernetes Service referencing the image.
B. Upload the image to Cloud Storage and create a Kubernetes Deployment referencing the image.
C. Upload the image to Container Registry and create a Kubernetes Service referencing the image.
D. Upload the image to Container Registry and create a Kubernetes Deployment referencing the image.

Answer
D

Detailed explanation
A. Incorrect - Kubernetes cannot pull images directly from Cloud Storage.
B. Incorrect - Deployment requires pulling from a container registry, not Cloud Storage.
C. Incorrect - a Service exposes workloads, but you need a Deployment to run containers.
D. Correct - Container Registry is the right place to store Docker images, and Kubernetes Deployments are used to run them.

Question 73
You are using Data Studio to visualize a table from your data warehouse that is built on top of BigQuery. Data is appended to the data warehouse during the day. At night, the daily summary is recalculated by overwriting the table. You just noticed that the charts in Data Studio are broken, and you want to analyze the problem. What should you do?

Options
A. Review the Error Reporting page in the Cloud Console to find any errors.
B. Use the BigQuery interface to review the nightly job and look for any errors.
C. Use Cloud Debugger to find out why the data was not refreshed correctly.
D. In Cloud Logging, create a filter for your Data Studio report.

Answer
B

Detailed explanation
A. Incorrect - Error Reporting is for application errors, not BigQuery job failures.
B. Correct - the nightly job overwrites the table; checking the BigQuery job logs reveals if it failed.
C. Incorrect - Cloud Debugger is for debugging applications, not SQL jobs.
D. Incorrect - filtering logs doesn’t directly show why the BigQuery job failed.

Question 74
You have been asked to set up the billing configuration for a new Google Cloud customer. Your customer wants to group resources that share common IAM policies. What should you do?

Options
A. Use labels to group resources that share common IAM policies.
B. Use folders to group resources that share common IAM policies.
C. Set up a proper billing account structure to group IAM policies.
D. Set up a proper project naming structure to group IAM policies.

Answer
B

Detailed explanation
A. Incorrect - labels are for organization/metadata, not IAM policy grouping.
B. Correct - folders allow grouping of projects/resources and can share IAM policies.
C. Incorrect - billing accounts control billing, not IAM policies.
D. Incorrect - naming structures are for identification, not IAM enforcement.

Question 75
You have been asked to create robust Virtual Private Network (VPN) connectivity between a new Virtual Private Cloud (VPC) and a remote site. Key requirements include dynamic routing, a shared address space of 10.19.0.1/22, and no overprovisioning of tunnels during a failover event. You want to follow Google-recommended practices to set up a high availability Cloud VPN. What should you do?

Options
A. Use a custom mode VPC network, configure static routes, and use active/passive routing.
B. Use an automatic mode VPC network, configure static routes, and use active/active routing.
C. Use an automatic mode VPC network, use Cloud Router border gateway protocol (BGP) routes, and use active/passive routing.
D. Use an automatic mode VPC network, use Cloud Router border gateway protocol (BGP) routes, and configure policy-based routing.

Answer
C

Detailed explanation
A. Incorrect - static routes don’t provide dynamic routing.
B. Incorrect - static routes with automatic mode is not best practice for HA VPN.
C. Correct - Google best practice is automatic VPC with dynamic routing (BGP) and active/passive setup for HA VPN.
D. Incorrect - policy-based routing is not the recommended HA VPN method.

Question 76
You are running multiple microservices in a Kubernetes Engine cluster. One microservice is rendering images. The microservice responsible for the image rendering requires a large amount of CPU time compared to the memory it requires. The other microservices are workloads optimized for n1-standard machine types. You need to optimize your cluster so all workloads use resources efficiently. What should you do?

Options
A. Assign the pods of the image rendering microservice a higher pod priority than the other microservices.
B. Create a node pool with compute-optimized machine type nodes for the image rendering microservice. Use the node pool with general-purpose machine type nodes for the other microservices.
C. Use the node pool with general-purpose machine type nodes for the image rendering microservice. Create a node pool with compute-optimized machine type nodes for the other microservices.
D. Configure the required amount of CPU and memory in the resource requests specification of the image rendering microservice deployment. Keep the resource requests for the other microservices at the default.

Answer
B

Detailed explanation
A. Incorrect - pod priority affects scheduling but not resource optimization.
B. Correct - compute-optimized node pools handle CPU-intensive workloads efficiently, while general-purpose nodes suit standard workloads.
C. Incorrect - this reverses the requirement; the rendering service needs compute-optimized, not general-purpose.
D. Incorrect - resource requests don't change underlying machine type optimization.

Question 77
Your organization has three existing Google Cloud projects. You need to bill the Marketing department for only their Google Cloud services for a new initiative within their group. What should you do?

Options
A. Verify that you are assigned the Billing Administrator IAM role for your organization's Google Cloud Project for the Marketing department. Link the new project to a Marketing Billing Account.
B. Verify that you are assigned the Billing Administrator IAM role for your organization's Google Cloud account. Create a new Google Cloud Project for the Marketing department. Set the default key-value project labels to department:marketing for all services in this project.
C. Verify that you are assigned the Organization Administrator IAM role for your organization's Google Cloud account. Create a new Google Cloud Project for the Marketing department. Link the new project to a Marketing Billing Account.
D. Verify that you are assigned the Organization Administrator IAM role for your organization's Google Cloud account. Create a new Google Cloud Project for the Marketing department. Set the default key-value project labels to department:marketing for all services in this project.

Answer
B

Detailed explanation
A. Incorrect - creating only a billing account link doesn't guarantee billing separation.
B. Correct - using a new project with labels ensures Marketing's costs are tracked separately and billed properly.
C. Incorrect - Org Admin is not necessary here; Billing Admin and labels suffice.
D. Incorrect - labels help for tracking but billing isolation still requires a separate project and account link.

Question 78
You deployed an application on a managed instance group in Compute Engine. The application accepts Transmission Control Protocol (TCP) traffic on port 389 and requires you to preserve the IP address of the client making a request. You want to expose the application to the internet by using a load balancer. What should you do?

Options
A. Expose the application by using an external TCP Network Load Balancer.
B. Expose the application by using a TCP Proxy Load Balancer.
C. Expose the application by using an SSL Proxy Load Balancer.
D. Expose the application by using an internal TCP Network Load Balancer.

Answer
B

Detailed explanation
A. Incorrect - TCP Network Load Balancer does not preserve client IP for global scale scenarios.
B. Correct - TCP Proxy Load Balancer supports preserving client IP and works for TCP traffic like port 389.
C. Incorrect - SSL Proxy is for HTTPS/SSL traffic, not raw TCP.
D. Incorrect - internal load balancer would not expose to the internet.

Question 79
You are building a multi-player gaming application that will store game information in a database. As popularity grows, you are concerned about delivering consistent performance. You need to ensure optimal gaming performance for global users without increasing management complexity. What should you do?

Options
A. Use Cloud SQL database with cross-region replication to store game statistics in the EU, US, and APAC regions.
B. Use Cloud Spanner to store user data mapped to the game statistics.
C. Use BigQuery to store game statistics with a Redis on Memorystore instance in the front to provide global consistency.
D. Store game statistics in a Bigtable database partitioned by username.

Answer
B

Detailed explanation
A. Incorrect - Cloud SQL replication is regional and not ideal for global consistency.
B. Correct - Cloud Spanner provides globally consistent relational storage, perfect for gaming workloads with global users.
C. Incorrect - BigQuery + Redis is for analytics, not transactional game data.
D. Incorrect - Bigtable is NoSQL and not suited for strong global consistency requirements.

Question 80
You are building an application that stores relational data from users. Users across the globe will use this application. Your CTO is concerned about scaling requirements because the user base size is unknown. You need to implement a database solution that can scale with user growth with minimum configuration changes. Which storage solution should you use?

Options
A. Cloud SQL
B. Firestore
C. Cloud Spanner
D. Bigtable

Answer
C

Detailed explanation
A. Incorrect - Cloud SQL is regional and cannot scale seamlessly with global workloads.
B. Incorrect - Firestore is NoSQL, not relational.
C. Correct - Cloud Spanner is a fully managed, globally scalable relational database, ideal for unpredictable growth.
D. Incorrect - Bigtable is also NoSQL and lacks relational structure.

Question 81
Your company has multiple projects linked to a single billing account in Google Cloud. You need to visualize the costs with specific metrics that should be dynamically calculated based on company-specific criteria. You want to automate the process. What should you do?

Options
A. In the Google Cloud console, visualize the costs related to the projects in the Reports section.
B. In the Google Cloud console, visualize the costs related to the projects in the Cost breakdown section.
C. In the Google Cloud console, use the export functionality of the Cost table. Create a Looker Studio dashboard on top of the CSV export.
D. Configure Cloud Billing data export to BigQuery for the billing account. Create a Looker Studio dashboard on top of the BigQuery export.

Answer
D

Detailed explanation
A. Incorrect - Reports only give static visualizations, not dynamic custom metrics.
B. Incorrect - Cost breakdown shows predefined categories, not flexible for custom automation.
C. Incorrect - CSV export is manual and not automated; it requires constant refresh.
D. Correct - exporting billing data to BigQuery enables automated, queryable billing datasets, and Looker Studio can visualize them dynamically.

Question 82
You have an application that runs on Compute Engine VM instances in a custom Virtual Private Cloud (VPC). Your company's security policies only allow the use of internal IP addresses on VM instances and do not let VM instances connect to the internet. You need to ensure that the application can access a file hosted in a Cloud Storage bucket within your project. What should you do?

Options
A. Enable Private Service Access on the Cloud Storage Bucket.
B. Add storage.googleapis.com to the list of restricted services in a VPC Service Controls perimeter and add your project to the list of protected projects.
C. Enable Private Google Access on the subnet within the custom VPC.
D. Deploy a Cloud NAT instance and route the traffic to the dedicated IP address of the Cloud Storage bucket.

Answer
C

Detailed explanation
A. Incorrect - Private Service Access is used for private IP connections to Google services like Cloud SQL, not Cloud Storage.
B. Incorrect - VPC Service Controls protect access but do not provide private connectivity.
C. Correct - enabling Private Google Access allows VMs without external IPs to access Google APIs (like Cloud Storage) over internal IPs.
D. Incorrect - Cloud NAT is for outbound internet access, not required if Private Google Access is enabled.

Question 83
Your company completed the acquisition of a startup and is now merging IT systems. The startup had a production Google Cloud project in their organization. You need to move this project into your organization and ensure it is billed to your organization. You want to accomplish this with minimal effort. What should you do?

Options
A. Use the projects.move method to move the project to your organization. Update the billing account of the project to that of your organization.
B. Ensure that you have an Organization Administrator IAM role assigned to you in both organizations. Navigate to the Resource Manager in the startup's Google Cloud organization and drag the project to your company's organization.
C. Create a Private Catalog for the Google Cloud Marketplace, and upload the resources of the startup's production project to the Catalog. Share the Catalog with your organization, and deploy the resources in your company's project.
D. Create an infrastructure-as-code template for all resources in the project by using Terraform, and deploy that template to a new project in your organization. Delete the project from the startup's Google Cloud organization.

Answer
A

Detailed explanation
A. Correct - the simplest solution is to use the projects.move API to transfer the project and update billing.
B. Incorrect - you cannot "drag" projects; migration must be done programmatically.
C. Incorrect - Private Catalog is for sharing solutions, not moving projects.
D. Incorrect - rebuilding infrastructure with Terraform is time-consuming and unnecessary.

Question 84
All development (dev) teams in your organization are located in the United States. Each dev team has its own Google Cloud project. You want to restrict access so that each dev team can only create resources in the United States. What should you do?

Options
A. Create a folder to contain all the dev projects. Create an organization policy to limit resources in US locations.
B. Create an organization to contain all the dev projects. Create an Identity and Access Management (IAM) policy to limit resources in US regions.
C. Create an Identity and Access Management (IAM) policy to restrict the resources locations in the US. Apply the policy to all dev projects.
D. Create an Identity and Access Management (IAM) policy to restrict the resources locations in all dev projects. Apply the policy to all dev roles.

Answer
A

Detailed explanation
A. Correct - organization policies allow restricting resource locations. Applying at the folder level covers all dev projects.
B. Incorrect - organizations already exist; IAM policies do not restrict resource locations.
C. Incorrect - IAM policies are about permissions, not geographic resource constraints.
D. Incorrect - again, IAM does not restrict where resources can be created.

Question 85
You are configuring Cloud DNS. You want to create DNS records to point home.mydomain.com, mydomain.com, and www.mydomain.com to the IP address of your Google Cloud load balancer. What should you do?

Options
A. Create one CNAME record to point mydomain.com to the load balancer, and create two A records to point WWW and HOME to mydomain.com respectively.
B. Create one CNAME record to point mydomain.com to the load balancer, and create two AAAA records to point WWW and HOME to mydomain.com respectively.
C. Create one A record to point mydomain.com to the load balancer, and create two CNAME records to point WWW and HOME to mydomain.com respectively.
D. Create one A record to point mydomain.com to the load balancer, and create two NS records to point WWW and HOME to mydomain.com respectively.

Answer
C

Detailed explanation
A. Incorrect - using CNAME for the root domain (mydomain.com) is not supported.
B. Incorrect - AAAA records are for IPv6, not the case here.
C. Correct - an A record maps the root domain to the load balancer, and CNAME records map subdomains (www, home) to the root.
D. Incorrect - NS records delegate DNS zones, not used for pointing domains.

Question 86
You have two subnets (subnet-a and subnet-b) in the default VPC. Your database servers are in subnet-a, and your application + web servers are in subnet-b. You want to configure a firewall rule that only allows traffic from the app servers to the database servers. What should you do?

Options
A. Create service accounts sa-app and sa-db. Associate service account sa-app with the app servers and sa-db with the database servers. Create an ingress firewall rule to allow traffic from sa-app to sa-db.
B. Create network tags app-server and db-server. Add tags to servers. Create an ingress firewall rule to allow traffic from app-server to db-server.
C. Create a service account sa-app and a network tag db-server. Associate sa-app with the app servers, tag db-server with the database servers. Create an ingress firewall rule to allow traffic from sa-app to db-server.
D. Create a network tag app-server and service account sa-db. Add tags and accounts to servers. Create an ingress firewall rule to allow traffic from app-server to sa-db.

Answer
A

Detailed explanation
A. Correct - using service accounts for firewall rules is the most secure and recommended method, ensuring only app servers can reach db servers.
B. Incorrect - network tags work but are less secure compared to service account–based rules.
C. Incorrect - mixing service account and tags adds complexity without extra benefit.
D. Incorrect - same issue, mixing identity/tagging unnecessarily.

Question 87
Your team wants to deploy a specific content management system (CMS) solution to Google Cloud. You need a quick and easy way to deploy and install the solution. What should you do?

Options
A. Search for the CMS solution in Google Cloud Marketplace. Use gcloud CLI to deploy the solution.
B. Search for the CMS solution in Google Cloud Marketplace. Deploy the solution directly from Cloud Marketplace.
C. Search for the CMS solution in Google Cloud Marketplace. Use Terraform and the Cloud Marketplace ID to deploy the solution with appropriate parameters.
D. Use the installation guide of the CMS provider. Perform the installation through your configuration management system.

Answer
B

Detailed explanation
A. Incorrect - CLI could be used but is not the simplest approach compared to Marketplace's direct deployment.
B. Correct - Cloud Marketplace allows one-click deployment of solutions, the fastest and easiest method.
C. Incorrect - Terraform is powerful but adds complexity, not "quick and easy."
D. Incorrect - manual installation requires extra steps and is not quick.

Question 88
You are working for a startup registered 6 months ago. As your customer base grows, your use of Google Cloud increases. You want to allow all engineers to create new projects without asking them for credit card information. What should you do?

Options
A. Create a Billing account, associate a payment method with it, and provide all project creators with permission to associate their projects with this billing account.
B. Grant all engineers permission to create their own billing accounts for each new project.
C. Apply for monthly invoiced billing, and have a single invoice paid by the finance team.
D. Create a billing account, associate it with a monthly purchase order (PO), and send the PO to Google Cloud.

Answer
A

Detailed explanation
A. Correct - by associating a billing account with payment, engineers can link projects without needing their own credit cards.
B. Incorrect - engineers should not create separate billing accounts.
C. Incorrect - invoiced billing may be an option but still requires billing account setup.
D. Incorrect - purchase orders are for enterprises; not relevant to enabling project creation.

Question 89
Your CI/CD server can't execute Google Cloud actions in a specific project due to permission issues. You need to validate whether the used service account has appropriate IAM roles. What should you do?

Options
A. Open the Google Cloud console, and check the IAM roles assigned to the service account at the project or inherited from folder/organization.
B. Open the Google Cloud console, and check the organization policies.
C. Open the Google Cloud console, and run a query to determine which resources this service account can access.
D. Open the Google Cloud console, and run a query of the audit logs to find permission denied errors.

Answer
A

Detailed explanation
A. Correct - checking IAM roles at all levels verifies if the service account has the required permissions.
B. Incorrect - org policies might restrict but do not directly show account roles.
C. Incorrect - querying accessible resources doesn't confirm IAM role assignments.
D. Incorrect - audit logs show errors, but the direct check is IAM roles.

Question 90
Your team is using Linux instances on Google Cloud. You need to ensure secure and cost-effective SSH access for your team. What should you do?

Options
A. Attach a public IP and allow incoming connections from the internet on port 22.
B. Use the gcloud compute ssh command with the --tunnel-through-iap flag. Allow ingress traffic from Google IP range on port 22.
C. Use a third-party tool to provide remote access.
D. Create a bastion host with public internet access. Use SSH tunneling from the bastion host.

Answer
B

Detailed explanation
A. Incorrect - exposing SSH directly over the internet is insecure.
B. Correct - IAP tunneling is secure, cost-effective, and avoids public IP exposure.
C. Incorrect - third-party tools add cost and complexity.
D. Incorrect - bastion hosts work but add operational overhead and cost compared to IAP.

Question 91
You want to send and consume Cloud Pub/Sub messages from your App Engine application. The Cloud Pub/Sub API is currently disabled. You will use a service account to authenticate your application to the API. You want to make sure your application can use Cloud Pub/Sub. What should you do?

Options
A. Enable the Cloud Pub/Sub API in the API Library on the GCP Console.
B. Rely on the automatic enablement of the Cloud Pub/Sub API when the Service Account accesses it.
C. Use Deployment Manager to deploy your application. Rely on the automatic enablement of all APIs used by the application being deployed.
D. Grant the App Engine Default service account the role of Cloud Pub/Sub Admin. Have your application enable the API on the first connection to Cloud Pub/Sub.

Answer
A

Detailed explanation
A. Correct - APIs in GCP must be explicitly enabled in the API Library. Without enabling Pub/Sub, the application cannot make API calls even if authenticated.
B. Incorrect - APIs are not automatically enabled by usage. Explicit action is required.
C. Incorrect - Deployment Manager does not auto-enable APIs; it only manages resources once APIs are active.
D. Incorrect - Even with correct roles, the API itself must first be enabled in the project.

Question 92
You need to monitor resources that are distributed over different projects in Google Cloud Platform. You want to consolidate reporting under the same Stackdriver Monitoring dashboard. What should you do?

Options
A. Use Shared VPC to connect all projects, and link Stackdriver to one of the projects.
B. For each project, create a Stackdriver account. In each project, create a service account for that project and grant it the role of Stackdriver Account Editor in all other projects.
C. Configure a single Stackdriver account, and link all projects to the same account.
D. Configure a single Stackdriver account for one of the projects. In Stackdriver, create a Group and add the other project names as criteria for that Group.

Answer
C

Detailed explanation
A. Incorrect - Shared VPC enables network connectivity but has no role in consolidating monitoring data.
B. Incorrect - This is unnecessarily complex. Multiple accounts make reporting fragmented.
C. Correct - One Stackdriver (now Cloud Monitoring) account can aggregate data across projects, enabling unified dashboards and alerts.
D. Incorrect - Groups organize monitored resources, but you still need one account linking all projects.

Question 93
You are deploying an application to a Compute Engine VM in a managed instance group. The application must be running at all times, but only a single instance of the VM should run per GCP project. How should you configure the instance group?

Options
A. Set autoscaling to On, set the minimum number of instances to 1, and then set the maximum number of instances to 1.
B. Set autoscaling to Off, set the minimum number of instances to 1, and then set the maximum number of instances to 1.
C. Set autoscaling to On, set the minimum number of instances to 1, and then set the maximum number of instances to 2.
D. Set autoscaling to Off, set the minimum number of instances to 1, and then set the maximum number of instances to 2.

Answer
A

Detailed explanation
A. Correct - Setting autoscaling on with min=1 and max=1 ensures one VM always runs, and the managed group auto-recreates it if it fails.
B. Incorrect - Without autoscaling, failures won't trigger automatic recreation.
C. Incorrect - Max=2 allows two instances, violating the requirement of only one VM.
D. Incorrect - Autoscaling off and max=2 doesn't guarantee self-healing or single-instance control.

Question 94
You want to verify the IAM users and roles assigned within a GCP project named my-project. What should you do?

Options
A. Run gcloud iam roles list. Review the output section.
B. Run gcloud iam service-accounts list. Review the output section.
C. Navigate to the project and then to the IAM section in the GCP Console. Review the members and roles.
D. Navigate to the project and then to the Roles section in the GCP Console. Review the roles and status.

Answer
C

Detailed explanation
A. Incorrect - This lists available roles but doesn't show assignments.
B. Incorrect - This lists service accounts only, not user-role mappings.
C. Correct - The IAM section shows which principals (users, groups, service accounts) are assigned which roles.
D. Incorrect - The Roles section shows definitions of roles, not assignments.

Question 95
You need to create a new billing account and then link it with an existing Google Cloud Platform project. What should you do?

Options
A. Verify that you are Project Billing Manager for the GCP project. Update the existing project to link it to the existing billing account.
B. Verify that you are Project Billing Manager for the GCP project. Create a new billing account and link the new billing account to the existing project.
C. Verify that you are Billing Administrator for the billing account. Create a new project and link the new project to the existing billing account.
D. Verify that you are Billing Administrator for the billing account. Update the existing project to link it to the existing billing account.

Answer
B

Detailed explanation
A. Incorrect - This links to an existing billing account, not a new one.
B. Correct - The Project Billing Manager can create a new billing account and associate the project with it.
C. Incorrect - This creates a new project, which is not required here.
D. Incorrect - Same as A, only works for existing accounts, not new.

Question 96
You have one project called proj-sa where you manage all your service accounts. You want to be able to use a service account from this project to take snapshots of VMs running in another project called proj-vm. What should you do?

Options
A. Download the private key from the service account, and add it to each VM's custom metadata.
B. Download the private key from the service account, and add the private key to each VM's SSH keys.
C. Grant the service account the IAM Role of Compute Storage Admin in the project called proj-vm.
D. When creating the VMs, set the service account's API scope for Compute Engine to read/write.

Answer
C

Detailed explanation
A. Incorrect - Storing keys in metadata is insecure and not Google-recommended.
B. Incorrect - SSH keys are unrelated to Cloud IAM permissions.
C. Correct - Cross-project IAM permissions are handled by assigning the appropriate role (roles/compute.storageAdmin) to the service account in the target project. This allows the service account from proj-sa to perform snapshot operations in proj-vm.
D. Incorrect - API scopes are legacy and not sufficient without proper IAM role assignments.

Question 97
You created a Google Cloud Platform project with an App Engine application inside the project. You initially configured the application to be served from the us-central region. Now you want the application to be served from the asia-northeast1 region. What should you do?

Options
A. Change the default region property setting in the existing GCP project to asia-northeast1.
B. Change the region property setting in the existing App Engine application from us-central to asia-northeast1.
C. Create a second App Engine application in the existing GCP project and specify asia-northeast1 as the region to serve your application.
D. Create a new GCP project and create an App Engine application inside this new project. Specify asia-northeast1 as the region to serve your application.

Answer
D

Detailed explanation
A. Incorrect - You cannot change the region of an App Engine app after creation.
B. Incorrect - Same limitation; region is fixed once set.
C. Incorrect - Each project supports only one App Engine application. You cannot create a second one.
D. Correct - To run App Engine in a new region, you must create a new GCP project and enable App Engine with the desired region.

Question 98
You need to grant access for three users so that they can view and edit table data on a Cloud Spanner instance. What should you do?

Options
A. Run gcloud iam roles describe roles/spanner.databaseUser. Add the users to the role.
B. Run gcloud iam roles describe roles/spanner.databaseUser. Add the users to a new group. Add the group to the role.
C. Run gcloud iam roles describe roles/spanner.viewer --project my-project. Add the users to the role.
D. Run gcloud iam roles describe roles/spanner.viewer --project my-project. Add the users to a new group. Add the group to the role.

Answer
B

Detailed explanation
A. Incorrect - Assigning roles directly to individuals works, but using groups is preferred for scalability and manageability.
B. Correct - Best practice is to create a Google group, add the three users, and assign the group the spanner.databaseUser role. This simplifies management if membership changes later.
C. Incorrect - spanner.viewer only allows read access, not editing table data.
D. Incorrect - Same as C, still insufficient permissions even if using a group.

Question 99
You create a new Google Kubernetes Engine (GKE) cluster and want to make sure that it always runs a supported and stable version of Kubernetes. What should you do?

Options
A. Enable the Node Auto-Repair feature for your GKE cluster.
B. Enable the Node Auto-Upgrades feature for your GKE cluster.
C. Select the latest available cluster version for your GKE cluster.
D. Select Container-Optimized OS (cos) as a node image for your GKE cluster.

Answer
B

Detailed explanation
A. Incorrect - Auto-Repair ensures unhealthy nodes are recreated, but doesn't keep Kubernetes updated.
B. Correct - Auto-Upgrades automatically updates GKE nodes to the latest supported stable version of Kubernetes. This ensures long-term security and compatibility.
C. Incorrect - Choosing the latest version at creation locks you in; you still need ongoing upgrades.
D. Incorrect - COS improves security and performance but does not handle Kubernetes upgrades.

Question 100
You have an instance group that you want to load balance. You want the load balancer to terminate the client SSL session. The instance group is used to serve a public web application over HTTPS. You want to follow Google-recommended practices. What should you do?

Options
A. Configure an HTTP(S) load balancer.
B. Configure an internal TCP load balancer.
C. Configure an external SSL proxy load balancer.
D. Configure an external TCP proxy load balancer.

Answer
A

Detailed explanation
A. Correct - HTTP(S) load balancers terminate SSL/TLS at the edge and distribute traffic to backend instances. They are recommended for web apps.
B. Incorrect - Internal load balancers are for private traffic inside a VPC, not public HTTPS traffic.
C. Incorrect - SSL proxy is used for non-HTTP TCP traffic (e.g., SMTP, custom TLS), not web apps.
D. Incorrect - TCP proxy load balancer is also for generic TCP, not HTTP(S).

Question 101
You have 32 GB of data in a single file that you need to upload to a Nearline Storage bucket. The WAN connection you are using is rated at 1 Gbps, and you are the only one on the connection. You want to use as much of the rated 1 Gbps as possible to transfer the file rapidly. How should you upload the file?

Options
A. Use the GCP Console to transfer the file instead of gsutil.
B. Enable parallel composite uploads using gsutil on the file transfer.
C. Decrease the TCP window size on the machine initiating the transfer.
D. Change the storage class of the bucket from Nearline to Multi-Regional.

Answer
B

Detailed explanation
A. Incorrect - The Console is not optimized for large single-file transfers.
B. Correct - Parallel composite uploads split the file into multiple components uploaded simultaneously, maximizing throughput and taking advantage of available bandwidth.
C. Incorrect - Decreasing TCP window size would reduce throughput, not increase it.
D. Incorrect - Storage class doesn’t impact upload speed; only cost and availability.

Question 102
You've deployed a microservice called myapp1 to a Google Kubernetes Engine cluster using the YAML file shown below. You need to refactor this configuration so that the database password is not stored in plain text. You want to follow Google-recommended practices. What should you do?

Options
A. Store the database password inside the Docker image of the container, not in the YAML file.
B. Store the database password inside a Secret object. Modify the YAML file to populate the DB_PASSWORD environment variable from the Secret.
C. Store the database password inside a ConfigMap object. Modify the YAML file to populate the DB_PASSWORD environment variable from the ConfigMap.
D. Store the database password in a file inside a Kubernetes persistent volume, and use a persistent volume claim to mount the volume to the container.

Answer
B

Detailed explanation
A. Incorrect - Baking passwords into images is insecure and not recommended.
B. Correct - Kubernetes Secrets are designed for storing sensitive data like passwords. The pod can reference the secret as an environment variable or mounted file.
C. Incorrect - ConfigMaps are intended for non-sensitive configuration (like flags, parameters), not secrets.
D. Incorrect - Persistent volumes can store secrets, but this is not the recommended or simplest method.

Question 103
You are running an application on multiple virtual machines within a managed instance group and have autoscaling enabled. The policy adds VMs when CPU utilization exceeds 80% and removes them when it falls below 80%. The initial delay for HTTP health checks is set to 30 seconds. It takes around 3 minutes for new instances to become available for users. You observe that when the instance group scales, it adds more instances than necessary to handle load. What should you do?

Options
A. Set the maximum number of instances to 1.
B. Decrease the maximum number of instances to 3.
C. Use a TCP health check instead of an HTTP health check.
D. Increase the initial delay of the HTTP health check to 200 seconds.

Answer
D

Detailed explanation
A. Incorrect - Limiting max instances to 1 defeats the purpose of autoscaling.
B. Incorrect - Arbitrarily lowering the max number doesn't fix the timing issue.
C. Incorrect - TCP vs. HTTP health check doesn't address boot time delays.
D. Correct - Since VMs take ~3 minutes to start, increasing the initial health check delay (to around 200s) prevents the system from marking them unhealthy prematurely and over-provisioning.

Question 104
You need to select and configure compute resources for a set of batch processing jobs. These jobs take around 2 hours to complete and are run nightly. You want to minimize service costs. What should you do?

Options
A. Select Google Kubernetes Engine. Use a single-node cluster with a small instance type.
B. Select Google Kubernetes Engine. Use a three-node cluster with micro instance types.
C. Select Compute Engine. Use preemptible VM instances of the appropriate standard machine type.
D. Select Compute Engine. Use VM instance types that support micro bursting.

Answer
C

Detailed explanation
A. Incorrect - A single-node GKE cluster doesn't scale well and incurs overhead from Kubernetes itself.
B. Incorrect - Three-node GKE clusters are more costly and unnecessary for batch jobs.
C. Correct - Preemptible VMs are 70–80% cheaper, perfect for fault-tolerant, time-flexible batch workloads. Even if preempted, jobs can be retried.
D. Incorrect - Micro-bursting is not guaranteed capacity and doesn't offer cost savings for predictable workloads.

Question 105
You recently deployed a new version of an application to App Engine and then discovered a bug. You need to immediately revert to the prior version. What should you do?

Options
A. Run gcloud app restore.
B. On the App Engine page of the GCP Console, select the application that needs to be reverted and click Revert.
C. On the App Engine Versions page of the GCP Console, route 100% of the traffic to the previous version.
D. Deploy the original version as a separate application, then split traffic between versions.

Answer
C

Detailed explanation
A. Incorrect - No gcloud app restore command exists.
B. Incorrect - The console has no one-click revert option for apps.
C. Correct - App Engine keeps prior versions. You can simply shift all traffic back to the stable version immediately.
D. Incorrect - Redeployment wastes time; traffic splitting is not necessary when you want 100% rollback.

Question 106
You deployed an App Engine application using gcloud app deploy, but it didn't deploy to the intended project. You want to find out why. What should you do?

Options
A. Check the app.yaml file for your application and check project settings.
B. Check the web-application.xml file for your application and check project settings.
C. Go to Deployment Manager and review settings for deployment of applications.
D. Go to Cloud Shell and run gcloud config list to review the Google Cloud configuration used for deployment.

Answer
D

Detailed explanation
A. Incorrect - app.yaml configures services, not project association.
B. Incorrect - web-application.xml is Java-specific deployment config, unrelated to GCP projects.
C. Incorrect - Deployment Manager manages infra, not App Engine project targets.
D. Correct - gcloud config list shows active project settings. If the wrong project was active, deployment goes there.

Question 107
You want to configure 10 Compute Engine instances for availability when maintenance occurs. Requirements: instances should auto-restart if they crash and be highly available during system maintenance. What should you do?

Options
A. Create an instance template. Set "Automatic Restart" on. Set "On-host maintenance" to Migrate VM. Add the template to an instance group.
B. Create an instance template. Set "Automatic Restart" off. Set "On-host maintenance" to Terminate VM.
C. Create an instance group. Set the Autohealing health check to HTTP.
D. Create an instance group. Verify "do not retry machine creation" is off.

Answer
A

Detailed explanation
A. Correct - Automatic restart ensures recovery after crash, "Migrate VM" keeps it running during host maintenance. Together, they maximize availability.
B. Incorrect - Turning restart off and terminating VMs reduces availability.
C. Incorrect - Autohealing restarts unhealthy VMs, but doesn't address system maintenance.
D. Incorrect - Advanced creation flags don't guarantee maintenance migration.

Question 108
You host a static website on Cloud Storage. Recently, when including PDF file links, browsers prompt users to download instead of viewing inline. You want PDFs to display in the browser. What should you do?

Options
A. Enable Cloud CDN on the website frontend.
B. Enable 'Share publicly' on the PDF file objects.
C. Set Content-Type metadata to application/pdf on the PDF file objects.
D. Add a label to the storage bucket with a key of Content-Type and value of application/pdf.

Answer
C

Detailed explanation
A. Incorrect - CDN caching doesn't affect file headers.
B. Incorrect - Public access doesn't change browser rendering.
C. Correct - Correct Content-Type metadata ensures browsers treat PDFs as inline viewable documents.
D. Incorrect - Labels are metadata for resource organization, not HTTP headers.

Question 109
You have a VM with 2 vCPUs and 4 GB memory. It is running out of memory, and you want to upgrade it to 8 GB. What should you do?

Options
A. Rely on live migration to move workload to a machine with more memory.
B. Use gcloud to add metadata to the VM. Set the key to required-memory-size = 8 GB.
C. Stop the VM, change the machine type to n1-standard-8, and start the VM.
D. Stop the VM, increase the memory to 8 GB, and start the VM.

Answer
D

Detailed explanation
A. Incorrect - Live migration preserves uptime but doesn't resize VMs.
B. Incorrect - Metadata keys don't change hardware specs.
C. Incorrect - Changing to n1-standard-8 adds CPUs too, not just memory.
D. Correct - Stopping the VM allows resizing memory independently (for custom machine types) and restarting with 8 GB.

Question 110
You have production and test workloads to deploy on Compute Engine. Production VMs must be in a different subnet than test VMs, but all must communicate internally. What should you do?

Options
A. Create a single custom VPC with 2 subnets. Place each subnet in a different region and with a different CIDR range.
B. Create a single custom VPC with 2 subnets. Place each subnet in the same region with the same CIDR range.
C. Create 2 custom VPCs, each with a single subnet. Place subnets in different regions.
D. Create 2 custom VPCs, each with a single subnet. Place subnets in the same region.

Answer
A

Detailed explanation
A. Correct - One VPC can have multiple subnets across regions. Different CIDR ensures no overlap. Internal communication still works over the VPC.
B. Incorrect - Same CIDR ranges cause IP conflicts.
C. Incorrect - Two VPCs don't allow automatic internal communication unless peered.
D. Incorrect - Same issue as C; separate VPCs isolate networks.

Question 111
You need to create an autoscaling managed instance group for an HTTPS web app. You want unhealthy VMs to be automatically recreated. What should you do?

Options
A. Create a health check on port 443 and use it with the Managed Instance Group.
B. Select Multi-Zone instead of Single-Zone when creating the Managed Instance Group.
C. In the Instance Template, add the label 'health-check'.
D. In the Instance Template, add a startup script that sends a heartbeat.

Answer
A

Detailed explanation
A. Correct - Health checks tied to MIGs enable autohealing. If an instance fails, MIG recreates it.
B. Incorrect - Multi-zone increases availability but doesn't recreate unhealthy VMs.
C. Incorrect - Labels don't enforce health monitoring.
D. Incorrect - Startup scripts can't replace health checks.

Question 112
Your company uses BigQuery for data warehousing. The team is small, changes often, and you want to let members query data while following best practices. What should you do?

Options
A. Create IAM entries for each user and assign bigQuery.jobUser.
B. Create IAM entries for each user and assign bigQuery.dataViewer.
C. Create a Google group. Add members to the group. Assign the group the bigQuery.jobUser role.
D. Create a Google group. Add members to the group. Assign the group the bigQuery.dataViewer role.

Answer
C

Detailed explanation
A. Incorrect - Individual assignments are harder to manage with frequent changes.
B. Incorrect - dataViewer allows browsing datasets but not running queries.
C. Correct - Groups centralize access management. jobUser role allows members to run queries. Adding/removing users from the group is simple.
D. Incorrect - Same as B, insufficient permissions.

Question 113
Your company has a 3-tier solution on Compute Engine. Each tier runs in its own subnet. You need to enable communication between tiers on TCP port 8080 as shown:

- Tier 1 ↔ Tier 2
- Tier 2 ↔ Tier 3

Options
A. Create firewall rules with source filter = subnet IP ranges, targets = all instances.
B. Create firewall rules with source filter = service accounts of tiers, targets = other tiers.
C. Create firewall rules allowing TCP:8080 for all instances in the VPC.
D. Create egress firewall rules for each subnet.

Answer
B

Detailed explanation
A. Incorrect - Using IP ranges works but is less secure and harder to maintain.
B. Correct - Using service accounts as source/target provides identity-based firewalling, the recommended approach.
C. Incorrect - Allowing all instances is overly permissive and insecure.
D. Incorrect - Egress rules alone don't ensure correct directional traffic control between tiers.

Question 114
You are given a project with a single VPC and subnet in us-central1. There's a Compute Engine instance in that subnet hosting an application. You need to deploy a new instance in europe-west1 that can access the same application privately. What should you do?

Options
A. Create a subnet in europe-west1 in the same VPC. Use the first instance's private address.
B. Create a new VPC in europe-west1. Expose the app with a load balancer.
C. Create a new subnet in europe-west1. Use Cloud VPN to connect the subnets.
D. Create a new VPC in europe-west1. Peer the two VPCs.

Answer
A

Detailed explanation
A. Correct - Subnets can be regional, but one VPC spans multiple regions. Adding a new subnet in europe-west1 within the same VPC allows internal communication.
B. Incorrect - Load balancing isn't needed if VMs are in the same VPC.
C. Incorrect - VPN is only required for different networks, not the same VPC.
D. Incorrect - VPC peering is only for distinct VPCs.

Question 115
Your projects incurred unexpected costs due to a development GKE cluster emitting excessive logs. You want to quickly disable logging to control costs. What should you do?

Options
A. Go to the Logs ingestion window in Stackdriver Logging, and disable the log source for the GKE container resource.
B. Go to the Logs ingestion window in Stackdriver Logging, and disable the log source for the GKE Cluster Operations resource.
C. Go to the GKE console, delete existing clusters, and recreate without logging enabled.
D. Go to the GKE console, delete existing clusters, and recreate without Stackdriver Monitoring.

Answer
A

Detailed explanation
A. Correct - Disabling ingestion for the GKE container resource immediately stops logs from being written and billed. Fastest option.
B. Incorrect - Disabling cluster operations logs doesn't stop pod/container logs.
C. Incorrect - Recreating clusters is disruptive and time-consuming.
D. Incorrect - Monitoring is separate from logging and won't stop the log costs.

Question 116
You have a website hosted on App Engine standard environment. You want 1% of your users to see a new test version of the website. You want to minimize complexity. What should you do?

Options
A. Deploy the new version in the same application and use the --migrate option.
B. Deploy the new version in the same application and use the --splits option to give a weight of 99 to the current version and a weight of 1 to the new version.
C. Create a new App Engine application in the same project. Deploy the new version in that application. Use the App Engine library to proxy 1% of the requests to the new version.
D. Create a new App Engine application in the same project. Deploy the new version in that application. Configure your network load balancer to send 1% of the traffic to that new application.

Answer
B

Detailed explanation

A. Incorrect - --migrate moves 100% of traffic to the new version, not partial.

B. Correct - The --splits option allows gradual traffic splitting between versions, perfect for A/B testing or canary releases (e.g., 99% old, 1% new).

C. Incorrect - A project can only have one App Engine application. You can't create multiple apps in one project.

D. Incorrect - Same limitation as C, and using load balancers adds unnecessary complexity.

Question 117
You have a web application deployed as a managed instance group. You have a new version of the application to gradually deploy. Your web application is currently receiving live traffic. You want to ensure available capacity does not decrease during deployment. What should you do?

Options
A. Perform a rolling-action start-update with maxSurge set to 0 and maxUnavailable set to 1.
B. Perform a rolling-action start-update with maxSurge set to 1 and maxUnavailable set to 0.
C. Create a new managed instance group with an updated instance template. Add the group to the backend service for the load balancer. When all instances in the new managed instance group are healthy, delete the old managed instance group.
D. Create a new instance template with the new application version. Update the existing managed instance group. Delete the old instances to recreate with the new template.

Answer
B

Detailed explanation

A. Incorrect - With maxSurge=0, no new instance is added during the update, so you risk downtime.

B. Correct - maxSurge=1 ensures a new instance comes up before any old instance is taken down, maintaining capacity. maxUnavailable=0 guarantees no reduction in serving capacity.

C. Incorrect - Creating a separate MIG works but is more complex than rolling updates.

D. Incorrect - This risks downtime because old instances are deleted before new ones are ready.

Question 118
You are building an application that stores relational data from users. Users across the globe will use this application. Your CTO is concerned about scaling because the user base size is unknown. You need a database solution that can scale with user growth with minimal configuration. Which storage solution should you use?

Options
A. Cloud SQL
B. Cloud Spanner
C. Cloud Firestore
D. Cloud Datastore

Answer
B

Detailed explanation

A. Incorrect - Cloud SQL is relational but not designed for global scale-out; scaling requires manual intervention.

B. Correct - Cloud Spanner is globally distributed, horizontally scalable, and relational. It handles unpredictable growth seamlessly with minimal management.

C. Incorrect - Firestore is NoSQL, not relational.

D. Incorrect - Datastore is also NoSQL and not suitable for relational workloads.

Question 119
You are the organization and billing administrator. The engineering team has Project Creator rights but should not be able to link projects to a billing account. Only the finance team should link projects to billing. What should you do?

Options
A. Assign the finance team only the Billing Account User role on the billing account.
B. Assign the engineering team only the Billing Account User role on the billing account.
C. Assign the finance team the Billing Account User role on the billing account and the Project Billing Manager role on the organization.
D. Assign the finance team the Billing Account User role and the Project Billing Manager role on the organization.

Answer
A

Detailed explanation

A. Correct - Billing Account User allows a team to link projects to a billing account but does not grant broader permissions. Restricting this to finance ensures engineers can create projects but not link billing.

B. Incorrect - This would give engineers the billing linking ability, which is not wanted.

C. Incorrect - Project Billing Manager allows broader changes, unnecessary for finance.

D. Incorrect - Same as C, too many permissions.

Question 120
You have an application running in GKE with cluster autoscaling enabled. The app exposes a TCP endpoint. You have another VM in a different VPC (overlapping IP ranges). That VM needs to connect to the GKE app. You want to minimize effort. What should you do?

Options
A. In GKE, create a Service of type LoadBalancer, set externalTrafficPolicy, and configure the VM to use the external load balancer.
B. In GKE, create a Service of type NodePort, set up a proxy VM, and configure iptables.
C. In GKE, create a Service of type LoadBalancer, add annotation for internal load balancer, peer the two VPCs, and configure the VM to use the load balancer.
D. In GKE, create a Service of type LoadBalancer, add Cloud Armor policy, and configure the VM to use the load balancer.

Answer
C

Detailed explanation

A. Incorrect - External load balancer works but sends traffic over the public internet, not private, which isn't ideal here.

B. Incorrect - Manual proxy with iptables is complex and fragile.

C. Correct - Internal load balancer with VPC peering allows private communication across VPCs even with overlapping ranges, minimal extra setup.

D. Incorrect - Cloud Armor is for security filtering, not connectivity.

Question 121

Your company's infrastructure is on-premises, but all machines are running at maximum capacity. You want to burst to Google Cloud. The workloads on Google Cloud must be able to directly communicate to the workloads on-premises using a private IP range. What should you do?

Options
A. In Google Cloud, configure the VPC as a host for Shared VPC.
B. In Google Cloud, configure the VPC for VPC Network Peering.
C. Create bastion hosts both in your on-premises environment and on Google Cloud. Configure both as proxy servers using their public IP addresses.
D. Set up Cloud VPN between the infrastructure on-premises and Google Cloud.

Answer
D

Detailed explanation

A (Incorrect): Shared VPC allows multiple projects in GCP to share the same VPC resources, but it does not connect on-premises infrastructure to Google Cloud.

B (Incorrect): VPC peering works between GCP VPC networks but not with on-premises infrastructure.

C (Incorrect): Bastion hosts use public IPs and proxy connections. This breaks the requirement to use *private IP ranges* for direct communication.

D (Correct): Cloud VPN securely connects your on-premises network to a GCP VPC over an encrypted tunnel, enabling private IP-to-private IP communication. This is the standard solution for hybrid connectivity when bursting workloads.

Question 122

You want to select and configure a solution for storing and archiving data on Google Cloud Platform. You need to support compliance objectives for data from one geographic location. This data is archived after 30 days and needs to be accessed annually. What should you do?

Options
A. Select Multi-Regional Storage. Add a bucket lifecycle rule that archives data after 30 days to Coldline Storage.
B. Select Multi-Regional Storage. Add a bucket lifecycle rule that archives data after 30 days to Nearline Storage.
C. Select Regional Storage. Add a bucket lifecycle rule that archives data after 30 days to Nearline Storage.
D. Select Regional Storage. Add a bucket lifecycle rule that archives data after 30 days to Coldline Storage.

Answer
D

Detailed explanation

A (Incorrect): Multi-Regional is for global availability and low-latency, not necessary when data is from a single geographic location.

B (Incorrect): Nearline is optimized for monthly access, but the requirement is *annual* access. Using Nearline increases unnecessary costs.

C (Incorrect): Regional + Nearline reduces cost but still meant for monthly access, not yearly.

D (Correct): Regional storage matches the compliance need (data stays in one geographic location). Coldline is the correct archival tier for data accessed once per year, optimized for long-term, infrequent access. Lifecycle rules automate the archival.

Question 123

Your company uses BigQuery for data warehousing. Over time, many different business units in your company have created 1000+ datasets across hundreds of projects. Your CIO wants you to examine all datasets to find tables that contain an employee_ssn column. You want to minimize effort in performing this task. What should you do?

Options
A. Go to Data Catalog and search for employee_ssn in the search box.
B. Write a shell script that uses the bq command line tool to loop through all the projects in your organization.
C. Write a script that loops through all the projects in your organization and runs a query on INFORMATION_SCHEMA.COLUMNS view to find the employee_ssn column.
D. Write a Cloud Dataflow job that loops through all the projects in your organization and runs a query on INFORMATION_SCHEMA.COLUMNS view to find employee_ssn column.

Answer
A

Detailed explanation

A (Correct): Data Catalog automatically indexes metadata across all datasets, projects, and tables. You can search for column names like *employee_ssn* centrally without scripting. This is the least effort.

B (Incorrect): Manually scripting with bq command line would be tedious and not scalable for 1000+ datasets.

C (Incorrect): INFORMATION_SCHEMA queries work per dataset, but you'd still need to manually iterate through projects and datasets, which is more effort.

D (Incorrect): Cloud Dataflow job for metadata scanning is overkill and unnecessarily complex compared to Data Catalog.

Question 124

You create a Deployment with 2 replicas in a Google Kubernetes Engine cluster that has a single preemptible node pool. After a few minutes, you use kubectl to examine the status of your Pod and observe that one of them is still in Pending status:

```
$ kubectl get pods -l app=myapp

NAME                          READY  STATUS   RESTART AGE

myapp-deployment-58ddbb995-lp86m     0/1   Pending  0     9m

myapp-deployment-58ddbb995-qjpkg     1/1   Running  0     9m
```

What is the most likely cause?

Options

A. The pending Pod's resource requests are too large to fit on a single node of the cluster.
B. Too many Pods are already running in the cluster, and there are not enough resources left to schedule the pending Pod.
C. The node pool is configured with a service account that does not have permission to pull the container image used by the pending Pod.
D. The pending Pod was originally scheduled on a node that has been preempted between the creation of the Deployment and your verification of the Pods' status. It is currently being rescheduled on a new node.

Answer

B

Detailed explanation

A (Incorrect): Resource mismatch could cause Pending, but nothing suggests oversized Pod requests.

B (Correct): The cluster uses a single preemptible node pool. If too many Pods are running, resources may not be sufficient to place new Pods, causing them to remain Pending. This matches the scenario.

C (Incorrect): Missing service account permissions typically cause ImagePullBackOff, not Pending.

D (Incorrect): Preemption affects running Pods, not scheduling. Pending happens due to resource unavailability, not rescheduling delays.

Question 125

You want to find out when users were added to Cloud Spanner Identity Access Management (IAM) roles on your Google Cloud Platform (GCP) project. What should you do in the GCP Console?

Options
A. Open the Cloud Spanner console to review configurations.
B. Open the IAM & admin console to review IAM policies for Cloud Spanner roles.
C. Go to the Stackdriver Monitoring console and review information for Cloud Spanner.
D. Go to the Stackdriver Logging console, review admin activity logs, and filter them for Cloud Spanner IAM roles.

Answer
D

Detailed explanation

A (Incorrect): The Cloud Spanner console only shows database configuration and queries, not IAM activity history.

B (Incorrect): IAM & admin shows current role assignments but not when the changes were made.

C (Incorrect): Monitoring provides metrics on performance and uptime, not IAM activity logs.

D (Correct): Admin activity logs in Cloud Logging (formerly Stackdriver Logging) record all IAM changes (who, what, when). Filtering for Cloud Spanner IAM role changes gives you the audit trail of when users were added.

Question 126

Your company implemented BigQuery as an enterprise data warehouse. Users from multiple business units run queries on this data warehouse. However, you notice that query costs for BigQuery are very high, and you need to control costs. Which two methods should you use? (Choose two.)

Options
A. Split the users from business units into multiple projects.
B. Apply a user- or project-level custom query quota for BigQuery data warehouse.
C. Create separate copies of your BigQuery data warehouse for each business unit.
D. Split your BigQuery data warehouse into multiple data warehouses for each business unit.
E. Change your BigQuery query model from on-demand to flat rate. Apply the appropriate number of slots to each Project.

Answer
B, E

Detailed explanation

A (Incorrect): Splitting into multiple projects doesn't control query costs by itself, since costs are tied to queries and storage.

B (Correct): Custom query quotas limit the amount of data scanned per user/project, directly controlling costs.

C (Incorrect): Creating separate warehouses increases duplication, storage cost, and maintenance overhead. Not cost-efficient.

D (Incorrect): Splitting the warehouse into multiple smaller ones adds complexity and doesn't inherently reduce query costs.

E (Correct): Switching to flat-rate pricing with slots allows predictable costs and prevents runaway on-demand query charges.

Question 127

You are building a product on top of Google Kubernetes Engine (GKE). You have a single GKE cluster. For each of your customers, a Pod is running in that cluster, and your customers can run arbitrary code inside their Pod. You want to maximize the isolation between your customers' Pods. What should you do?

Options
A. Use Binary Authorization and whitelist only the container images used by your customers' Pods.
B. Use the Container Analysis API to detect vulnerabilities in the containers used by your customers' Pods.
C. Create a GKE node pool with a sandbox type configured to gVisor. Add the parameter runtimeClassName: gvisor to the specification of your customers' Pods.
D. Use the cos_containerd image for your GKE nodes. Add a nodeSelector with the value cloud.google.com/gke-os-distribution: cos_containerd to the specification of your customers' Pods.

Answer
C

Detailed explanation

A (Incorrect): Binary Authorization ensures image provenance but doesn't isolate Pods from each other.

B (Incorrect): Vulnerability scanning detects issues but does not enforce runtime isolation.

C (Correct): gVisor provides strong runtime isolation between Pods by creating a sandbox environment. This minimizes the risk of one customer's Pod affecting another's.

D (Incorrect): Container-Optimized OS with containerd improves security and performance but does not provide Pod-level runtime isolation like gVisor.

Question 128

Your customer has implemented a solution that uses Cloud Spanner and notices some read latency-related performance issues on one table. This table is accessed only by their users using a primary key. The table schema is shown below:

```
CREATE TABLE Persons (
  person_id INT64 NOT NULL,   -- sequential number based on number of registration
  account_creation_date DATE, -- system date
  birthdate DATE,          -- customer birthdate
  firstname STRING(255),     -- first name
  lastname STRING(255),      -- last name
  profile_picture BYTES (255) -- profile picture
) PRIMARY KEY (person_id);
```

You want to resolve the issue. What should you do?

Options

A. Remove the profile_picture field from the table.
B. Add a secondary index on the person_id column.
C. Change the primary key to not have monotonically increasing values.
D. Create a secondary index using the following Data Definition Language (DDL):

```
CREATE INDEX person_id_ix
ON Persons (person_id, firstname, lastname)
STORING (profile_picture);
```

Answer

C

Detailed explanation

A (Incorrect): Removing the profile_picture column reduces storage but does not solve latency caused by the primary key design.

B (Incorrect): The primary key already has an index; adding another on the same column doesn't help.

C (Correct): Cloud Spanner distributes data by primary key. A monotonically increasing key like person_id causes hotspotting and latency. Using a non-sequential primary key improves performance.

D (Incorrect): Adding a secondary index with storing does not fix hotspotting caused by sequential primary keys.

Question 129

Your finance team wants to view the billing report for your projects. You want to make sure that the finance team does not get additional permissions to the project. What should you do?

Options
A. Add the group for the finance team to roles/billing user role.
B. Add the group for the finance team to roles/billing admin role.
C. Add the group for the finance team to roles/billing viewer role.
D. Add the group for the finance team to roles/billing project/Manager role.

Answer
C

Detailed explanation

A (Incorrect): Billing user allows linking/unlinking projects to billing accounts, which is more than just viewing.

B (Incorrect): Billing admin grants full control, not appropriate for read-only reporting.

C (Correct): Billing viewer provides read-only access to billing data and reports without granting project-level permissions.

D (Incorrect): Billing project/Manager role doesn't exist; misleading option.

Question 130

Your organization has strict requirements to control access to Google Cloud projects. You need to enable your Site Reliability Engineers (SREs) to approve requests from the Google Cloud support team when an SRE opens a support case. You want to follow Google-recommended practices. What should you do?

Options
A. Add your SREs to roles/iam.roleAdmin role.
B. Add your SREs to roles/accessapproval.approver role.
C. Add your SREs to a group and then add this group to roles/iam.roleAdmin role.
D. Add your SREs to a group and then add this group to roles/accessapproval.approver role.

Answer
D

Detailed explanation

A (Incorrect): IAM roleAdmin is too broad and unnecessary for this use case.

B (Incorrect): Adding individuals directly makes access management harder to maintain.

C (Incorrect): Same as A, still grants excessive permissions.

D (Correct): Best practice is to use groups for IAM management. Adding the SRE group to the Access Approval Approver role lets them approve support access requests while maintaining least privilege and easier management.

Question 131
You need to host an application on a Compute Engine instance in a project shared with other teams. You want to prevent the other teams from accidentally causing downtime on that application. Which feature should you use?

Options
A. Use a Shielded VM.
B. Use a Preemptible VM.
C. Use a sole-tenant node.
D. Enable deletion protection on the instance.

Answer
D

Detailed explanation
A. Incorrect - Shielded VMs protect the VM from firmware/rootkit attacks and provide integrity verification, but they do not prevent other teams from deleting the instance.
B. Incorrect - Preemptible VMs are short-lived and can be preempted by Google; they increase the chance of downtime rather than prevent it.
C. Incorrect - A sole-tenant node provides hardware isolation but does not stop other project members with permissions from deleting or modifying the VM.
D. Correct - Enabling deletion protection prevents accidental deletion of the VM, which directly prevents other teams from accidentally causing downtime by deleting the instance.

Question 132
Your organization needs to grant users access to query datasets in BigQuery but prevent them from accidentally deleting the datasets. You want a solution that follows Google-recommended practices. What should you do?

Options
A. Add users to roles/bigquery user role only, instead of roles/bigquery dataOwner.
B. Add users to roles/bigquery dataEditor role only, instead of roles/bigquery dataOwner.
C. Create a custom role by removing delete permissions, and add users to that role only.
D. Create a custom role by removing delete permissions. Add users to the group, and then add the group to the custom role.

Answer
D

Detailed explanation
A. Incorrect - The bigquery.user role is primarily for running jobs and creating datasets in the project scope and does not directly map to dataset-level query permissions; it’s not the precise least-privilege solution for dataset query access without delete.
B. Incorrect - bigquery.dataEditor still grants broad data modification permissions (including deletes) and is therefore not safe for preventing accidental dataset deletion.
C. Incorrect - Creating a custom role that removes delete permissions is appropriate, but assigning users individually instead of via a group is less manageable and not recommended for scalable IAM administration.
D. Correct - Create a custom role that removes delete permissions and assign that role to a group; adding users to the group gives them the intended query access while preventing accidental deletes and follows best practices (use groups for IAM assignments).

Question 133
You have a developer laptop with the Cloud SDK installed on Ubuntu. The Cloud SDK was installed from the Google Cloud Ubuntu package repository. You want to test your application locally on your laptop with Cloud Datastore. What should you do?

Options
A. Export Cloud Datastore data using gcloud datastore export.
B. Create a Cloud Datastore index using gcloud datastore indexes create.
C. Install the google-cloud-sdk-datastore-emulator component using the apt get install command.
D. Install the cloud-datastore-emulator component using the gcloud components install command.

Answer
C

Detailed explanation
A. Incorrect - Exporting data is for moving data out of Datastore, not for running a local emulator to test locally.
B. Incorrect - Creating indexes configures production Datastore, not a local testing environment.
C. Correct - When Cloud SDK is installed from the OS package repository, the Datastore emulator is provided as a system package (google-cloud-sdk-datastore-emulator) and should be installed via apt-get install to run locally.
D. Incorrect - gcloud components install manages components for SDK installations done via the downloadable installer; it typically does not work for SDK installed via the OS package manager, so apt-get is the correct method here.

Question 134
Your company set up a complex organizational structure on Google Cloud. The structure includes hundreds of folders and projects. Only a few team members should be able to view the hierarchical structure. You need to assign minimum permissions to these team members, and you want to follow Google-recommended practices. What should you do?

Options
A. Add the users to roles/browser role.
B. Add the users to roles/iam.roleViewer role.
C. Add the users to a group, and add this group to roles/browser.
D. Add the users to a group, and add this group to roles/iam.roleViewer role.

Answer
C

Detailed explanation
A. Incorrect - Assigning users individually to roles/browser works but does not follow best practice of using groups for easier management.
B. Incorrect - roles/iam.roleViewer is focused on role definitions and IAM policy viewing rather than browsing the resource hierarchy.
C. Correct - Adding the users to a group and granting the group the roles/browser role provides the minimal permission to view the resource hierarchy and follows best practice by managing access through a group.
D. Incorrect - roles/iam.roleViewer is not the appropriate role for simply viewing the organizational/folder/project hierarchy; roles/browser is the correct minimal role for that purpose.

Question 135
Your company has a single sign-on (SSO) identity provider that supports Security Assertion Markup Language (SAML) integration with service providers. Your company has users in Cloud Identity. You would like users to authenticate using your company's SSO provider. What should you do?

Options
A. In Cloud Identity, set up SSO with Google as an identity provider to access custom SAML apps.
B. In Cloud Identity, set up SSO with a third-party identity provider with Google as a service provider.
C. Obtain OAuth 2.0 credentials, configure the user consent screen, and set up OAuth 2.0 for Mobile & Desktop Apps.
D. Obtain OAuth 2.0 credentials, configure the user consent screen, and set up OAuth 2.0 for Web Server Applications.

Answer
B

Detailed explanation
A. Incorrect - Setting up Google as an identity provider is for cases where Google acts as the SSO provider; here the company's third-party SSO should act as the identity provider.
B. Correct - Configure Cloud Identity to use your company's third-party SAML-capable identity provider (Google acts as the service provider). This lets Cloud Identity users authenticate via the corporate SSO provider.
C. Incorrect - OAuth 2.0 for mobile/desktop apps is unrelated to SAML-based SSO for Cloud Identity.
D. Incorrect - OAuth 2.0 web server flows are not the mechanism to integrate a SAML SSO identity provider with Cloud Identity for organizational SSO.

Question 136
Your organization has a dedicated person who creates and manages all service accounts for Google Cloud projects. You need to assign this person the minimum role for projects. What should you do?

Options
A. Add the user to roles/iam.roleAdmin role.
B. Add the user to roles/iam.securityAdmin role.
C. Add the user to roles/iam.serviceAccountUser role.
D. Add the user to roles/iam.serviceAccountAdmin role.

Answer
D

Detailed explanation
A. Incorrect - roleAdmin grants excessive permissions to manage all IAM roles and is broader than required.
B. Incorrect - securityAdmin focuses on security-related resources and is not scoped to service account administration.
C. Incorrect - serviceAccountUser allows using service accounts but not creating or managing them.
D. Correct - serviceAccountAdmin is the least privilege role that allows creating, updating, and managing service accounts across projects, which matches the requirement.

Question 137
You are building an archival solution for your data warehouse and have selected Cloud Storage to archive your data. Your users need to be able to access this archived data once a quarter for some regulatory requirements. You want to select a cost-efficient option. Which storage option should you use?

Options
A. Cold Storage
B. Nearline Storage
C. Regional Storage
D. Multi-Regional Storage

Answer
A

Detailed explanation
A. Correct - Coldline Storage is designed for infrequent access (once a quarter or less) and provides the lowest cost option for long-term archival while still meeting regulatory access requirements.
B. Incorrect - Nearline Storage is cheaper than standard but intended for monthly access, not quarterly, so costs will be higher for this scenario.
C. Incorrect - Regional Storage is optimized for frequently accessed data within a specific region, making it more expensive for archival workloads.
D. Incorrect - Multi-Regional Storage is the most expensive option, intended for highly available, frequently accessed global data, not quarterly regulatory access.

Question 138
A team of data scientists infrequently needs to use a Google Kubernetes Engine (GKE) cluster that you manage. They require GPUs for some long-running, non-restartable jobs. You want to minimize cost. What should you do?

Options
A. Enable node auto-provisioning on the GKE cluster.
B. Create a VerticalPodAutoscaler for those workloads.
C. Create a node pool with preemptible VMs and GPUs attached to those VMs.
D. Create a node pool of instances with GPUs, and enable autoscaling on this node pool with a minimum size of 1.

Answer
D

Detailed explanation
A. Incorrect - Node auto-provisioning automatically adds node pools but does not ensure GPUs are provisioned for specific workloads.
B. Incorrect - VerticalPodAutoscaler adjusts CPU and memory for Pods but does not manage GPU scaling or costs effectively.
C. Incorrect - Preemptible GPU VMs are not suitable for long-running, non-restartable jobs because they can be terminated at any time.
D. Correct - A GPU-enabled node pool with autoscaling and minimum size 1 ensures GPUs are available when needed, scaling down when not, which balances cost and workload reliability.

Question 139
Your organization has user identities in Active Directory. Your organization wants to use Active Directory as their source of truth for identities. Your organization wants to have full control over the Google accounts used by employees for all Google services, including your Google Cloud Platform (GCP) organization. What should you do?

Options
A. Use Google Cloud Directory Sync (GCDS) to synchronize users into Cloud Identity.
B. Use the cloud Identity APIs and write a script to synchronize users to Cloud Identity.
C. Export users from Active Directory as a CSV and import them to Cloud Identity via the Admin Console.
D. Ask each employee to create a Google account using self signup. Require that each employee use their company email address and password.

Answer
A

Detailed explanation
A. Correct - Google Cloud Directory Sync integrates Active Directory with Cloud Identity, keeping user accounts synchronized automatically and allowing AD to remain the source of truth.
B. Incorrect - Writing custom sync scripts with Identity APIs is possible but not recommended since GCDS already provides a secure, supported solution.
C. Incorrect - Importing CSVs is a manual process, not scalable for continuous synchronization.
D. Incorrect - Asking employees to self-signup does not provide central management or enforce organizational control.

Question 140
You have successfully created a development environment in a project for an application. This application uses Compute Engine and Cloud SQL. Now you need to create a production environment for this application. The security team has forbidden the existence of network routes between these two environments and has asked you to follow Google-recommended practices. What should you do?

Options
A. Create a new project, enable the Compute Engine and Cloud SQL APIs in that project, and replicate the setup you have created in the development environment.
B. Create a new production subnet in the existing VPC and a new production Cloud SQL instance in your existing project, and deploy your application using those resources.
C. Create a new project, modify your existing VPC to be a Shared VPC, share that VPC with your new project, and replicate the setup you have in the development environment in that new project in the Shared VPC.
D. Ask the security team to grant you the Project Editor role in an existing production project used by another division of your company. Once they grant you that role, replicate the setup you have in the development environment in that project.

Answer
A

Detailed explanation
A. Correct - Creating a separate project for production isolates it from development, satisfies the requirement for no shared routes, and follows the Google-recommended multi-project separation of environments.

B. Incorrect - A subnet in the same VPC would still share network routes with development, which violates the security team's requirement.
C. Incorrect - A Shared VPC allows communication between projects, again violating the isolation requirement.
D. Incorrect - Using another division's production project introduces risks, governance issues, and does not ensure isolation or best practices.

Question 141

Your management has asked an external auditor to review all the resources in a specific project. The security team has enabled the Organization Policy called Domain Restricted Sharing on the organization node by specifying only your Cloud Identity domain. You want the auditor to only be able to view, but not modify, the resources in that project. What should you do?

Options
A. Ask the auditor for their Google account, and give them the Viewer role on the project.
B. Ask the auditor for their Google account, and give them the Security Reviewer role on the project.
C. Create a temporary account for the auditor in Cloud Identity, and give that account the Viewer role on the project.
D. Create a temporary account for the auditor in Cloud Identity, and give that account the Security Reviewer role on the project.

Answer
C

Detailed explanation
A. Incorrect: External auditors will not have accounts under your Cloud Identity domain, so you cannot simply assign them roles directly.
B. Incorrect: Same limitation as A, external auditors' own Google accounts won't pass the Domain Restricted Sharing policy.
C. Correct: You must create a temporary Cloud Identity account inside your domain (which complies with Domain Restricted Sharing) and then assign the Viewer role. This allows read-only access without modification rights.
D. Incorrect: Security Reviewer provides more privileges than required for a standard audit. Viewer is the recommended minimal role.

Question 142

You have a workload running on Compute Engine that is critical to your business. You want to ensure that the data on the boot disk of this workload is backed up regularly. You need to be able to restore a backup as quickly as possible in case of disaster. You also want older backups to be cleaned automatically to save on cost. You want to follow Google-recommended practices. What should you do?

Options
A. Create a Cloud Function to create an instance template.
B. Create a snapshot schedule for the disk using the desired interval.
C. Create a cron job to create a new disk from the disk using gcloud.
D. Create a Cloud Task to create an image and export it to Cloud Storage.

Answer
B

Detailed explanation
A. Incorrect: Instance templates are for launching instances, not for scheduling disk backups.
B. Correct: A snapshot schedule automates disk backups at regular intervals and supports retention policies to delete old snapshots, which saves cost while meeting recovery objectives.
C. Incorrect: Manual cron job scripting is error-prone and not recommended when snapshot schedules exist as a built-in feature.
D. Incorrect: Exporting images to Cloud Storage adds unnecessary overhead and slows down recovery compared to snapshots.

Question 143

You need to assign a Cloud Identity and Access Management (Cloud IAM) role to an external auditor. The auditor needs to have permissions to review your Google Cloud Platform (GCP) Audit Logs and also to review your Data Access logs. What should you do?

Options
A. Assign the auditor the IAM role roles/logging.privateLogViewer. Perform the export of logs to Cloud Storage.
B. Assign the auditor the IAM role roles/logging.privateLogViewer. Direct the auditor to also review the logs for changes to Cloud IAM policy.
C. Assign the auditor's IAM user to a custom role that has logging.privateLogEntries.list permission. Perform the export of logs to Cloud Storage.
D. Assign the auditor's IAM user to a custom role that has logging.privateLogEntries.list permission. Direct the auditor to also review the logs for changes to Cloud IAM policy.

Answer
B

Detailed explanation
A. Incorrect: Exporting logs to Cloud Storage is unnecessary; the role itself provides viewing access.
B. Correct: The roles/logging.privateLogViewer IAM role grants access to Audit Logs and Data Access logs. Directing the auditor to use this role is the proper Google-recommended practice.
C. Incorrect: A custom role is unnecessary; Google provides a predefined role (privateLogViewer) for exactly this use case.
D. Incorrect: Same issue as C, custom roles add complexity when a predefined role already satisfies requirements.

Question 144

You are managing several Google Cloud Platform (GCP) projects and need access to all logs for the past 60 days. You want to be able to explore and quickly analyze the log contents. You want to follow Google-recommended practices to obtain the combined logs for all projects. What should you do?

Options
A. Navigate to Stackdriver Logging and select resource.labels.project_id="*".
B. Create a Stackdriver Logging Export with a Sink destination to a BigQuery dataset. Configure the table expiration to 60 days.
C. Create a Stackdriver Logging Export with a Sink destination to Cloud Storage. Create a lifecycle rule to delete objects after 60 days.
D. Configure a Cloud Scheduler job to read from Stackdriver and store the logs in BigQuery. Configure the table expiration to 60 days.

Answer
B

Detailed explanation
A. Incorrect: The Logging console does not support querying across projects with a wildcard like *.
B. Correct: Exporting logs to BigQuery via a Logging sink allows you to query logs across multiple projects efficiently with SQL and set automatic expiration to manage retention costs.
C. Incorrect: Exporting to Cloud Storage is cheaper but not efficient for querying and analyzing logs quickly.
D. Incorrect: Cloud Scheduler isn't needed. Logging exports already provide a direct integration with BigQuery.

Question 145

You need to reduce GCP service costs for a division of your company using the fewest possible steps. You need to turn off all configured services in an existing GCP project. What should you do?

Options
A. Verify that you are assigned the Project Owners IAM role for this project. Locate the project in the GCP console, click Shut down and then enter the project ID.
B. Verify that you are assigned the Project Owners IAM role for this project. Switch to the project in the GCP console, locate the resources and delete them.
C. Verify that you are assigned the Organizational Administrator IAM role for this project. Locate the project in the GCP console, enter the project ID and then click Shut down.
D. Verify that you are assigned the Organizational Administrators IAM role for this project. Switch to the project in the GCP console, locate the resources and delete them.

Answer
A

Detailed explanation
A. Correct: Shutting down the project is the fastest and cleanest way to stop all resources and billing. It requires Project Owner role and confirmation of project ID.
B. Incorrect: Manually deleting resources is slower and error-prone. Some services might still remain active.
C. Incorrect: Organizational Administrator role is not required to shut down a project. Project Owner is sufficient.
D. Incorrect: Same issue as C — too broad a role and unnecessary steps.

Question 146

You are configuring service accounts for an application that spans multiple projects. Virtual machines (VMs) running in the web-applications project need access to BigQuery datasets in crm-databases-proj. You want to follow Google-recommended practices to give access to the service account in the web-applications project. What should you do?

Options
A. Incorrect: Give project owner for web-applications appropriate roles to crm-databases-proj. This gives excessive privileges, violating the principle of least privilege.
B. Incorrect: Give project owner role to crm-databases-proj and the web-applications project. Again, project owner role is unnecessary and insecure.
C. Incorrect: Give project owner role to crm-databases-proj and bigquery.dataViewer role to web-applications. Project owner is not required at all here.
D. Correct: Give bigquery.dataViewer role to crm-databases-proj and appropriate roles to web-applications. This provides only the necessary dataset access following the principle of least privilege.

Answer
D

Detailed explanation
A. Incorrect because project owner role provides full administrative control instead of just dataset access.
B. Incorrect because granting owner roles to both projects exposes unnecessary permissions and risks.
C. Incorrect because the service account does not need ownership at the project level, only dataset-level permissions.
D. Correct because assigning bigquery.dataViewer on crm-databases-proj allows the VM's service account to query datasets, and appropriate roles in web-applications project handle VM execution. This is the minimal and recommended approach.

Question 147

An employee was terminated, but their access to Google Cloud Platform (GCP) was not removed until 2 weeks later. You need to find out if this employee accessed any sensitive customer information after their termination. What should you do?

Options
A. Incorrect: View System Event Logs in Stackdriver. Search for the user's email as the principal. System Event Logs show configuration events, not data access.
B. Incorrect: View System Event Logs in Stackdriver. Search for the service account associated with the user. Service accounts don't track human user data access directly.
C. Correct: View Data Access audit logs in Stackdriver. Search for the user's email as the principal. Data Access logs show who read or modified data.
D. Incorrect: View the Admin Activity log in Stackdriver. Search for the service account associated with the user. Admin logs show resource changes, not customer data reads.

Answer
C

Detailed explanation
A. Incorrect because System Event Logs focus on infrastructure actions like VM start/stop, not sensitive data access.
B. Incorrect because using service account info doesn't confirm personal user actions after termination.

C. Correct because Data Access audit logs capture who accessed or modified data. Searching by the employee's email confirms whether sensitive information was accessed.
D. Incorrect because Admin Activity logs record administrative actions such as IAM policy changes, not data queries.

Question 148

You need to create a custom IAM role for use with a GCP service. All permissions in the role must be suitable for production use. You also want to clearly share with your organization the status of the custom role. This will be the first version of the custom role. What should you do?

Options
A. Correct: Use permissions in your role that use the 'supported' support level for role permissions. Set the role stage to ALPHA while testing the role permissions. This ensures production-safe permissions and clear role maturity.
B. Incorrect: Use permissions with 'supported' level but set the role stage to BETA. BETA is used later, not for the first version.
C. Incorrect: Use permissions with 'testing' support level. Testing permissions are not recommended for production roles.
D. Incorrect: Same issue as C, using 'testing' permissions is unsafe for production.

Answer
A

Detailed explanation
A. Correct because "supported" permissions are production-ready. Marking the role stage as ALPHA signals that the role is under testing but still uses stable permissions.
B. Incorrect because BETA is for more mature versions of roles. For first rollout, ALPHA is appropriate.
C. Incorrect because "testing" permissions are experimental and not production-grade.
D. Incorrect for the same reason as C — unsafe permissions for a production role.

Question 149

Your company has a large quantity of unstructured data in different file formats. You want to perform ETL transformations on the data. You need to make the data accessible on Google Cloud so it can be processed by a Dataflow job. What should you do?

Options
A. Incorrect: Upload the data to BigQuery using the bq command line tool. BigQuery does not ingest raw unstructured files directly.
B. Correct: Upload the data to Cloud Storage using the gsutil command line tool. Cloud Storage is the staging area for Dataflow to process unstructured data.
C. Incorrect: Upload the data into Cloud SQL using the import function in the console. Cloud SQL is designed for structured relational data only.
D. Incorrect: Upload the data into Cloud Spanner using the import function in the console. Cloud Spanner also handles structured relational databases, not raw unstructured files.

Answer
B

Detailed explanation
A. Incorrect because BigQuery expects structured data or external tables referencing storage, not raw uploads of files.

B. Correct because Cloud Storage is the recommended, scalable storage solution for staging unstructured data. Dataflow pipelines can directly read from it for transformations.
C. Incorrect because Cloud SQL does not support raw file ingestion.
D. Incorrect because Cloud Spanner is not suited for raw unstructured files, only structured database workloads.

Question 150

You need to manage multiple Google Cloud projects in the fewest steps possible. You want to configure the Google Cloud SDK command line interface (CLI) so that you can easily manage multiple projects. What should you do?

Options
A. Correct: Create a configuration for each project you need to manage. Activate the appropriate configuration when you work with each of your assigned Google Cloud projects. This is efficient and recommended.
B. Incorrect: Create a configuration for each project, then repeatedly run gcloud init to update values. This is inefficient compared to using configurations.
C. Incorrect: Use the default configuration for one project only. This doesn't scale when managing multiple projects.
D. Incorrect: Same inefficiency as B, using gcloud init each time is not practical.

Answer
A

Detailed explanation
A. Correct because creating and switching named configurations (gcloud config configurations create/activate) is the recommended practice for multi-project management.
B. Incorrect because using gcloud init repeatedly is time-consuming and not scalable.
C. Incorrect because relying only on the default configuration restricts you to a single project at a time.
D. Incorrect because it again requires running gcloud init frequently instead of using configurations.

Question 151
You used the gcloud container clusters command to create two Google Cloud Kubernetes (GKE) clusters: prod-cluster and dev-cluster.

- prod-cluster is a standard cluster.
- dev-cluster is an auto-pilot cluster.

When you run the kubectl get nodes command, you only see the nodes from prod-cluster. Which commands should you run to check the node status for dev-cluster?

Options
A. gcloud container clusters get-credentials dev-cluster kubectl get nodes
B. gcloud container clusters update -generate-password dev-cluster kubectl get nodes
C. kubectl config set-context dev-cluster kubectl cluster-info
D. kubectl config set-credentials dev-cluster kubectl cluster-info

Answer
A

Detailed explanation
A. - correct. Running gcloud container clusters get-credentials dev-cluster updates kubeconfig to point to dev-cluster, and then kubectl get nodes retrieves the dev-cluster's node list.
B. - incorrect. Updating cluster password is not the right way to switch context or check nodes.
C. - incorrect. kubectl config set-context alone won't configure kubeconfig with proper credentials from GKE.
D. - incorrect. set-credentials does not fetch GKE authentication details and won't allow node listing.

Question 152
You recently discovered that your developers are using many service account keys during their development process. While you work on a long-term improvement, you need to quickly implement a process to enforce short-lived service account credentials in your company. You have the following requirements:

- All service accounts that require a key should be created in a centralized project called pj-sa.
- Service account keys should only be valid for one day.

You need a Google-recommended solution that minimizes cost. What should you do?

Options
A. Implement a Cloud Run job to rotate all service account keys periodically in pj-sa. Enforce an org policy to deny service account key creation with an exception to pj-sa.
B. Implement a Kubernetes CronJob to rotate all service account keys periodically. Disable attachment of service accounts to resources in all projects with an exception to pj-sa.
C. Enforce an org policy constraint allowing the lifetime of service account keys to be 24 hours. Enforce an org policy constraint denying service account key creation with an exception on pj-sa.
D. Enforce a DENY org policy constraint over the lifetime of service account keys for 24 hours. Disable attachment of service accounts to resources in all projects with an exception to pj-sa.

Answer
C

Detailed explanation
A. - incorrect. Cloud Run rotation jobs add unnecessary complexity and cost compared to enforcing org policies.
B. - incorrect. Kubernetes CronJob rotation is not a Google-recommended or cost-minimizing solution.
C. - correct. Google recommends enforcing org policies for service account key lifetime and key creation restrictions. This minimizes cost and ensures compliance.
D. - incorrect. DENY policies here would prevent service accounts from functioning instead of just limiting key lifetime.

Question 153
Your company is running a three-tier web application on virtual machines that use a MySQL database. You need to create an estimated total cost of cloud infrastructure to run this application on Google Cloud instances and Cloud SQL. What should you do?

Options
A. Create a Google spreadsheet with multiple Google Cloud resource combinations. On a separate sheet, import the current Google Cloud prices and use these prices for the calculations within formulas.
B. Use the Google Cloud Pricing Calculator and select the Cloud Operations template to define your web application with as much detail as possible.
C. Implement a similar architecture on Google Cloud, and run a reasonable load test on a smaller scale. Check the billing information, and calculate the estimated costs based on the real load your system usually handles.
D. Use the Google Cloud Pricing Calculator to determine the cost of every Google Cloud resource you expect to use. Use similar size instances for the web server, and use your current on-premises machines as a comparison for Cloud SQL.

Answer
D

Detailed explanation
A. - incorrect. Manual spreadsheets with imported pricing are inefficient and not recommended.
B. - incorrect. The Cloud Operations template is not the right fit for cost estimation of a 3-tier app.
C. - incorrect. Running real workloads for estimation is time-consuming and costly.
D. - correct. The Google Cloud Pricing Calculator is the recommended way to estimate costs. Using current on-premises machines for SQL comparison ensures realistic projections.

Question 154
You have a Bigtable instance that consists of three nodes that store personally identifiable information (PII) data. You need to log all read or write operations, including any metadata or configuration reads of this database table, in your company's Security Information and Event Management (SIEM) system. What should you do?

Options
A. Navigate to Cloud Monitoring in the Google Cloud console, and create a custom monitoring job for the Bigtable instance to track all changes. Create an alert by using webhook endpoints, with the SIEM endpoint as a receiver.
B. Navigate to the Audit Logs page in the Google Cloud console, and enable Admin Write logs for the Bigtable instance. Create a Cloud Functions instance to export logs from Cloud Logging to your SIEM.
C. Navigate to the Audit Logs page in the Google Cloud console, and enable Data Read, Data Write, and Admin Read logs for the Bigtable instance. Create a Pub/Sub topic as a Cloud Logging sink destination, and add your SIEM as a subscriber to the topic.
D. Install the Ops Agent on the Bigtable instance during configuration. Create a service account with read permissions for the Bigtable instance. Create a custom Dataflow job with this service account to export logs to the company's SIEM system.

Answer
C

Detailed explanation
A. - incorrect. Monitoring cannot capture full audit logs for PII-sensitive operations.
B. - incorrect. Admin Write logs only capture a subset of operations, not full audit logging.
C. - correct. Audit Logs with Data Read, Data Write, and Admin Read combined with Pub/Sub export to SIEM provide a complete, recommended solution.
D. - incorrect. Ops Agent and Dataflow are not the right tools for centralized SIEM export of Bigtable logs.

Question 155
You want to set up a Google Kubernetes Engine cluster. Verifiable node identity and integrity are required for the cluster, and nodes cannot be accessed from the internet. You want to reduce the operational cost of managing your cluster, and you want to follow Google-recommended practices. What should you do?

Options
A. Deploy a private autopilot cluster.
B. Deploy a public autopilot cluster.
C. Deploy a standard public cluster and enable shielded nodes.
D. Deploy a standard private cluster and enable shielded nodes.

Answer
A

Detailed explanation
A. - correct. A private autopilot cluster ensures nodes are not internet-accessible, enforces strong security with verifiable node identity, and reduces operational cost since Autopilot manages infrastructure automatically.
B. - incorrect. Public clusters allow internet exposure, which is against the requirement.
C. - incorrect. A public cluster, even with shielded nodes, still exposes nodes to the internet.
D. - incorrect. A private standard cluster meets security needs but has higher operational cost than Autopilot, making it less recommended.

Question 156
Your company wants to migrate their on-premises workloads to Google Cloud. The current on-premises workloads consist of:

- A Flask web application
- A backend API
- A scheduled long-running background job for ETL and reporting

You need to keep operational costs low. You want to follow Google-recommended practices to migrate these workloads to serverless solutions on Google Cloud. What should you do?

Options
A. Migrate the web application to App Engine and the backend API to Cloud Run. Use Cloud Tasks to run your background job on Compute Engine.
B. Migrate the web application to App Engine and the backend API to Cloud Run. Use Cloud Tasks to run your background job on Cloud Run.
C. Run the web application on a Cloud Storage bucket and the backend API on Cloud Run. Use Cloud Tasks to run your background job on Cloud Run.
D. Run the web application on a Cloud Storage bucket and the backend API on Cloud Run. Use Cloud Tasks to run your background job on Compute Engine.

Answer
B

Detailed explanation
A. - incorrect. Using Compute Engine for background jobs increases cost and operational overhead.
B. - correct. Migrating to App Engine (Flask app) + Cloud Run (API and ETL jobs via Cloud Tasks) is the serverless, cost-efficient, and Google-recommended approach.
C. - incorrect. Hosting a Flask app on Cloud Storage bucket is not possible since it requires a compute environment, not just static hosting.
D. - incorrect. Similar to A, Compute Engine adds unnecessary costs compared to Cloud Run.

Question 157
Your company is moving its continuous integration and delivery (CI/CD) pipeline to Compute Engine instances. The pipeline will manage the entire cloud infrastructure through code. How can you ensure that the pipeline has appropriate permissions while your system is following security best practices?

Options
A. Attach a single service account to the compute instances. Add minimal rights to the service account. Allow the service account to impersonate a Cloud Identity user with elevated permissions to create, update, or delete resources.
B. Add a step for human approval to the CI/CD pipeline before the execution of the infrastructure provisioning. Use the human approvals IAM account for the provisioning.
C. Attach a single service account to the compute instances. Add all required Identity and Access Management (IAM) permissions to this service account to create, update, or delete resources.
D. Create multiple service accounts, one for each pipeline with the appropriate minimal Identity and Access Management (IAM) permissions. Use a secret manager service to store the key files of the service accounts. Allow the CI/CD pipeline to request the appropriate secrets during the execution of the pipeline.

Answer
D

Detailed explanation
A. - incorrect. Impersonating a Cloud Identity user is not recommended and risks privilege escalation.
B. - incorrect. Manual human approvals reduce automation and are not aligned with CI/CD best practices.
C. - incorrect. A single account with broad permissions violates the principle of least privilege.
D. - correct. Using multiple service accounts with minimal IAM permissions and integrating Secret Manager ensures security, least privilege, and automation — the recommended Google approach.

Question 158
Your application stores files on Cloud Storage by using the Standard Storage class. The application only requires access to files created in the last 30 days. You want to automatically save costs on files that are no longer accessed by the application. What should you do?

Options
A. Create an object lifecycle on the storage bucket to change the storage class to Archive Storage for objects with an age over 30 days.
B. Create a cron job in Cloud Scheduler to call a Cloud Functions instance every day to delete files older than 30 days.
C. Create a retention policy on the storage bucket of 30 days, and lock the bucket by using a retention policy lock.
D. Enable object versioning on the storage bucket and add lifecycle rules to expire non-current versions after 30 days.

Answer
A

Detailed explanation
A. - correct. Lifecycle rules are the recommended, automated, cost-saving way to move older files to cheaper storage classes like Archive.
B. - incorrect. Cloud Scheduler + Cloud Functions introduces unnecessary complexity and cost.
C. - incorrect. Retention policies enforce immutability; they don't reduce storage costs automatically.
D. - incorrect. Object versioning increases cost by storing multiple versions, opposite of the requirement.

Question 159
Your manager asks you to deploy a workload to a Kubernetes cluster. You are not sure of the workload's resource requirements or how the requirements might vary depending on usage patterns, external dependencies, or other factors. You need a solution that makes cost-effective recommendations regarding CPU and memory requirements, and allows the workload to function consistently in any situation. You want to follow Google-recommended practices. What should you do?

Options
A. Configure the Horizontal Pod Autoscaler for availability, and configure the cluster autoscaler for suggestions.
B. Configure the Horizontal Pod Autoscaler for availability, and configure the Vertical Pod Autoscaler recommendations for suggestions.
C. Configure the Vertical Pod Autoscaler recommendations for availability, and configure the Cluster autoscaler for suggestions.
D. Configure the Vertical Pod Autoscaler recommendations for availability, and configure the Horizontal Pod Autoscaler for suggestions.

Answer
B

Detailed explanation
A. - incorrect. The cluster autoscaler manages nodes, not CPU/memory recommendations for pods.
B. - correct. Horizontal Pod Autoscaler ensures availability by scaling pods, while Vertical Pod Autoscaler gives cost-effective CPU/memory recommendations. This is the recommended combination.

C. - incorrect. Vertical Pod Autoscaler does not handle availability — HPA is needed for that.
D. - incorrect. Reversing the roles is misleading; VPA does not manage availability scaling.

Question 160
You need to migrate invoice documents stored on-premises to Cloud Storage. The documents have the following storage requirements:

- Documents must be kept for five years.
- Up to five revisions of the same invoice document must be stored, to allow for corrections.
- Documents older than 365 days should be moved to lower cost storage tiers.

You want to follow Google-recommended practices to minimize your operational and development costs. What should you do?

Options
A. Enable retention policies on the bucket, and use Cloud Scheduler to invoke a Cloud Function to move or delete your documents based on their metadata.
B. Enable retention policies on the bucket, use lifecycle rules to change the storage classes of the objects, set the number of versions, and delete old files.
C. Enable object versioning on the bucket, and use Cloud Scheduler to invoke a Cloud Functions instance to move or delete your documents based on their metadata.
D. Enable object versioning on the bucket, use lifecycle conditions to change the storage class of the objects, set the number of versions, and delete old files.

Answer
D

Detailed explanation
A. - incorrect. Cloud Scheduler + Cloud Functions adds operational complexity; retention policies alone don't manage versions or class transitions.
B. - incorrect. Retention policies enforce immutability, which does not meet the revision requirement.
C. - incorrect. Cloud Scheduler with Functions is not the cost-effective or recommended solution.
D. - correct. Object versioning maintains up to 5 revisions, lifecycle rules automate cost-saving transitions to lower storage classes, and deletion of old files meets all requirements.

Question 161
You installed the Google Cloud CLI on your workstation and set the proxy configuration. However, you are worried that your proxy credentials will be recorded in the gcloud CLI logs. You want to prevent your proxy credential from being logged. What should you do?

Options
A. Configure username and password by using gcloud config set proxy/username and gcloud config set proxy/password commands.
B. Encode username and password in sha256 encoding, and save it to a text file. Use filename as a value in the gcloud config set core/custom_ca_certs_file command.
C. Provide values for CLOUDSDK_PROXY_USERNAME and CLOUDSDK_PROXY_PASSWORD in the gcloud CLI tool configuration file.
D. Set the CLOUDSDK_PROXY_USERNAME and CLOUDSDK_PROXY_PASSWORD properties by using environment variables in your command line tool.

Answer
D

Detailed explanation
A. - incorrect. Using gcloud config set will store credentials in CLI config and can appear in logs.
B. - incorrect. Encoding credentials does not prevent logging, it just obfuscates.
C. - incorrect. Storing credentials in the config file still risks logging and exposure.
D. - correct. Environment variables are the recommended and secure method since they prevent credentials from being written to logs or config files.

Question 162
Your company developed an application to deploy on Google Kubernetes Engine. Certain parts of the application are not fault-tolerant and are allowed to have downtime. Other parts of the application are critical and must always be available. You need to configure a Google Kubernetes Engine cluster while optimizing for cost. What should you do?

Options
A. Create a cluster with a single node-pool by using standard VMs. Label the fault-tolerant Deployments as spot_true.
B. Create a cluster with a single node-pool by using Spot VMs. Label the critical Deployments as spot_false.
C. Create a cluster with both a Spot VM node pool and a node pool by using standard VMs. Deploy the critical deployments on the Spot VM node pool and the fault-tolerant deployments on the node pool by using standard VMs.
D. Create a cluster with both a Spot VM node pool and a node pool by using standard VMs. Deploy the critical deployments on the node pool by using standard VMs and the fault-tolerant deployments on the Spot VM node pool.

Answer
D

Detailed explanation
A. - incorrect. Fault-tolerant workloads should go to Spot VMs, not standard.
B. - incorrect. Critical workloads should not be run on Spot VMs because they can be preempted.
C. - incorrect. Reverses the correct placement — critical workloads on Spot VMs is wrong.
D. - correct. Standard VMs ensure critical workloads are always available, and Spot VMs reduce cost for fault-tolerant workloads.

Question 163
You need to deploy an application in Google Cloud using serverless technology. You want to test a new version of the application with a small percentage of production traffic. What should you do?

Options
A. Deploy the application to Cloud Run. Use gradual rollouts for traffic splitting.
B. Deploy the application to Google Kubernetes Engine. Use Anthos Service Mesh for traffic splitting.
C. Deploy the application to Cloud Functions. Specify the version number in the functions name.
D. Deploy the application to App Engine. For each new version, create a new service.

Answer
A

Detailed explanation
A. - correct. Cloud Run supports gradual rollouts with percentage-based traffic splitting between revisions.
B. - incorrect. GKE + Anthos Service Mesh works but is more complex than required for a serverless solution.
C. - incorrect. Cloud Functions does not support fine-grained traffic splitting.
D. - incorrect. App Engine creates separate services per version, not efficient for gradual rollout testing.

Question 164
Your company's security vulnerability management policy wants a member of the security team to have visibility into vulnerabilities and other OS metadata for a specific Compute Engine instance. This Compute Engine instance hosts a critical application in your Google Cloud project. You need to implement your company's security vulnerability management policy. What should you do?

Options
A. Ensure that the Ops Agent is installed on the Compute Engine instance. Create a custom metric in the Cloud Monitoring dashboard. Provide the security team member with access to this dashboard.
B. Ensure that the Ops Agent is installed on the Compute Engine instance. Provide the security team member roles/osconfig.inventoryViewer permission.
C. Ensure that the OS Config agent is installed on the Compute Engine instance. Provide the security team member roles/osconfig.vulnerabilityReportViewer permission.
D. Ensure that the OS Config agent is installed on the Compute Engine instance. Create a log sink to BigQuery dataset. Provide the security team member with access to this dataset.

Answer
C

Detailed explanation
A. - incorrect. Ops Agent + Monitoring only provides metrics, not detailed vulnerability visibility.
B. - incorrect. InventoryViewer only gives package inventory visibility, not vulnerability reporting.
C. - correct. OS Config agent with vulnerabilityReportViewer role gives precise vulnerability insights.
D. - incorrect. Exporting logs to BigQuery is unnecessary and not the recommended approach.

Question 165
You want to enable your development team to deploy new features to an existing Cloud Run service in production. To minimize the risk associated with a new revision, you want to reduce the number of customers who might be affected by an outage without introducing any development or operational costs to your customers. You want to follow Google-recommended practices for managing revisions to a service. What should you do?

Options
A. Ask your customers to retry access to your service with exponential backoff to mitigate any potential problems after the new revision is deployed.
B. Gradually roll out the new revision and split customer traffic between the revisions to allow rollback in case a problem occurs.
C. Send all customer traffic to the new revision, and roll back to a previous revision if you witness any problems in production.
D. Deploy your application to a second Cloud Run service, and ask your customers to use the second Cloud Run service.

Answer
B

Detailed explanation
A. - incorrect. Asking customers to retry is not a reliable rollout strategy.
B. - correct. Cloud Run supports gradual rollout and traffic splitting, allowing safe testing and rollback.
C. - incorrect. Sending all traffic at once increases risk of customer impact.
D. - incorrect. Running a duplicate service doubles cost and effort unnecessarily.

Question 166
You have deployed an application on a Compute Engine instance. An external consultant needs to access the Linux-based instance. The consultant is connected to your corporate network through a VPN connection, but the consultant has no Google account. What should you do?

Options
A. Instruct the external consultant to use the gcloud compute ssh command line tool by using Identity-Aware Proxy to access the instance.
B. Instruct the external consultant to use the gcloud compute ssh command line tool by using the public IP address of the instance to access it.
C. Instruct the external consultant to generate an SSH key pair, and request the public key from the consultant. Add the public key to the instance yourself, and have the consultant access the instance through SSH with their private key.
D. Instruct the external consultant to generate an SSH key pair, and request the private key from the consultant. Add the private key to the instance yourself, and have the consultant access the instance through SSH with their public key.

Answer
C

Detailed explanation
A. - incorrect. Identity-Aware Proxy requires a Google account, which the consultant does not have.
B. - incorrect. Public IP access works but still requires credentials or a Google account.
C. - correct. The proper way is to use SSH key pairs. Consultant generates keys, you add the public key to instance, they use their private key for login.
D. - incorrect. Private keys should never be shared for security reasons.

Question 167
After a recent security incident, your startup company wants better insight into what is happening in the Google Cloud environment. You need to monitor unexpected firewall changes and instance creation. Your company prefers simple solutions. What should you do?

Options
A. Create a log sink to forward Cloud Audit Logs filtered for firewalls and compute instances to Cloud Storage. Use BigQuery to periodically analyze log events in the storage bucket.
B. Use Cloud Logging filters to create log-based metrics for firewall and instance actions. Monitor the changes and set up reasonable alerts.
C. Install Kibana on a compute instance. Create a log sink to forward Cloud Audit Logs filtered for firewalls and compute instances to Pub/Sub. Target the Pub/Sub topic to push messages to the Kibana instance. Analyze the logs on Kibana in real time.
D. Turn on Google Cloud firewall rules logging, and set up alerts for any insert, update, or delete events.

Answer
B

Detailed explanation
A. - incorrect. BigQuery analysis is not simple and adds cost/complexity.
B. - correct. Cloud Logging + log-based metrics + alerts is a simple, cost-effective way to monitor unexpected changes.
C. - incorrect. Kibana setup is complex and unnecessary for this requirement.
D. - incorrect. Firewall rule logging only tracks firewall rules, not instance creations, so it's incomplete.

Question 168
You are configuring service accounts for an application that spans multiple projects. Virtual machines (VMs) running in the web-applications project need access to BigQuery datasets in the crm-databases project. You want to follow Google-recommended practices to grant access to the service account in the web-applications project. What should you do?

Options
A. Grant "project owner" for web-applications appropriate roles to crm-databases.
B. Grant "project owner" role to crm-databases and the web-applications project.
C. Grant "project owner" role to crm-databases and roles/bigquery.dataViewer role to web-applications.
D. Grant roles/bigquery.dataViewer role to crm-databases and appropriate roles to web-applications.

Answer
D

Detailed explanation
A. - incorrect. Granting project owner is overly permissive and not best practice.
B. - incorrect. Granting project owner to both projects breaks principle of least privilege.
C. - incorrect. Still assigns project owner role unnecessarily.
D. - correct. Assigning only required roles (e.g., BigQuery Data Viewer) follows least privilege and best practices.

Question 169
Your Dataproc cluster runs in a single Virtual Private Cloud (VPC) network in a single subnetwork with range 172.16.20.128/25. There are no private IP addresses available in the subnetwork. You want to add new VMs to communicate with your cluster using the minimum number of steps. What should you do?

Options
A. Modify the existing subnet range to 172.16.20.0/24.
B. Create a new Secondary IP Range in the VPC and configure the VMs to use that range.
C. Create a new VPC network for the VMs. Enable VPC Peering between the VMs' VPC network and the Dataproc cluster VPC network.
D. Create a new VPC network for the VMs with a subnet of 172.32.0.0/16. Enable VPC network Peering between the Dataproc VPC network and the VMs VPC network. Configure a custom Route exchange.

Answer
A

Detailed explanation
A. - correct. Expanding the subnet to /24 increases IP availability with minimal effort.
B. - incorrect. Secondary ranges are for alias IPs, not adding IPs for new VMs.
C. - incorrect. Creating a new VPC with peering is more complex than needed.
D. - incorrect. Similar to C, adds unnecessary complexity compared to subnet expansion.

Question 170
You are building a backend service for an ecommerce platform that will persist transaction data from mobile and web clients. After the platform is launched, you expect a large volume of global transactions. Your business team wants to run SQL queries to analyze the data. You need to build a highly available and scalable data store for the platform. What should you do?

Options
A. Create a multi-region Cloud Spanner instance with an optimized schema.
B. Create a multi-region Firestore database with aggregation query enabled.
C. Create a multi-region Cloud SQL for PostgreSQL database with optimized indexes.
D. Create a multi-region BigQuery dataset with optimized tables.

Answer
A

Detailed explanation
A. - correct. Cloud Spanner provides global availability, strong consistency, horizontal scalability, and supports SQL queries, making it ideal for transactional workloads.
B. - incorrect. Firestore is NoSQL and not designed for global-scale relational transactions.
C. - incorrect. Cloud SQL does not scale horizontally for high global transactional workloads.
D. - incorrect. BigQuery is for analytics, not transactional persistence.

Question 171
You are in charge of provisioning access for all Google Cloud users in your organization. Your company recently acquired a startup company that has their own Google Cloud organization. You need to ensure that your Site Reliability Engineers (SREs) have the same project permissions in the startup company's organization as in your own organization. What should you do?

Options
A. In the Google Cloud console for your organization, select Create role from selection, and choose destination as the startup company's organization.
B. In the Google Cloud console for the startup company, select Create role from selection and choose source as the startup company's Google Cloud organization.
C. Use the gcloud iam roles copy command, and provide the Organization ID of the startup company's Google Cloud Organization as the destination.
D. Use the gcloud iam roles copy command, and provide the project IDs of all projects in the startup company's organization as the destination.

Answer
C

Detailed explanation
A. - incorrect. You cannot directly copy roles between organizations in the console.
B. - incorrect. Same issue, console does not allow cross-organization role copy.
C. - correct. The gcloud iam roles copy command allows role copying between organizations, specifying the startup org ID as the destination.
D. - incorrect. Copying roles project by project is inefficient compared to using the organization ID.

Question 172
You need to extract text from audio files by using the Speech-to-Text API. The audio files are pushed to a Cloud Storage bucket. You need to implement a fully managed, serverless compute solution that requires authentication and aligns with Google-recommended practices. You want to automate the call to the API by submitting each file to the API as the audio file arrives in the bucket. What should you do?

Options
A. Create an App Engine standard environment triggered by Cloud Storage bucket events to submit the file URI to the Google Speech-to-Text API.
B. Run a Kubernetes job to scan the bucket regularly for incoming files, and call the Speech-to-Text API for each unprocessed file.
C. Run a Python script by using a Linux cron job in Compute Engine to scan the bucket regularly for incoming files, and call the Speech-to-Text API for each unprocessed file.
D. Create a Cloud Function triggered by Cloud Storage bucket events to submit the file URI to the Google Speech-to-Text API.

Answer
D

Detailed explanation
A. - incorrect. App Engine is not event-driven directly by bucket events.
B. - incorrect. Kubernetes adds unnecessary operational overhead.
C. - incorrect. Cron jobs on Compute Engine are not serverless and not cost-efficient.
D. - correct. Cloud Functions + Cloud Storage event trigger is fully managed, serverless, and recommended for this workflow.

Question 173
Your customer wants you to create a secure website with autoscaling based on the compute instance CPU load. You want to enhance performance by storing static content in Cloud Storage. Which resources are needed to distribute the user traffic?

Options
A. An external HTTPS load balancer with a managed SSL certificate to distribute the load and a URL map to target the requests for the static content to the Cloud Storage backend.
B. An external network load balancer pointing to the backend instances to distribute the load evenly. The web servers will forward the request to the Cloud Storage as needed.
C. An internal HTTPS load balancer together with Identity-Aware Proxy to allow only HTTPS traffic.
D. An external HTTPS load balancer to distribute the load and a URL map to target the requests for the static content to the Cloud Storage backend. Install the HTTPS certificates on the instance.

Answer
A

Detailed explanation
A. - correct. External HTTPS load balancer + managed SSL + URL map → ensures secure access and proper routing to Cloud Storage backend.
B. - incorrect. Network load balancer does not support HTTPS or direct Cloud Storage backend.
C. - incorrect. Internal HTTPS load balancer is not for public-facing websites.
D. - incorrect. Installing certs on instances adds complexity; managed SSL certs are recommended.

Question 174
The core business of your company is to rent out construction equipment at large scale. All the equipment is equipped with multiple sensors that send event information every few seconds. These signals can vary from engine status, distance traveled, fuel level, etc. Customers are billed based on this data. You expect high throughput (up to thousands of events per hour per device) and need to retrieve consistent data based on the event time. Storing and retrieving individual signals should be atomic. What should you do?

Options
A. Create files in Cloud Storage as data comes in.
B. Create a file in Filestore per device, and append new data to that file.
C. Ingest the data into Cloud SQL. Use multiple read replicas to match the throughput.
D. Ingest the data into Bigtable. Create a row key based on the event timestamp.

Answer
D

Detailed explanation
A. - incorrect. Cloud Storage is for object storage, not high-throughput event data ingestion.
B. - incorrect. Filestore is not designed for handling streaming IoT-scale writes.
C. - incorrect. Cloud SQL cannot handle such high write throughput and atomicity requirements.
D. - correct. Bigtable is built for high-throughput, time-series IoT data with atomic row operations.

Question 175
You just installed the Google Cloud CLI on your new corporate laptop. You need to list the existing instances of your company on Google Cloud. What must you do before you run the gcloud compute instances list command? (Choose two.)

Options
A. Run gcloud auth login, enter your login credentials in the dialog window, and paste the received login token to gcloud CLI.
B. Create a Google Cloud service account, and download the service account key. Place the key file in a folder on your machine where gcloud CLI can find it.
C. Download your Cloud Identity user account key. Place the key file in a folder on your machine where gcloud CLI can find it.
D. Run gcloud config set compute/zone $my_zone to set the default zone for gcloud CLI.
E. Run gcloud config set project $my_project to set the default project for gcloud CLI.

Answer
A, E

Detailed explanation
A. - correct. Authentication via gcloud auth login is needed.
B. - incorrect. Service accounts are not required for basic CLI usage as a user.
C. - incorrect. There's no user account key for Cloud Identity like this.
D. - incorrect. Zone is optional for listing instances.
E. - correct. You must set the default project for the CLI to know where to list instances.

Question 176
You are planning to migrate your on-premises data to Google Cloud. The data includes:

- 200 TB of video files in SAN storage
- Data warehouse data stored on Amazon Redshift
- 20 GB of PNG files stored on an S3 bucket

You need to load the video files into a Cloud Storage bucket, transfer the data warehouse data into BigQuery, and load the PNG files into a second Cloud Storage bucket. You want to follow Google-recommended practices and avoid writing any code for the migration. What should you do?

Options
A. Use gcloud storage for the video files, Dataflow for the data warehouse data, and Storage Transfer Service for the PNG files.
B. Use Transfer Appliance for the videos, BigQuery Data Transfer Service for the data warehouse data, and Storage Transfer Service for the PNG files.
C. Use Storage Transfer Service for the videos, BigQuery Data Transfer Service for the data warehouse data, and Storage Transfer Service for the PNG files.
D. Use Cloud Data Fusion for the videos, Dataflow for the data warehouse data, and Storage Transfer Service for the PNG files.

Answer
B

Detailed explanation
A. - incorrect. Gcloud storage and Dataflow require manual intervention.
B. - correct. Transfer Appliance handles large-scale (200 TB) migration, BigQuery Data Transfer supports Redshift, and Storage Transfer Service moves S3 data easily without code.
C. - incorrect. Storage Transfer Service is not suitable for massive (200 TB) on-prem SAN files.
D. - incorrect. Cloud Data Fusion is for data pipelines, not bulk file migration.

Question 177
You want to deploy a new containerized application into Google Cloud by using a Kubernetes manifest. You want to have full control over the Kubernetes deployment, and at the same time, you want to minimize configuring infrastructure. What should you do?

Options
A. Deploy the application on GKE Autopilot.
B. Deploy the application on Cloud Run.
C. Deploy the application on GKE Standard.
D. Deploy the application on Cloud Functions.

Answer
A

Detailed explanation
A. - correct. GKE Autopilot lets you deploy manifests with full Kubernetes control while minimizing infra management.
B. - incorrect. Cloud Run does not use Kubernetes manifests.
C. - incorrect. GKE Standard requires managing infrastructure nodes.
D. - incorrect. Cloud Functions does not support Kubernetes manifests.

Question 178
Your team is building a website that handles votes from a large user population. The incoming votes will arrive at various rates. You want to optimize the storage and processing of the votes. What should you do?

Options
A. Save the incoming votes to Firestore. Use Cloud Scheduler to trigger a Cloud Functions instance to periodically process the votes.
B. Use a dedicated instance to process the incoming votes. Send the votes directly to this instance.
C. Save the incoming votes to a JSON file on Cloud Storage. Process the votes in a batch at the end of the day.
D. Save the incoming votes to Pub/Sub. Use the Pub/Sub topic to trigger a Cloud Functions instance to process the votes.

Answer
D

Detailed explanation
A. - incorrect. Firestore + Scheduler adds unnecessary delay and overhead.
B. - incorrect. Dedicated instance is not scalable or cost-effective.
C. - incorrect. Batch processing on Cloud Storage is inefficient for real-time needs.
D. - correct. Pub/Sub provides scalable ingestion, and Cloud Functions process votes serverlessly in real-time.

Question 179
You are deploying an application on Google Cloud that requires a relational database for storage. To satisfy your company's security policies, your application must connect to your database through an encrypted and authenticated connection that requires minimal management and integrates with Identity and Access Management (IAM). What should you do?

Options
A. Deploy a Cloud SQL database with the SSL mode set to encrypted only, configure SSL/TLS client certificates, and configure a database user and password.
B. Deploy a Cloud SQL database with the SSL mode set to encrypted only, configure SSL/TLS client certificates, and configure IAM database authentication.
C. Deploy a Cloud SQL database and configure IAM database authentication. Access the database through the Cloud SQL Auth Proxy.
D. Deploy a Cloud SQL database and configure a database user and password. Access the database through the Cloud SQL Auth Proxy.

Answer
C

Detailed explanation
A. - incorrect. Database user/password does not integrate with IAM.
B. - incorrect. Certificates still require management overhead.
C. - correct. IAM database authentication + Cloud SQL Auth Proxy ensures encryption, authentication, and minimal management.
D. - incorrect. Database passwords don't leverage IAM integration.

Question 180
You have two Google Cloud projects: project-a with VPC vpc-a (10.0.0.0/16) and project-b with VPC vpc-b (10.8.0.0/16). Your frontend application resides in vpc-a and the backend API services are deployed in vpc-b. You need to efficiently and cost-effectively enable communication between these Google Cloud projects. You also want to follow Google-recommended practices. What should you do?

Options
A. Create an OpenVPN connection between vpc-a and vpc-b.
B. Create VPC Network Peering between vpc-a and vpc-b.
C. Configure a Cloud Router in vpc-a and another Cloud Router in vpc-b.
D. Configure a Cloud Interconnect connection between vpc-a and vpc-b.

Answer
B

Detailed explanation
A. - incorrect. VPN adds complexity and cost, not needed within Google Cloud.
B. - correct. VPC Network Peering is the simplest and recommended method for inter-project communication within GCP.
C. - incorrect. Cloud Routers are for dynamic routing with interconnect or VPN, not needed here.
D. - incorrect. Cloud Interconnect is for on-prem to cloud connectivity, not between projects.

Question 181
Every employee of your company has a Google account. Your operational team needs to manage a large number of instances on Compute Engine. Each member of this team needs only administrative access to the servers. Your security team wants to ensure that the deployment of credentials is operationally efficient and must be able to determine who accessed a given instance. What should you do?

Options
A. Generate a new SSH key pair. Give the private key to each member of your team. Configure the public key in the metadata of each instance.
B. Ask each member of the team to generate a new SSH key pair and to send you their public key. Use a configuration management tool to deploy those keys on each instance.
C. Ask each member of the team to generate a new SSH key pair and to add the public key to their Google account. Grant the compute.osAdminLogin role to the Google group corresponding to this team.
D. Generate a new SSH key pair. Give the private key to each member of your team. Configure the public key as a project-wide public SSH key in your Cloud Platform project and allow project-wide public SSH keys on each instance.

Answer
C

Detailed explanation
A. Incorrect - Sharing one private key among multiple users removes accountability and is insecure.
B. Partially correct - Individual keys provide accountability, but manual distribution doesn't scale.
C. Correct - Each user has their own SSH key linked to their Google account, with group-based role assignment. This ensures both efficiency and accountability.
D. Incorrect - A shared project-wide key repeats the accountability issue from option A.

Question 182
You need to create a custom VPC with a single subnet. The subnet's range must be as large as possible. Which range should you use?

Options
A. 0.0.0.0/0
B. 10.0.0.0/8
C. 172.16.0.0/12
D. 192.168.0.0/16

Answer
B

Detailed explanation
A. Incorrect - 0.0.0.0/0 represents the entire internet, not valid for subnets.
B. Correct - 10.0.0.0/8 provides the largest private address space (16M+ IPs).
C. Incorrect - 172.16.0.0/12 is smaller (1M IPs).
D. Incorrect - 192.168.0.0/16 is the smallest (65K IPs).

Question 183
You want to select and configure a cost-effective solution for relational data on Google Cloud Platform. You are working with a small set of operational data in one geographic location. You need to support point-in-time recovery. What should you do?

Options
A. Select Cloud SQL (MySQL). Verify that the enable binary logging option is selected.
B. Select Cloud SQL (MySQL). Select the create failover replicas option.
C. Select Cloud Spanner. Set up your instance with 2 nodes.
D. Select Cloud Spanner. Set up your instance as multi-regional.

Answer
A

Detailed explanation
A. Correct - Binary logging in Cloud SQL allows point-in-time recovery and is cost-effective.
B. Incorrect - Failover replicas improve availability, not point-in-time recovery.
C. Incorrect - Spanner is for large-scale distributed workloads, overkill here.
D. Incorrect - Multi-regional Spanner is the most expensive choice.

Question 184
You want to configure autohealing for network load balancing for a group of Compute Engine instances that run in multiple zones, using the fewest possible steps. You need to configure re-creation of VMs if they are unresponsive after 3 attempts of 10 seconds each. What should you do?

Options
A. Create an HTTP load balancer with a backend configuration that references an existing instance group. Set the health check to healthy (HTTP).
B. Create an HTTP load balancer with a backend configuration that references an existing instance group. Define a balancing mode and set the maximum RPS to 10.
C. Create a managed instance group. Set the Autohealing health check to healthy (HTTP).
D. Create a managed instance group. Verify that the autoscaling setting is on.

Answer
C

Detailed explanation
A. Incorrect - Load balancer health checks only manage traffic routing, not VM recreation.
B. Incorrect - Backend balancing limits requests, doesn't handle autohealing.
C. Correct - Managed instance groups support autohealing with health checks, recreating VMs after failure.
D. Incorrect - Autoscaling handles capacity changes, not autohealing.

Question 185
You are using multiple configurations for gcloud. You want to review the configured Kubernetes Engine cluster of an inactive configuration using the fewest possible steps. What should you do?

Options
A. Use gcloud config configurations describe to review the output.
B. Use gcloud config configurations activate and gcloud config list to review the output.
C. Use kubectl config get-contexts to review the output.
D. Use kubectl config use-context and kubectl config view to review the output.

Answer
D

Detailed explanation
A. Incorrect - describe shows config properties, not cluster details.
B. Incorrect - Switching configs works, but requires multiple steps.
C. Incorrect - get-contexts lists contexts but doesn't show cluster config.
D. Correct - kubectl config use-context with view allows inspecting inactive configs with minimal steps.

Question 186
Your company uses Cloud Storage to store application backup files for disaster recovery purposes. You want to follow Google's recommended practices. Which storage option should you use?

Options
A. Multi-Regional Storage
B. Regional Storage
C. Nearline Storage
D. Coldline Storage

Answer
D

Detailed explanation
A. Incorrect - Multi-Regional is for frequently accessed global data, more expensive.
B. Incorrect - Regional is for data frequently accessed in one region.
C. Incorrect - Nearline is cost-effective but for monthly access, not long-term.
D. Correct - Coldline is designed for long-term storage and rare access, ideal for backups.

Question 187
Several employees at your company have been creating projects with Cloud Platform and paying for it with their personal credit cards, which the company reimburses. The company wants to centralize all these projects under a single, new billing account. What should you do?

Options
A. Contact support with your bank account details and request a corporate billing account.
B. Create a ticket with Google Support and wait for them to share your credit card details over the phone.
C. In the Google Platform Console, go to Resource Manager and move all projects to the root Organization.
D. In the Google Cloud Platform Console, create a new billing account and set up a payment method.

Answer
D

Detailed explanation
A. Incorrect - You don't contact support directly for billing account creation.
B. Incorrect - Google never shares credit card details this way.
C. Incorrect - Moving projects to Organization doesn't change billing.
D. Correct - Create a new billing account with proper corporate payment, then link projects to it.

Question 188
You have an application that looks for its licensing server on the IP 10.0.3.21. You need to deploy the licensing server on Compute Engine. You do not want to change the configuration of the application and want the application to be able to reach the licensing server. What should you do?

Options
A. Reserve the IP 10.0.3.21 as a static internal IP address using gcloud and assign it to the licensing server.
B. Reserve the IP 10.0.3.21 as a static public IP address using gcloud and assign it to the licensing server.
C. Use the IP 10.0.3.21 as a custom ephemeral IP address and assign it to the licensing server.
D. Start the licensing server with an automatic ephemeral IP address, and then promote it to a static internal IP address.

Answer
A

Detailed explanation
A. Correct - The application is configured to look for the internal IP 10.0.3.21. Reserving this exact address as a static internal IP ensures it always maps to the licensing server. Internal IPs are permanent when reserved and guarantee that the server is reachable without reconfiguring the application.
B. Incorrect - A static public IP is external and not relevant here, because the application expects a private/internal IP.
C. Incorrect - Ephemeral IPs are temporary. Even if you specify a custom ephemeral, it could change after VM restart, breaking the licensing application.
D. Incorrect - Promoting an ephemeral IP to static works for existing VMs but cannot guarantee pre-defined internal IP like 10.0.3.21.

Question 189
You are deploying an application to App Engine. You want the number of instances to scale based on request rate. You need at least 3 unoccupied instances at all times. Which scaling type should you use?

Options
A. Manual Scaling with 3 instances.
B. Basic Scaling with min_instances set to 3.
C. Basic Scaling with max_instances set to 3.
D. Automatic Scaling with min_idle_instances set to 3.

Answer
D

Detailed explanation
A. Incorrect - Manual scaling fixes the number of instances but does not auto-scale based on traffic. It stays static.
B. Incorrect - Basic scaling spins up instances only when requests arrive. The min_instances flag isn't supported; it won't guarantee idle instances.
C. Incorrect - Setting max_instances only caps scaling; it doesn't guarantee a minimum.
D. Correct - Automatic scaling with min_idle_instances=3 ensures that at least 3 instances are always running and ready to handle traffic, while still scaling up as demand increases.

Question 190
You have a development project with appropriate IAM roles defined. You are creating a production project and want to have the same IAM roles on the new project, using the fewest possible steps. What should you do?

Options
A. Use gcloud iam roles copy and specify the production project as the destination project.
B. Use gcloud iam roles copy and specify your organization as the destination organization.
C. In the Google Cloud Platform Console, use the 'create role from role' functionality.
D. In the Google Cloud Platform Console, use the 'create role' functionality and select all applicable permissions.

Answer
A

Detailed explanation
A. Correct - gcloud iam roles copy allows cloning IAM roles from one project to another with a single command. This is the fastest and least error-prone method.
B. Incorrect - Copying to the organization level applies roles org-wide, which is unnecessary and not requested here.
C. Incorrect - "Create role from role" in the console requires manual action and is slower than a direct gcloud copy.
D. Incorrect - Creating roles manually requires selecting permissions again, more work and prone to errors.

Question 191
You need a dynamic way of provisioning VMs on Compute Engine. The exact specifications will be in a dedicated configuration file. You want to follow Google's recommended practices. Which method should you use?

Options
A. Deployment Manager
B. Cloud Composer
C. Managed Instance Group
D. Unmanaged Instance Group

Answer
A

Detailed explanation
A. Correct - Deployment Manager allows you to define infrastructure as code in YAML/JSON configuration files. It automates provisioning and is the Google-recommended IaC tool.
B. Incorrect - Cloud Composer is for workflow orchestration, not VM provisioning.
C. Incorrect - Managed instance groups are good for scaling identical VMs, but not for dynamic specification via config files.
D. Incorrect - Unmanaged instance groups require manual handling and don't support declarative provisioning.

Question 192
You have a Dockerfile that you need to deploy on Kubernetes Engine. What should you do?

Options
A. Use kubectl app deploy <dockerfilename>.
B. Use gcloud app deploy <dockerfilename>.
C. Create a docker image from the Dockerfile and upload it to Container Registry. Create a Deployment YAML file to point to that image. Use kubectl to create the deployment with that file.
D. Create a docker image from the Dockerfile and upload it to Cloud Storage. Create a Deployment YAML file to point to that image. Use kubectl to create the deployment with that file.

Answer
C

Detailed explanation
A. Incorrect - There's no such kubectl command for directly deploying Dockerfiles.
B. Incorrect - gcloud app deploy is for App Engine, not Kubernetes.
C. Correct - Standard Kubernetes workflow is to build the Docker image, push it to Container Registry, then deploy using a YAML manifest and kubectl.
D. Incorrect - Cloud Storage is not used as a container image registry. Container Registry or Artifact Registry must be used.

Question 193
Your development team needs a new Jenkins server for their project. You need to deploy the server using the fewest steps possible. What should you do?

Options
A. Download and deploy the Jenkins Java WAR to App Engine Standard.
B. Create a new Compute Engine instance and install Jenkins through the command line interface.
C. Create a Kubernetes cluster on Compute Engine and create a deployment with the Jenkins Docker image.
D. Use GCP Marketplace to launch the Jenkins solution.

Answer
D

Detailed explanation
A. Incorrect - App Engine Standard doesn't support arbitrary WAR deployment like Jenkins.
B. Incorrect - Installing manually on a VM works, but requires many steps (OS, dependencies, config).
C. Incorrect - Running Jenkins in Kubernetes is possible but complex for a simple setup.
D. Correct - Marketplace offers pre-packaged Jenkins images, which can be deployed in a few clicks. This is the fastest, recommended option.

Question 194
You need to update a deployment in Deployment Manager without any resource downtime in the deployment. Which command should you use?

Options
A. gcloud deployment-manager deployments create --config <deployment-config-path>
B. gcloud deployment-manager deployments update --config <deployment-config-path>
C. gcloud deployment-manager resources create --config <deployment-config-path>
D. gcloud deployment-manager resources update --config <deployment-config-path>

Answer
B

Detailed explanation
A. Incorrect - create is only for new deployments, not updates.
B. Correct - update safely modifies an existing deployment using the new config file, without destroying resources.
C. Incorrect - Resources create is not a valid gcloud command.
D. Incorrect - Same issue, not applicable at deployment level.

Question 195
You need to run an important query in BigQuery but expect it to return a lot of records. You want to find out how much it will cost to run the query. You are using on-demand pricing. What should you do?

Options
A. Arrange to switch to Flat-Rate pricing for this query, then move back to on-demand.
B. Use the command line to run a dry run query to estimate the number of bytes read. Then convert that bytes estimate to dollars using the Pricing Calculator.
C. Use the command line to run a dry run query to estimate the number of rows returned. Then convert that rows estimate to dollars using the Pricing Calculator.
D. Run a select count(*) query to estimate how many rows your query will look through. Then convert that number of rows to dollars using the Pricing Calculator.

Answer
B

Detailed explanation
A. Incorrect - Flat-rate pricing is for consistent workloads and can't be switched temporarily.
B. Correct - A dry run in BigQuery calculates how many bytes will be scanned. Pricing is based on bytes read, not rows returned, so this is the correct way to estimate cost.
C. Incorrect - Rows returned don't determine cost. Only data scanned matters.
D. Incorrect - Counting rows doesn't reflect storage bytes scanned, so estimate will be wrong.

Question 196
You have a single binary application that you want to run on Google Cloud Platform. You decided to automatically scale the application based on underlying infrastructure CPU usage. Your organizational policies require you to use virtual machines directly. You need to ensure that the application scaling is operationally efficient and completed as quickly as possible. What should you do?

Options
A. Create a Google Kubernetes Engine cluster, and use horizontal pod autoscaling to scale the application.
B. Create an instance template, and use the template in a managed instance group with autoscaling configured.
C. Create an instance template, and use the template in a managed instance group that scales up and down based on the time of day.
D. Use a set of third-party tools to build automation around scaling the application up and down, based on Stackdriver CPU usage monitoring.

Answer
B

Detailed explanation
A. Incorrect - Kubernetes is unnecessary here since the policy requires direct VMs, not containers.
B. Correct - Managed instance groups with autoscaling automatically add/remove VMs based on CPU usage. This is efficient, scalable, and directly supports VM workloads.
C. Incorrect - Scaling by time is static scheduling, not responsive to real CPU load.
D. Incorrect - Third-party tooling increases complexity. GCP autoscaling is native and recommended.

Question 197
You are analyzing Google Cloud Platform service costs from three separate projects. You want to use this information to create service cost estimates by service type, daily and monthly, for the next six months using standard query syntax. What should you do?

Options
A. Export your bill to a Cloud Storage bucket, and then import into Cloud Bigtable for analysis.
B. Export your bill to a Cloud Storage bucket, and then import into Google Sheets for analysis.
C. Export your transactions to a local file, and perform analysis with a desktop tool.
D. Export your bill to a BigQuery dataset, and then write time window–based SQL queries for analysis.

Answer
D

Detailed explanation
A. Incorrect - Bigtable is a NoSQL store for low-latency queries, not ideal for analytical cost queries.
B. Incorrect - Google Sheets cannot efficiently handle large billing data with SQL-like aggregations.
C. Incorrect - Exporting locally works but doesn't scale and misses GCP-native analytics.
D. Correct - Exporting billing data to BigQuery allows using SQL queries with time windows, grouping by service type, daily/monthly. This is Google's recommended method.

Question 198
You need to set up a policy so that videos stored in a specific Cloud Storage Regional bucket are moved to Coldline after 90 days, and then deleted after one year from their creation. How should you set up the policy?

Options
A. Use Cloud Storage Object Lifecycle Management using Age conditions with SetStorageClass and Delete actions. Set the SetStorageClass action to 90 days and the Delete action to 275 days (365 – 90).
B. Use Cloud Storage Object Lifecycle Management using Age conditions with SetStorageClass and Delete actions. Set the SetStorageClass action to 90 days and the Delete action to 365 days.
C. Use gsutil rewrite and set the Delete action to 275 days (365 – 90).
D. Use gsutil rewrite and set the Delete action to 365 days.

Answer
B

Detailed explanation
A. Incorrect - Miscalculated days. Delete should be 365, not 275.
B. Correct - Lifecycle policy supports multiple rules: after 90 days move to Coldline, after 365 days delete. This matches the requirement exactly.
C. Incorrect - gsutil rewrite is manual, not a policy.
D. Incorrect - Same issue, gsutil rewrite isn't used for automated lifecycle policies.

Question 199
You have a Linux VM that must connect to Cloud SQL. You created a service account with the appropriate access rights. You want to make sure that the VM uses this service account instead of the default Compute Engine service account. What should you do?

Options
A. When creating the VM via the web console, specify the service account under the 'Identity and API Access' section.
B. Download a JSON Private Key for the service account. On the Project Metadata, add that JSON as the value for the key compute-engine-service-account.
C. Download a JSON Private Key for the service account. On the Custom Metadata of the VM, add that JSON as the value for the key compute-engine-service-account.
D. Download a JSON Private Key for the service account. After creating the VM, ssh into the VM and save the JSON under ~/.gcloud/compute-engine-service-account.json.

Answer
A

Detailed explanation
A. Correct - The recommended way is to assign the custom service account directly when creating the VM. GCP ensures the VM uses this identity for API access.
B. Incorrect - Adding JSON to project metadata is not how service accounts are configured.
C. Incorrect - Same issue, metadata isn't intended for key injection.
D. Incorrect - Manually saving JSON keys is insecure and not recommended.

Question 200
You created an instance of SQL Server 2017 on Compute Engine to test features in the new version. You want to connect to this instance using the fewest number of steps. What should you do?

Options
A. Install an RDP client on your desktop. Verify that a firewall rule for port 3389 exists.
B. Install an RDP client on your desktop. Set a Windows username and password in the GCP Console. Use the credentials to log in to the instance.
C. Set a Windows password in the GCP Console. Verify that a firewall rule for port 22 exists. Click the RDP button in the GCP Console and supply the credentials to log in.
D. Set a Windows username and password in the GCP Console. Verify that a firewall rule for port 3389 exists. Click the RDP button in the GCP Console and supply the credentials to log in.

Answer
B

Detailed explanation
A. Incorrect - Simply checking firewall rules isn't enough; you need credentials too.
B. Correct - Setting Windows credentials in the Console, then using them with RDP is the minimum required set of steps.
C. Incorrect - Port 22 is for SSH, not RDP.
D. Incorrect - This includes extra steps (setting both username and password manually). Option B is Simler.

Question 201
You have one GCP account running in your default region and zone and another account running in a non-default region and zone. You want to start a new Compute Engine instance in these two Google Cloud Platform accounts using the command line interface. What should you do?

Options
A. Create two configurations using gcloud config configurations create [NAME]. Run gcloud config configurations activate [NAME] to switch between accounts when running the commands to start the Compute Engine instances.
B. Create two configurations using gcloud config configurations create [NAME]. Run gcloud config configurations list to start the Compute Engine instances.
C. Activate two configurations using gcloud config configurations activate [NAME]. Run gcloud config list to start the Compute Engine instances.
D. Activate two configurations using gcloud config configurations activate [NAME]. Run gcloud config configurations list to start the Compute Engine instances.

Answer
A

Detailed explanation
A. Correct - gcloud supports multiple named configurations. Activating the right config switches credentials/region before running commands.
B. Incorrect - list shows configs, it doesn't start instances.
C. Incorrect - Activating is correct, but gcloud config list just displays properties, not create instances.
D. Incorrect - Same as above, list doesn't create instances.

Question 202
You significantly changed a complex Deployment Manager template and want to confirm that the dependencies of all defined resources are properly met before committing it to the project. You want the most rapid feedback on your changes. What should you do?

Options
A. Use granular logging statements within a Deployment Manager template authored in Python.
B. Monitor activity of the Deployment Manager execution on the Stackdriver Logging page of the GCP Console.
C. Execute the Deployment Manager template against a separate project with the same configuration, and monitor for failures.
D. Execute the Deployment Manager template using the "-preview" option in the same project, and observe the state of interdependent resources.

Answer
D

Detailed explanation
A. Incorrect - Logging helps debug but doesn't validate dependencies upfront.
B. Incorrect - Monitoring logs shows issues only after execution begins.
C. Incorrect - Using a separate project takes more time and resources.
D. Correct - The preview option validates changes and shows planned state without creating resources. Fastest and safest way.

Question 203
You are building a pipeline to process time-series data. Which Google Cloud Platform services should you put in boxes 1, 2, 3, and 4?

Options
A. Cloud Pub/Sub, Cloud Dataflow, Cloud Datastore, BigQuery
B. Firebase Messages, Cloud Pub/Sub, Cloud Spanner, BigQuery
C. Cloud Pub/Sub, Cloud Storage, BigQuery, Cloud Bigtable
D. Cloud Pub/Sub, Cloud Dataflow, Cloud Bigtable, BigQuery

Answer
D

Detailed explanation
A. Incorrect - Datastore is not ideal for time-series.
B. Incorrect - Firebase is for mobile messaging, not streaming pipelines.
C. Incorrect - Cloud Storage is not a processing stage, only storage.
D. Correct - Pub/Sub ingests, Dataflow processes, Bigtable stores time-series, BigQuery analyzes.

Question 204
You have a project for your App Engine application that serves a development environment. The required testing has succeeded and you want to create a new project to serve as your production environment. What should you do?

Options
A. Use gcloud to create the new project, and then deploy your application to the new project.
B. Use gcloud to create the new project and to copy the deployed application to the new project.
C. Create a Deployment Manager configuration file that copies the current App Engine deployment into a new project.
D. Deploy your application again using gcloud and specify the project parameter with the new project name to create the new project.

Answer
A

Detailed explanation
A. Correct - Best practice is to create a fresh project and redeploy. Keeps dev/prod separated.
B. Incorrect - There’s no command to copy deployments between projects.
C. Incorrect - Deployment Manager doesn’t replicate entire App Engine environments.
D. Incorrect - gcloud deploy commands don’t create projects directly.

Question 205
You need to configure IAM access audit logging in BigQuery for external auditors. You want to follow Google-recommended practices. What should you do?

Options
A. Add the auditors group to the logging.viewer and bigQuery.dataViewer predefined IAM roles.
B. Add the auditors group to two new custom IAM roles.
C. Add the auditor user accounts to the logging.viewer and bigQuery.dataViewer predefined IAM roles.
D. Add the auditor user accounts to two new custom IAM roles.

Answer
A

Detailed explanation
A. Correct - Using groups with predefined roles is scalable and minimal privilege. Recommended approach.
B. Incorrect - Custom roles add overhead, unnecessary here.
C. Incorrect - Adding individual accounts works but isn't scalable.
D. Incorrect - Custom roles again add complexity.

Question 206
You need to set up permissions for a set of Compute Engine instances to enable them to write data into a particular Cloud Storage bucket. You want to follow Google-recommended practices. What should you do?

Options
A. Create a service account with an access scope. Use the access scope https://www.googleapis.com/auth/devstorage.write_only.
B. Create a service account with an access scope. Use the access scope https://www.googleapis.com/auth/cloud-platform.
C. Create a service account and add it to the IAM role storage.objectCreator for that bucket.
D. Create a service account and add it to the IAM role storage.objectAdmin for that bucket.

Answer
C

Detailed explanation
A. Incorrect - Access scopes are legacy and too broad.
B. Incorrect - Cloud-platform scope gives excessive permissions.
C. Correct - Best practice is to grant least-privilege IAM role storage.objectCreator which allows writing objects but not deleting.
D. Incorrect - objectAdmin gives unnecessary delete/manage rights.

Question 207
You have sensitive data stored in three Cloud Storage buckets and have enabled data access logging. You want to verify activities for a particular user for these buckets, using the fewest possible steps. You need to verify addition of metadata labels and which files have been viewed. What should you do?

Options
A. Using the GCP Console, filter the Activity log to view the information.
B. Using the GCP Console, filter the Stackdriver log to view the information.
C. View the bucket in the Storage section of the GCP Console.
D. Create a trace in Stackdriver to view the information.

Answer
B

Detailed explanation
A. Incorrect - Activity log shows admin actions (like creating buckets), not object-level access.
B. Correct - Stackdriver (Cloud Logging) captures detailed data access logs including label changes and object reads. Filtering by user gives exactly what's needed.
C. Incorrect - Viewing the bucket doesn't give historical access logs.
D. Incorrect - Stackdriver Trace is for request latency tracing, not storage activity.

Question 208
You are the project owner of a GCP project and want to delegate control to colleagues to manage buckets and files in Cloud Storage. You want to follow Google-recommended practices. Which IAM roles should you grant your colleagues?

Options
A. Project Editor
B. Storage Admin
C. Storage Object Admin
D. Storage Object Creator

Answer
B

Detailed explanation
A. Incorrect - Project Editor gives broad permissions beyond storage.
B. Correct - Storage Admin role allows full control over buckets and objects, without broader project access.
C. Incorrect - Storage Object Admin only manages objects, not buckets.
D. Incorrect - Object Creator allows uploads only.

Question 209
You have an object in a Cloud Storage bucket that you want to share with an external company. The object contains sensitive data. You want access to the content to be removed after four hours. The external company does not have a Google account. What should you do?

Options
A. Create a signed URL with a four-hour expiration and share the URL with the company.
B. Set object access to 'public' and use object lifecycle management to remove the object after four hours.
C. Configure the storage bucket as a static website and furnish the object's URL to the company. Delete the object from the storage bucket after four hours.
D. Create a new Cloud Storage bucket specifically for the external company to access. Copy the object to that bucket. Delete the bucket after four hours.

Answer
A

Detailed explanation
A. Correct - Signed URLs allow time-limited access without needing a Google account. Four-hour expiry fits perfectly.
B. Incorrect - Public access exposes data to everyone, insecure.
C. Incorrect - Manual deletion required, not automated, less secure.
D. Incorrect - Creating new buckets is unnecessary overhead.

Question 210
You are creating a Google Kubernetes Engine (GKE) cluster with a cluster autoscaler feature enabled. You need to make sure that each node of the cluster will run a monitoring pod that sends container metrics to a third-party monitoring solution. What should you do?

Options
A. Deploy the monitoring pod in a StatefulSet object.
B. Deploy the monitoring pod in a DaemonSet object.
C. Reference the monitoring pod in a Deployment object.
D. Reference the monitoring pod in a cluster initializer at the GKE cluster creation time.

Answer
B

Detailed explanation
A. Incorrect - StatefulSets are used when pods need persistent identity/storage, not for monitoring.
B. Correct - DaemonSets ensure that one pod runs on each node in the cluster, perfect for node-level monitoring agents.
C. Incorrect - Deployments scale arbitrarily, not per-node.
D. Incorrect - Cluster initializers are not used for pod scheduling.

Question 211
Your company is running a critical workload on a single Compute Engine VM instance. Your company's disaster recovery policies require you to back up the entire instance's disk data every day. The backups must be retained for 7 days. You must configure a backup solution that complies with your company's security policies and requires minimal setup and configuration. What should you do?

Options
A. Configure the instance to use persistent disk asynchronous replication.
B. Configure daily scheduled persistent disk snapshots with a retention period of 7 days.
C. Configure Cloud Scheduler to trigger a Cloud Function each day that creates a new machine image and deletes machine images that are older than 7 days.
D. Configure a bash script using gutil to run daily through a cron job. Copy the disk's files to a Cloud Storage bucket with archive storage class and an object lifecycle rule to delete the objects after 7 days.

Answer
B

Detailed explanation
A. - incorrect. Persistent disk replication is not intended as a backup solution.
B. - correct. Daily snapshots with retention of 7 days are the simplest and Google-recommended backup method.
C. - incorrect. Cloud Scheduler + Cloud Function is more complex than required.
D. - incorrect. Manual scripts add operational overhead and are less secure.

Question 212
Your company requires that Google Cloud products are created with a specific configuration to comply with your company's security policies. You need to implement a mechanism that will allow software engineers at your company to deploy and update Google Cloud products in a preconfigured and approved manner. What should you do?

Options
A. Create Java packages that utilize the Google Cloud Client Libraries for Java to configure Google Cloud products. Store and share the packages in a source code repository.
B. Create bash scripts that utilize the Google Cloud CLI to configure Google Cloud products. Store and share the bash scripts in a source code repository.
C. Use the Google Cloud APIs by using curl to configure Google Cloud products. Store and share the curl commands in a source code repository.
D. Create Terraform modules that utilize the Google Cloud Terraform Provider to configure Google Cloud products. Store and share the modules in a source code repository.

Answer
D

Detailed explanation
A. - incorrect. Java libraries are not suitable for standardized infrastructure provisioning.
B. - incorrect. Bash scripts are not scalable or easily reusable.
C. - incorrect. Curl commands lack structure and approval mechanisms.
D. - correct. Terraform modules with the Google Cloud provider enforce preconfigured policies and allow consistent, approved deployments.

Question 213
You are a Google Cloud organization administrator. You need to configure organization policies and log sinks on Google Cloud projects that cannot be removed by project users to comply with your company's security policies. The security policies are different for each company department. Each department has a user with the Project Owner role assigned to their projects. What should you do?

Options
A. Use a standard naming convention for projects that includes the department name. Configure organization policies on the organization and log sinks on the projects.
B. Use a standard naming convention for projects that includes the department name. Configure both organization policies and log sinks on the projects.
C. Organize projects under folders for each department. Configure both organization policies and log sinks on the folders.
D. Organize projects under folders for each department. Configure organization policies on the organization and log sinks on the folders.

Answer
C

Detailed explanation
A. - incorrect. Policies at the project level can be modified by project owners.
B. - incorrect. Same issue, project owners can override.
C. - correct. Organizing by folders allows applying org policies and log sinks at the folder level, which project owners cannot override.
D. - incorrect. Organization-level policies apply globally, not department-specific.

Question 214
You are deploying a web application using Compute Engine. You created a managed instance group (MIG) to host the application. You want to follow Google-recommended practices to implement a secure and highly available solution. What should you do?

Options
A. Use SSL proxy load balancing for the MIG and an A record in your DNS private zone with the load balancer's IP address.
B. Use SSL proxy load balancing for the MIG and a CNAME record in your DNS public zone with the load balancer's IP address.
C. Use HTTPS load balancing for the MIG and a CNAME record in your DNS private zone with the load balancer's IP address.
D. Use HTTPS load balancing for the MIG and an A record in your DNS public zone with the load balancer's IP address.

Answer
D

Detailed explanation
A. - incorrect. SSL proxy load balancer is not the recommended option.
B. - incorrect. Public CNAME mapping is less secure.
C. - incorrect. Using a private zone does not expose the application securely to the internet.
D. - correct. HTTPS load balancer with public A record ensures security and high availability.

Question 215
You have several hundred microservice applications running in a Google Kubernetes Engine (GKE) cluster. Each microservice is a deployment with resource limits configured for each container in the deployment. You've observed that the resource limits for memory and CPU are not appropriately set for many of the microservices. You want to ensure that each microservice has right sized limits for memory and CPU. What should you do?

Options
A. Configure a Vertical Pod Autoscaler for each microservice.
B. Modify the cluster's node pool machine type and choose a machine type with more memory and CPU.
C. Configure a Horizontal Pod Autoscaler for each microservice.
D. Configure GKE cluster autoscaling.

Answer
A

Detailed explanation
A. - correct. Vertical Pod Autoscaler automatically adjusts resource requests and limits for containers.
B. - incorrect. Increasing node size does not fix per-microservice right sizing.
C. - incorrect. HPA scales pods, not resource requests/limits.
D. - incorrect. Cluster autoscaling adds/removes nodes, not pod resource tuning.

Question 216
Your company uses BigQuery to store and analyze data. Upon submitting your query in BigQuery, the query fails with a quotaExceeded error. You need to diagnose the issue causing the error. What should you do? (Choose two.)

Options
A. Use BigQuery BI Engine to analyze the issue.
B. Use the INFORMATION_SCHEMA views to analyze the underlying issue.
C. Configure Cloud Trace to analyze the issue.
D. Search errors in Cloud Audit Logs to analyze the issue.
E. View errors in Cloud Monitoring to analyze the issue.

Answer
B, D

Detailed explanation
A. - incorrect. BI Engine is for acceleration, not quota diagnostics.
B. - correct. INFORMATION_SCHEMA views expose query usage and quota info.
C. - incorrect. Cloud Trace is for latency analysis.
D. - correct. Cloud Audit Logs capture BigQuery errors, including quotaExceeded.
E. - incorrect. Monitoring tracks metrics, not detailed query errors.

Question 217
Your team has developed a stateless application which requires it to be run directly on virtual machines. The application is expected to receive a fluctuating amount of traffic and needs to scale automatically. You need to deploy the application. What should you do?

Options
A. Deploy the application on a managed instance group and configure autoscaling.
B. Deploy the application on a Kubernetes Engine cluster and configure node pool autoscaling.
C. Deploy the application on Cloud Functions and configure the maximum number of instances.
D. Deploy the application on Cloud Run and configure autoscaling.

Answer
A

Detailed explanation
A. - correct. Managed instance groups with autoscaling are best for VM-based stateless apps.
B. - incorrect. Kubernetes is unnecessary complexity for direct VM apps.
C. - incorrect. Cloud Functions is not for VM workloads.
D. - incorrect. Cloud Run is container-based, not direct VM-based.

Question 218
Your web application is hosted on Cloud Run and needs to query a Cloud SQL database. Every morning during a traffic spike, you notice API quota errors in Cloud SQL logs. The project has already reached the maximum API quota. You want to make a configuration change to mitigate the issue. What should you do?

Options
A. Modify the minimum number of Cloud Run instances.
B. Use traffic splitting.
C. Modify the maximum number of Cloud Run instances.
D. Set a minimum concurrent requests environment variable for the application.

Answer
A

Detailed explanation
A. - correct. Increasing minimum instances keeps connections warm, reducing quota spikes.
B. - incorrect. Traffic splitting is for rollout, not quota issues.
C. - incorrect. Increasing max instances may worsen quota usage.
D. - incorrect. Concurrent requests help efficiency but do not fix quota spikes.

Question 219
You need to deploy a single stateless web application with a web interface and multiple endpoints. For security reasons, the web application must be reachable from an internal IP address from your company's private VPC and on-premises network. You also need to update the web application multiple times per day with minimal effort and want to manage a minimal amount of cloud infrastructure. What should you do?

Options
A. Deploy the web application on Google Kubernetes Engine standard edition with an internal ingress.
B. Deploy the web application on Cloud Run with Private Google Access configured.
C. Deploy the web application on Cloud Run with Private Service Connect configured.
D. Deploy the web application to GKE Autopilot with Private Google Access configured.

Answer
B

Detailed explanation
A. - incorrect. GKE requires more infra management.
B. - correct. Cloud Run with Private Google Access allows private VPC access with minimal infra management.
C. - incorrect. Private Service Connect is for service-to-service connections, not app hosting.
D. - incorrect. GKE Autopilot still requires more infra management than Cloud Run.

Question 220
You use Cloud Logging to capture application logs. You now need to use SQL to analyze the application logs in Cloud Logging, and you want to follow Google-recommended practices. What should you do?

Options
A. Develop SQL queries by using Gemini for Google Cloud.
B. Enable Log Analytics for the log bucket and create a linked dataset in BigQuery.
C. Create a schema for the storage bucket and run SQL queries for the data in the bucket.
D. Export logs to a storage bucket and create an external view in BigQuery.

Answer
B

Detailed explanation
A. - incorrect. Gemini is not the tool for log SQL queries.
B. - correct. Enabling Log Analytics integrates logging with BigQuery for SQL analysis.
C. - incorrect. Storage buckets are not designed for SQL queries directly.
D. - incorrect. Export + external view is possible but not the recommended approach.

Question 221
You need to deploy a third-party software application onto a single Compute Engine VM instance. The application requires the highest speed read and write disk access for the internal database. You need to ensure the instance will recover on failure. What should you do?

Options
A. Create an instance template. Set the disk type to be an SSD Persistent Disk. Launch the instance template as part of a stateful managed instance group.
B. Create an instance template. Set the disk type to be an SSD Persistent Disk. Launch the instance template as part of a stateless managed instance group.
C. Create an instance template. Set the disk type to be Hyperdisk Extreme. Launch the instance template as part of a stateful managed instance group.
D. Create an instance template. Set the disk type to be Hyperdisk Extreme. Launch the instance template as part of a stateless managed instance group.

Answer
C

Detailed explanation
A. - incorrect. SSD persistent disks are performant but not as fast as Hyperdisk Extreme for high IOPS workloads.
B. - incorrect. Stateless groups don't persist VM state and won't recover properly on failure.
C. - correct. Hyperdisk Extreme provides ultra-high IOPS and throughput, and stateful managed instance group ensures VM recovery with attached disks.
D. - incorrect. Stateless groups do not persist application state.

Question 222
You have a VM instance running in a VPC with single-stack subnets. You need to ensure that the VM instance has a fixed IP address so that other services hosted in the same VPC can communicate with the VM. You want to follow Google-recommended practices while minimizing cost. What should you do?

Options
A. Promote the existing IP address of the VM to become a static external IP address.
B. Promote the existing IP address of the VM to become a static internal IP address.
C. Reserve a new static external IPv6 address and assign the new IP address to the VM.
D. Reserve a new static external IP address and assign the new IP address to the VM.

Answer
B

Detailed explanation
A. - incorrect. External IPs are not needed for internal-only communication.
B. - correct. Promoting the internal IP to static ensures the VM keeps its private IP permanently at minimal cost.
C. - incorrect. IPv6 external addresses are unnecessary and costlier.
D. - incorrect. External IPs increase exposure and cost.

Question 223
Your preview application, deployed on a single-zone Google Kubernetes Engine (GKE) cluster in us-central1, has gained popularity. You are now ready to make the application generally available. You need to deploy the application to production while ensuring high availability and resilience. You also want to follow Google-recommended practices. What should you do?

Options
A. Use the gcloud container clusters create command with the options --enable-multi-networking and --enable-autoscaling to create an autoscaling zonal cluster and deploy the application to it.
B. Use the gcloud container clusters create-auto command to create an autopilot cluster and deploy the application to it.
C. Use the gcloud container clusters update command with the option --region us-central1 to update the cluster and deploy the application to it.
D. Use the gcloud container clusters update command with the option --node-locations us-central1-a,us-central1-b to update the cluster and deploy the application to the nodes.

Answer
D

Detailed explanation
A. - incorrect. Zonal clusters are not highly available; HA requires regional or multi-zone.
B. - incorrect. Autopilot would require recreating the cluster, not updating.
C. - incorrect. You cannot change a zonal cluster to regional with update.
D. - correct. Adding multiple node locations in the region turns the cluster into multi-zone, ensuring resilience and HA.

Question 224
You are developing an application that will be deployed on Google Cloud. The application will use a service account to retrieve data from BigQuery. Before you deploy your application, you want to test the permissions of this service account from your local machine to ensure there will be no authentication issues. You want to ensure that you use the most secure method while following Google-recommended practices. What should you do?

Options
A. Generate a service account key, and configure the gcloud CLI to use this key. Issue a relevant BigQuery request through the gcloud CLI to test the access.
B. Grant the service account the BigQuery Administrator IAM role to ensure the service account has all required access.
C. Configure the gcloud CLI to use service account impersonation. Issue a relevant BigQuery request through the gcloud CLI to test the access.
D. Configure the gcloud CLI with Application Default Credentials using your user account. Issue a relevant BigQuery request through the gcloud CLI to test the access.

Answer
C

Detailed explanation
A. - incorrect. Using service account keys is insecure and not recommended.
B. - incorrect. Over-privileging the account violates least privilege principle.
C. - correct. Service account impersonation avoids key management and follows best practices.
D. - incorrect. Using your user credentials doesn’t validate the service account permissions.

Question 225
Your organization is migrating to Google Cloud. You want only users with company-issued Google accounts to access your Google Cloud environment. You must ensure that users of the same department can only access resources within their own department. You want to minimize operational costs while following Google-recommended practices. What should you do?

Options
A. Assign users to the relevant Google Groups, and provide access to cloud resources through Identity and Access Management (IAM) roles. Periodically identify and remove non-company issued Google accounts.
B. Assign users to the relevant Google Groups, and provide access to cloud resources through Identity and Access Management (IAM) roles. Use organization policies to block non-company issued emails.
C. Create a folder for each department in Resource Manager. Grant the users of each department the Folder Admin role on the folder of their department.
D. Create a folder for each department in Resource Manager. Grant all company users the Folder Admin role on the organization level.

Answer
B

Detailed explanation
A. - incorrect. Manual removal of accounts is inefficient and error-prone.
B. - correct. Google Groups + IAM roles + org policies to block external emails ensures security with low operational cost.
C. - incorrect. Folder Admin rights at department level do not block external accounts.
D. - incorrect. Granting all users Folder Admin at org level is insecure.

Question 226
You are deploying an application to Cloud Run. Your application requires the use of an API that runs on Google Kubernetes Engine (GKE). You need to ensure that your Cloud Run service can privately reach the API on GKE, and you want to follow Google-recommended practices. What should you do?

Options
A. Deploy an ingress resource on the GKE cluster to expose the API to the internet. Use Cloud Armor to filter for IPs. On the Cloud Run service, configure the application to fetch its public IP address and update the Cloud Armor policy on startup.
B. Create an ingress firewall rule on the VPC to allow connections from 0.0.0.0/0 on ports 80 and 443.
C. Create an egress firewall rule on the VPC to allow connections to 0.0.0.0/0 on ports 80 and 443.
D. Deploy an internal Application Load Balancer to expose the API on GKE to the VPC. Configure Cloud DNS with the IP address of the internal Application Load Balancer. Deploy a Serverless VPC Access connector to allow the Cloud Run service to call the API through the FQDN on Cloud DNS.

Answer
D

Detailed explanation
A. - incorrect. Exposing API to the internet is insecure.
B. - incorrect. Allowing all inbound traffic (0.0.0.0/0) is insecure.
C. - incorrect. Allowing all outbound traffic is insecure.

D. - correct. Internal Load Balancer + DNS + VPC Access Connector ensures secure private connectivity.

Question 227
Your company uses a multi-cloud strategy that includes Google Cloud. You want to centralize application logs in a third-party SaaS tool from all environments. You need to integrate logs originating from Cloud Logging, and you want to ensure the export occurs with the least amount of delay possible. What should you do?

Options
A. Create a Cloud Logging sink and configure BigQuery as the destination. Configure the SaaS tool to query BigQuery to retrieve the logs.
B. Create a Cloud Logging sink and configure Pub/Sub as the destination. Configure the SaaS tool to subscribe to the Pub/Sub topic to retrieve the logs.
C. Create a Cloud Logging sink and configure Cloud Storage as the destination. Configure the SaaS tool to read the Cloud Storage bucket to retrieve the logs.
D. Use a Cloud Scheduler cron job to trigger a Cloud Function that queries Cloud Logging and sends the logs to the SaaS tool.

Answer
B

Detailed explanation
A. - incorrect. BigQuery introduces query latency.
B. - correct. Pub/Sub provides near real-time log export with minimal delay.
C. - incorrect. Cloud Storage introduces ingestion delays.
D. - incorrect. Scheduler + Cloud Function adds unnecessary overhead.

Question 228
You are planning to migrate a database and a backend application to a Standard Google Kubernetes Engine (GKE) cluster. You need to prevent data loss and make sure there are enough nodes available for your backend application based on workload demands. You want to follow Google-recommended practices and minimize manual work. What should you do?

Options
A. Run your database as a StatefulSet. Configure cluster autoscaling to handle changes in workloads.
B. Run your database as a single Pod. Run the resize command when you notice changes in workloads.
C. Run your database as a DaemonSet. Run the resize command when you notice changes in workloads.
D. Run your database as a Deployment. Configure cluster autoscaling to handle changes in workloads.

Answer
A

Detailed explanation
A. - correct. StatefulSets provide stable network IDs and storage persistence, while cluster autoscaling adjusts node count automatically.
B. - incorrect. Single Pods are not reliable for database persistence.
C. - incorrect. DaemonSets are for per-node tasks, not databases.
D. - incorrect. Deployment is not recommended for databases requiring persistent state.

Question 229
You are the Organization Administrator for your company's Google Cloud resources. Your company has strict compliance rules requiring you to be notified about modifications to files and documents in Cloud Storage. In a recent incident, one of your team members modified files but you did not receive alerts. You must ensure you receive notifications for all changes while minimizing overhead. What should you do?

Options
A. View Cloud Audit logs for all Cloud Storage files in Logs Explorer. Filter by Admin Activity logs.
B. Enable Cloud Storage object versioning on your bucket. Configure Pub/Sub notifications for your Cloud Storage buckets.
C. Enable versioning on the Cloud Storage bucket. Set up a custom script that scans versions of Cloud Storage objects being modified and alert the admin.
D. Configure Object change notifications on the Cloud Storage buckets. Send the events to Pub/Sub.

Answer
B

Detailed explanation
A. - incorrect. Viewing logs is reactive, not proactive notifications.
B. - correct. Object versioning + Pub/Sub notifications provide proactive alerts for changes.
C. - incorrect. Custom scripts add overhead and complexity.
D. - incorrect. Object change notifications are deprecated in favor of Pub/Sub.

Question 230
Your company would like to store invoices and other financial documents in Google Cloud. You must ensure documents are kept for three years. Analysts need frequent access to invoices from the past six months. After six months, invoices should be archived but still accessible for audit purposes. You want to minimize costs and follow Google-recommended practices. What should you do?

Options
A. Use Cloud Storage with Object Lifecycle Management to change the object storage class to Coldline after six months.
B. Use Cloud Storage with Object Lifecycle Management to change the object storage class to Standard after six months.
C. Store your documents on Filestore, and move them to Cloud Storage with object storage class set to Coldline after six months.
D. Store your documents on Filestore, and move them to Cloud Storage with object storage class set to Standard after six months.

Answer
A

Detailed explanation
A. - correct. Standard → Coldline transition after 6 months saves cost while keeping data accessible.
B. - incorrect. Standard is the default, no cost optimization.
C. - incorrect. Filestore is not cost-efficient for document archival.
D. - incorrect. Same as B, moving to Standard does not save costs.

Question 241
Your managed instance group raised an alert stating that new instance creation has failed to create new instances. You need to maintain the number of running instances specified by the template to be able to process expected application traffic. What should you do?

Options
A. Create an instance template that contains valid syntax which will be used by the instance group. Delete any persistent disks with the same name as instance names.
B. Create an instance template that contains valid syntax that will be used by the instance group. Verify that the instance name and persistent disk name values are not the same in the template.
C. Verify that the instance template being used by the instance group contains valid syntax. Delete any persistent disks with the same name as instance names. Set the disks.autoDelete property to true in the instance template.
D. Delete the current instance template and replace it with a new instance template. Verify that the instance name and persistent disk name values are not the same in the template. Set the disks.autoDelete property to true in the instance template.

Answer
A

Detailed explanation
A. Correct - The failure occurs because instance names conflict with persistent disk names. The correct approach is to create a new instance template with valid syntax and delete any persistent disks that have the same names as the instances. This ensures the managed instance group can successfully create new instances.
B. Incorrect - While verifying that instance name and disk name are not the same is important, this option alone doesn't address the existing conflict from persistent disks with duplicate names.
C. Incorrect - Setting the disks.autoDelete property ensures disks are cleaned up when instances are deleted, but it doesn't resolve the problem of existing conflicting persistent disks that already block new instance creation.
D. Incorrect - Replacing the current template and setting disks.autoDelete is an additional step, but again, it doesn't solve the underlying issue of already existing persistent disks with duplicate names.

Question 242

Your company is moving from an on-premises environment to Google Cloud. You have multiple development teams that use Cassandra environments as backend databases. They all need a development environment that is isolated from other Cassandra instances. You want to move to Google Cloud quickly and with minimal support effort. What should you do?

Options
A. Build an instruction guide to install Cassandra on Google Cloud. Make the instruction guide accessible to your developers.
B. Advise your developers to go to Cloud Marketplace. Ask the developers to launch a Cassandra image for their development work.
C. Build a Cassandra Compute Engine instance and take a snapshot of it. Use the snapshot to create instances for your developers.
D. Build a Cassandra Compute Engine instance and take a snapshot of it. Upload the snapshot to Cloud Storage and make it accessible to your developers. Build instructions to create a Compute Engine instance from the snapshot so that developers can do it themselves.

Answer
B

Detailed explanation
A. Incorrect - Writing an instruction guide takes more effort and maintenance. Developers must manually install Cassandra, which is not efficient.
B. Correct - Cloud Marketplace provides pre-built Cassandra images. Developers can easily launch isolated instances without extra setup. This ensures quick setup with minimal support overhead.
C. Incorrect - Taking and sharing snapshots can work, but it requires ongoing maintenance and manual creation of instances for each developer. Not the most efficient.
D. Incorrect - Uploading snapshots and creating instructions increases manual work for developers and admins. Cloud Marketplace is faster and requires less effort.

Question 243

You have a Compute Engine instance hosting a production application. You want to receive an email if the instance consumes more than 90% of its CPU resources for more than 15 minutes. You want to use Google services. What should you do?

Options
A. Create a consumer Gmail account. Write a script that monitors the CPU usage. When the CPU usage exceeds the threshold, have that script send an email using the Gmail account and smtp.gmail.com on port 25 as SMTP server.
B. Create a Stackdriver Workspace, and associate your Google Cloud Platform (GCP) project with it. Create an Alerting Policy in Stackdriver that uses the threshold as a trigger condition. Configure your email address in the notification channel.
C. Create a Stackdriver Workspace, and associate your GCP project with it. Write a script that monitors the CPU usage and sends it as a custom metric to Stackdriver. Create an uptime check for the instance in Stackdriver.
D. In Stackdriver Logging, create a logs-based metric to extract the CPU usage by using this regular expression: CPU Usage: ([0-9].13) % . In Stackdriver Monitoring, create an Alerting Policy based on this metric. Configure your email address in the notification channel.

Answer
B

Detailed explanation
A. Incorrect - Writing custom scripts and using Gmail SMTP is not recommended. It’s insecure and not scalable.
B. Correct - Stackdriver Monitoring (now Cloud Monitoring) is designed for this use case. It allows you to set thresholds (CPU > 90% for 15 minutes) and configure email notifications.
C. Incorrect - Sending custom metrics works but is unnecessary since CPU metrics are already available natively in Monitoring.
D. Incorrect - Parsing logs to extract CPU usage is complicated and not the recommended method, since Monitoring provides direct CPU metrics.

Question 244

You have an application that uses Cloud Spanner as a backend database. The application has a very predictable traffic pattern. You want to automatically scale up or down the number of Spanner nodes depending on traffic. What should you do?

Options
A. Create a cron job that runs on a scheduled basis to review Cloud Monitoring metrics, and then resize the Spanner instance accordingly.
B. Create a Cloud Monitoring alerting policy to send an alert to an SRE when Cloud Spanner CPU exceeds the threshold. SREs would scale resources up or down accordingly.
C. Create a Cloud Monitoring alerting policy to send an alert to Google Cloud Support email when Cloud Spanner CPU exceeds your threshold. Google support would scale resources up or down accordingly.
D. Create a Cloud Monitoring alerting policy to send an alert to webhook when Cloud Spanner CPU is over or under your threshold. Create a Cloud Function that listens to HTTP and resizes Spanner resources accordingly.

Answer
D

Detailed explanation
A. Incorrect - A cron job with manual resizing works but does not fully automate the scaling process.
B. Incorrect - Sending alerts to SREs requires manual intervention, which defeats the purpose of automation.
C. Incorrect - Google Cloud Support does not automatically scale your resources for you.
D. Correct - This is the recommended approach. Monitoring triggers an alert → webhook invokes a Cloud Function → Cloud Function programmatically resizes the Spanner instance. It's fully automated and scalable.

Question 245

Your company publishes large files on an Apache web server that runs on a Compute Engine instance. The Apache web server is not the only application running in the project. You want to receive an email when the egress network costs for the server exceed 100 dollars for the current month as measured by Google Cloud. What should you do?

Options
A. Set up a budget alert on the project with an amount of 100 dollars, a threshold of 100%, and notification type of email.
B. Set up a budget alert on the billing account with an amount of 100 dollars, a threshold of 100%, and notification type of email.
C. Export the billing data to BigQuery. Create a Cloud Function that uses BigQuery to sum the egress network costs of the exported billing data for the Apache web server for the current month and sends an email if it is over 100 dollars. Schedule the Cloud Function using Cloud Scheduler to run hourly.
D. Use the Cloud Logging Agent to export the Apache web server logs to Cloud Logging. Create a Cloud Function that uses BigQuery to parse the HTTP response log data in Cloud Logging for the current month and sends an email if the size of all HTTP responses, multiplied by current Google Cloud egress prices, totals over 100 dollars. Schedule the Cloud Function using Cloud Scheduler to run hourly.

Answer
C

Detailed explanation
A. Incorrect - Budget alerts on projects track total project spend, not application-specific egress costs.
B. Incorrect - Same issue as A. Billing account budget alerts cannot target a specific VM or service.
C. Correct - Exporting billing data to BigQuery allows filtering by resource (Apache web server) and querying for egress costs. A Cloud Function can then send alerts based on actual egress costs.
D. Incorrect - Parsing logs to estimate egress costs is overly complex and prone to inaccuracies compared to billing data.

Question 246

You have designed a solution on Google Cloud that uses multiple Google Cloud products. Your company has asked you to estimate the costs of the solution. You need to provide estimates for the monthly total cost. What should you do?

Options
A. For each Google Cloud product in the solution, review the pricing details on the products pricing page. Use the pricing calculator to total the monthly costs for each Google Cloud product.
B. For each Google Cloud product in the solution, review the pricing details on the products pricing page. Create a Google Sheet that summarizes the expected monthly costs for each product.
C. Provision the solution on Google Cloud. Leave the solution provisioned for 1 week. Navigate to the Billing Report page in the Cloud Console. Multiply the 1 week cost to determine the monthly costs.
D. Provision the solution on Google Cloud. Leave the solution provisioned for 1 week. Use Cloud Monitoring to determine the provisioned and used resource amounts. Multiply the 1 week cost to determine the monthly costs.

Answer
A

Detailed explanation
A. Correct - Google Cloud Pricing Calculator is the recommended tool to estimate costs. It uses official pricing details and allows you to model different services and usage.
B. Incorrect - Creating a Google Sheet is manual and prone to human error. It doesn't provide accurate estimates.
C. Incorrect - Estimating costs by provisioning for a week and extrapolating may not reflect actual monthly usage. It's also wasteful and costly.
D. Incorrect - Similar to C, using Monitoring for cost estimation is inefficient and not precise.

Question 247

You have an application that receives SSL-encrypted TCP traffic on port 443. Clients for this application are located all over the world. You want to minimize latency for the clients. Which load balancing option should you use?

Options
A. HTTPS Load Balancer
B. Network Load Balancer
C. SSL Proxy Load Balancer
D. Internal TCP/UDP Load Balancer. Add a firewall rule allowing ingress traffic from 0.0.0.0/0 on the target instances.

Answer
C

Detailed explanation
A. Incorrect - HTTPS Load Balancer is for HTTP/HTTPS traffic (layer 7). It is not suitable for generic SSL/TCP traffic.
B. Incorrect - Network Load Balancer is regional, not global. It does not minimize latency for global clients.
C. Correct - SSL Proxy Load Balancer is designed for non-HTTP(S) SSL/TCP traffic (like port 443 with custom SSL apps). It provides a global, anycast IP and routes clients to the closest backend, minimizing latency.
D. Incorrect - Internal TCP/UDP Load Balancer is for internal traffic, not global client traffic. Allowing ingress from 0.0.0.0/0 is insecure and does not provide global load balancing.

Question 248

You have an application on a general-purpose Compute Engine instance that is experiencing excessive disk read throttling on its Zonal SSD Persistent Disk. The application primarily reads large files from disk. The disk size is currently 500 GB. You want to provide the maximum amount of throughput while minimizing costs. What should you do?

Options
A. Increase the size of the disk to 1 TB.
B. Increase the allocated CPU to the instance.
C. Migrate to use a Local SSD on the instance.
D. Migrate to use a Regional SSD on the instance.

Answer
C

Detailed explanation
A. Incorrect - Increasing disk size increases IOPS/throughput, but Local SSDs provide much higher throughput and lower latency.
B. Incorrect - Adding more CPU does not solve disk read throttling. CPU is not the bottleneck here.
C. Correct - Local SSDs provide extremely high throughput and low latency for disk I/O compared to Persistent Disks. This is the best solution for workloads with high read requirements.
D. Incorrect - Regional SSDs provide redundancy across zones but don’t improve throughput compared to a Zonal SSD.

Question 249

Your Dataproc cluster runs in a single Virtual Private Cloud (VPC) network in a single subnet with range 172.16.20.128/25. There are no private IP addresses available in the VPC network. You want to add new VMs to communicate with your cluster using the minimum number of steps. What should you do?

Options
A. Modify the existing subnet range to 172.16.20.0/24.
B. Create a new Secondary IP Range in the VPC and configure the VMs to use that range.
C. Create a new VPC network for the VMs. Enable VPC Peering between the VMs' VPC network and the Dataproc cluster VPC network.
D. Create a new VPC network for the VMs with a subnet of 172.32.0.0/16. Enable VPC Peering between the Dataproc VPC network and the VMs VPC network. Configure a custom Route exchange.

Answer
A

Detailed explanation
A. Correct - Expanding the existing subnet range gives you more usable private IP addresses with the least configuration change. It's the most efficient solution.
B. Incorrect - Secondary IP ranges are used for alias IPs, not for extending the subnet's primary IP range.
C. Incorrect - Creating a new VPC and setting up VPC Peering adds unnecessary complexity compared to simply expanding the subnet.
D. Incorrect - Same as C, creating a new VPC and custom routes is more complex than extending the existing subnet.

Question 250

You manage an App Engine Service that aggregates and visualizes data from BigQuery. The application is deployed with the default App Engine Service account. The data that needs to be visualized resides in a different project managed by another team. You do not have access to this project, but you want your application to be able to read data from the BigQuery dataset. What should you do?

Options
A. Ask the other team to grant your default App Engine Service account the role of BigQuery Job User.
B. Ask the other team to grant your default App Engine Service account the role of BigQuery Data Viewer.
C. In Cloud IAM of your project, ensure that the default App Engine service account has the role of BigQuery Data Viewer.
D. In Cloud IAM of your project, grant a newly created service account from the other team the role of BigQuery Job User in your project.

Answer
B

Detailed explanation
A. Incorrect - BigQuery Job User allows running queries, but it does not grant permission to read the dataset itself.
B. Correct - BigQuery Data Viewer role grants read-only access to datasets, tables, and views. This is the least-privilege, correct role for accessing another project's BigQuery data.
C. Incorrect - Granting roles in your own project does not provide access to datasets in another project. Permissions must be set in the dataset's project.
D. Incorrect - The other team's service account being added in your project does not give your App Engine access to their BigQuery dataset.

Question 251
You need to create a copy of a custom Compute Engine virtual machine (VM) to facilitate an expected increase in application traffic due to a business acquisition. What should you do?

Options
A. Create a Compute Engine snapshot of your base VM. Create your images from that snapshot.
B. Create a Compute Engine snapshot of your base VM. Create your instances from that snapshot.
C. Create a custom Compute Engine image from a snapshot. Create your images from that image.
D. Create a custom Compute Engine image from a snapshot. Create your instances from that image.

Answer
D

Detailed explanation
A. Incorrect - A snapshot is designed for persistent disks, not for creating VM images that are reusable. Images are the correct choice for duplication.
B. Incorrect - You cannot directly create instances from snapshots. Snapshots restore disks but are not suitable for scaling multiple VMs.
C. Incorrect - This option creates another image from an image, which adds unnecessary steps.
D. Correct - The recommended way is to create a custom image from a snapshot and then use that image to launch consistent new VM instances. This ensures reliability and easy replication of the original VM.

Question 252
You have deployed an application on a single Compute Engine instance. The application writes logs to disk. Users start reporting errors with the application. You want to diagnose the problem. What should you do?

Options
A. Navigate to Cloud Logging and view the application logs.
B. Connect to the instance's serial console and read the application logs.
C. Configure a Health Check on the instance and set a Low Healthy Threshold value.
D. Install and configure the Cloud Logging Agent and view the logs from Cloud Logging.

Answer
D

Detailed explanation
A. Incorrect - Cloud Logging won't show logs unless the logging agent is installed on the VM.
B. Incorrect - Serial console helps in debugging boot and OS-level issues, not application logs.
C. Incorrect - A health check only tells you whether the instance is up or down, it does not show the logs.
D. Correct - The Cloud Logging agent must be installed and configured to stream logs from the VM to Cloud Logging. This enables centralized log viewing and debugging.

Question 253
An application generates daily reports in a Compute Engine virtual machine (VM). The VM is in the project corp-iot-insights. Your team operates only in the project corp-aggregate-reports and needs a copy of the daily exports in the bucket corp-aggregate-reports-storage. You want to configure access so that the daily reports from the VM are available in the bucket corp-aggregate-reports-storage and use as few steps as possible while following Google-recommended practices. What should you do?

Options
A. Move both projects under the same folder.
B. Grant the VM Service Account the role Storage Object Creator on corp-aggregate-reports-storage.
C. Create a Shared VPC network between both projects. Grant the VM Service Account the role Storage Object Creator on corp-iot-insights.
D. Make corp-aggregate-reports-storage public and create a folder with a pseudo-randomized suffix name. Share the folder with the IoT team.

Answer
B

Detailed explanation
A. Incorrect - Moving projects under the same folder does not solve cross-project permissions for Cloud Storage.
B. Correct - The VM service account needs Storage Object Creator on the destination bucket (corp-aggregate-reports-storage). This follows the principle of least privilege and is the recommended approach.
C. Incorrect - Shared VPC is for networking, not for enabling object writes to Cloud Storage across projects.
D. Incorrect - Making the bucket public violates security best practices.

Question 254
You built an application on your development laptop that uses Google Cloud services. Your application uses Application Default Credentials for authentication and works fine on your development laptop. You want to migrate this application to a Compute Engine virtual machine (VM) and set up authentication using Google-recommended practices and minimal changes. What should you do?

Options
A. Assign appropriate access for Google services to the service account used by the Compute Engine VM.
B. Create a service account with appropriate access for Google services, and configure the application to use this account.
C. Store credentials for service accounts with appropriate access for Google services in a config file, and deploy this config file with your application.
D. Store credentials for your user account with appropriate access for Google services in a config file, and deploy this config file with your application.

Answer
B

Detailed explanation
A. Incorrect - Using the default VM service account is not ideal unless scoped properly. Creating a dedicated service account is more secure.
B. Correct - The best practice is to create a dedicated service account, grant it the required roles, and attach it to the VM. This allows the application to authenticate using Application Default Credentials without code changes.
C. Incorrect - Storing service account credentials in a config file poses a security risk.
D. Incorrect - User credentials should not be stored or distributed in applications.

Question 255
You need to create a Compute Engine instance in a new project that doesn't exist yet. What should you do?

Options
A. Using the Cloud SDK, create a new project, enable the Compute Engine API in that project, and then create the instance specifying your new project.
B. Enable the Compute Engine API in the Cloud Console, use the Cloud SDK to create the instance, and then use the --project flag to specify a new project. Answer yes when prompted by Cloud SDK to enable the Compute Engine API.
C. Using the Cloud SDK, create the new instance, and use the --project flag to specify the new project. Answer yes when prompted by Cloud SDK to enable the Compute Engine API.
D. Enable the Compute Engine API in the Cloud Console. Go to the Compute Engine section of the Console to create a new instance, and look for the Create In A New Project option in the creation form.

Answer
A

Detailed explanation
A. Correct - You must first create the project, enable the API, and then create resources like Compute Engine instances. This follows the correct sequence.
B. Incorrect - You cannot specify a project that does not exist yet. The project must exist before enabling APIs.
C. Incorrect - Same issue as above; the project must be created first.
D. Incorrect - There is no "Create in a New Project" option directly in the Compute Engine console.

Question 256
Your company runs one batch process in an on-premises server that takes around 30 hours to complete. The task runs monthly, can be performed offline, and must be restarted if interrupted. You want to migrate this workload to the cloud while minimizing cost. What should you do?

Options
A. Migrate the workload to a Compute Engine Preemptible VM.
B. Migrate the workload to a Google Kubernetes Engine cluster with Preemptible nodes.
C. Migrate the workload to a Compute Engine VM. Start and stop the instance as needed.
D. Create an Instance Template with Preemptible VMs On. Create a Managed Instance Group from the template and adjust Target CPU Utilization. Migrate the workload.

Answer
C

Detailed explanation
A. Incorrect - Preemptible VMs are not suitable because the workload takes 30 hours and preemptible VMs can be stopped after 24 hours.
B. Incorrect - Same limitation as A, since preemptible nodes will not last long enough for the batch job.
C. Correct - A standard Compute Engine VM that you can start and stop as needed is the best option for long workloads. It minimizes cost because you only pay while it runs, and it won't be interrupted.
D. Incorrect - Managed Instance Groups with preemptibles are good for scalable, fault-tolerant tasks, but not for a long, restart-required batch job.

Question 257
You are developing a new application and are looking for a Jenkins installation to build and deploy your source code. You want to automate the installation as quickly and easily as possible. What should you do?

Options
A. Deploy Jenkins through the Google Cloud Marketplace.
B. Create a new Compute Engine instance. Run the Jenkins executable.
C. Create a new Kubernetes Engine cluster. Create a deployment for the Jenkins image.
D. Create an instance template with the Jenkins executable. Create a managed instance group with this template.

Answer
A

Detailed explanation
A. Correct - Cloud Marketplace provides preconfigured Jenkins solutions that can be deployed quickly with minimal setup. This saves time and effort.
B. Incorrect - Installing Jenkins manually on a Compute Engine instance requires configuration and setup effort.
C. Incorrect - Kubernetes deployment works but is overcomplicated for a simple Jenkins setup.
D. Incorrect - Managed instance groups are for scaling applications, not ideal for a Jenkins master deployment.

Question 258
You have downloaded and installed the gcloud command line interface (CLI) and have authenticated with your Google Account. Most of your Compute Engine instances in your project run in the europe-west1-d zone. You want to avoid having to specify this zone with each CLI command when managing these instances. What should you do?

Options
A. Set the europe-west1-d zone as the default zone using the gcloud config subcommand.
B. In the Settings page for Compute Engine under Default location, set the zone to europe-west1-d.
C. In the CLI installation directory, create a file called default.conf containing zone=europe-west1-d.
D. Create a Metadata entry on the Compute Engine page with key compute/zone and value europe-west1-d.

Answer
A

Detailed explanation
A. Correct - The correct approach is to set the default zone with gcloud config set compute/zone europe-west1-d. This ensures you don't have to specify the zone each time.
B. Incorrect - There is no "Default location" setting for Compute Engine instances in the Console.
C. Incorrect - The CLI does not use a default.conf file.
D. Incorrect - Metadata entries are not used for setting default CLI behavior.

Question 259
The core business of your company is to rent out construction equipment at large scale. All the equipment has been equipped with multiple sensors that send event information every few seconds. These signals can vary from engine status, distance traveled, fuel level, and more. Customers are billed based on the consumption monitored by these sensors. You expect high throughput — up to thousands of events per hour per device — and need to retrieve consistent data based on the time of the event. Storing and retrieving individual signals should be atomic. What should you do?

Options
A. Create a file in Cloud Storage per device and append new data to that file.
B. Create a file in Cloud Filestore per device and append new data to that file.
C. Ingest the data into Datastore. Store data in an entity group based on the device.
D. Ingest the data into Cloud Bigtable. Create a row key based on the event timestamp.

Answer
D

Detailed explanation
A. Incorrect - Cloud Storage is not designed for atomic writes at high throughput or frequent updates.
B. Incorrect - Cloud Filestore is for NFS file storage, not optimized for high-throughput event ingestion.
C. Incorrect - Datastore is not ideal for very high throughput with atomic event data.
D. Correct - Cloud Bigtable is built for high-throughput, low-latency writes and can store sensor data efficiently with row keys designed around timestamps.

Question 260
You are asked to set up application performance monitoring on Google Cloud projects A, B, and C as a single pane of glass. You want to monitor CPU, memory, and disk. What should you do?

Options
A. Enable API and then share charts from project A, B, and C.
B. Enable API and then give the metrics.reader role to projects A, B, and C.
C. Enable API and then use default dashboards to view all projects in sequence.
D. Enable API, create a workspace under project A, and then add projects B and C.

Answer
D

Detailed explanation
A. Incorrect - Sharing charts from multiple projects separately doesn't give a single pane of glass.
B. Incorrect - The metrics.reader role allows access but does not provide cross-project visibility in one place.
C. Incorrect - Viewing dashboards project by project does not meet the "single pane of glass" requirement.
D. Correct - Stackdriver (now Cloud Monitoring) uses Workspaces. You can create one workspace in project A and add projects B and C for unified monitoring.

Question 261
You created several resources in multiple Google Cloud projects. All projects are linked to different billing accounts. To better estimate future charges, you want to have a single visual representation of all costs incurred. You want to include new cost data as soon as possible. What should you do?

Options
A. Configure Billing Data Export to BigQuery and visualize the data in Data Studio.
B. Visit the Cost Table page to get a CSV export and visualize it using Data Studio.

Answer
A

Detailed explanation
A. Correct - Exporting billing data to BigQuery gives you near real-time updates and allows you to visualize and analyze costs across multiple billing accounts in Data Studio. This is the recommended method for scalability, automation, and accuracy.
B. Incorrect - Exporting CSV from the Cost Table provides only point-in-time static data, requires manual effort, and does not include near real-time updates. It is not suitable for ongoing monitoring.

Question 262
Your company has workloads running on Compute Engine and on-premises. The Google Cloud Virtual Private Cloud (VPC) is connected to your WAN over a Virtual Private Network (VPN). You need to deploy a new Compute Engine instance and ensure that no public Internet traffic can be routed to it. What should you do?

Options
A. Create the instance without a public IP address.
B. Create the instance with Private Google Access enabled.
C. Create a deny-all egress firewall rule on the VPC network.
D. Create a route on the VPC to route all traffic to the instance over the VPN tunnel.

Answer
A

Detailed explanation
A. Correct - By not assigning a public IP, the instance will only be reachable through internal IP addresses over the VPN, preventing exposure to the public Internet.
B. Incorrect - Private Google Access allows access to Google APIs without a public IP but does not prevent other public traffic.
C. Incorrect - A deny-all egress firewall rule stops outbound traffic but doesn't prevent inbound Internet traffic to a public IP if assigned.
D. Incorrect - Adding a custom route over the VPN doesn't remove Internet exposure if the VM has a public IP.

Question 263
Your team maintains the infrastructure for your organization. The current infrastructure requires changes. You need to share your proposed changes with the rest of the team. You want to follow Google's recommended best practices. What should you do?

Options
A. Use Deployment Manager templates to describe the proposed changes and store them in a Cloud Storage bucket.
B. Use Deployment Manager templates to describe the proposed changes and store them in Cloud Source Repositories.
C. Apply the changes in a development environment, run gcloud compute instances list, and then save the output in a shared Storage bucket.
D. Apply the changes in a development environment, run gcloud compute instances list, and then save the output in Cloud Source Repositories.

Answer
B

Detailed explanation
A. Incorrect - Storing in Cloud Storage provides persistence but not proper version control or collaboration features.
B. Correct - Cloud Source Repositories provide version control, collaboration, and history tracking. This aligns with Google-recommended best practices for infrastructure-as-code.
C. Incorrect - Running gcloud compute instances list shows only a snapshot of resources, not infrastructure changes.
D. Incorrect - Same issue as C, saving lists of instances doesn't describe infrastructure changes properly.

Question 264
You have a Compute Engine instance hosting an application used between 9 AM and 6 PM on weekdays. You want to back up this instance daily for disaster recovery purposes. You want to keep the backups for 30 days. You want the Google-recommended solution with the least management overhead and the least number of services. What should you do?

Options
A. Update your instances' metadata to add snapshot schedule and retention values.
B. In the Cloud Console, go to the Compute Engine Disks page, configure snapshot schedule with daily frequency, auto-delete after 30 days.
C. Create a Cloud Function that creates snapshots, deletes old ones, and trigger it daily with Cloud Scheduler.
D. Create a bash script in the instance to copy content of disk to Cloud Storage, delete old backups, and run with cron.

Answer
B

Detailed explanation
A. Incorrect - Setting metadata alone does not configure automated snapshot schedules. It requires explicit snapshot schedules in the console or API.
B. Correct - Using the built-in snapshot schedule in the Console is the recommended, managed, and simplest way. It automates snapshot creation and retention, with minimal overhead.
C. Incorrect - A custom Cloud Function + Scheduler works but adds unnecessary complexity when Google provides a built-in snapshot scheduling service.
D. Incorrect - Bash scripts and cron jobs are manual, error-prone, and not scalable.

Question 265
Your existing application running in Google Kubernetes Engine (GKE) consists of multiple pods running on four GKE n1-standard-2 nodes. You need to deploy additional pods requiring n2-highmem-16 nodes without any downtime. What should you do?

Options
A. Use gcloud container clusters upgrade. Deploy the new services.
B. Create a new Node Pool and specify machine type n2-highmem-16. Deploy the new pods.
C. Create a new cluster with n2-highmem-16 nodes. Redeploy the pods and delete the old cluster.
D. Create a new cluster with both n1-standard-2 and n2-highmem-16 nodes. Redeploy the pods and delete the old cluster.

Answer
B

Detailed explanation
A. Incorrect - Upgrading the cluster does not add new node types or handle different machine requirements.
B. Correct - Adding a new node pool with the required machine type (n2-highmem-16) allows scheduling pods on larger nodes while keeping existing nodes running, ensuring zero downtime.
C. Incorrect - Creating a new cluster and redeploying causes downtime and extra migration work.
D. Incorrect - Same issue as C, managing multiple clusters complicates things unnecessarily and risks downtime.

Question 266
You have an application that uses Cloud Spanner as a database backend to keep current state information about users. Cloud Bigtable logs all events triggered by users. You export Cloud Spanner data to Cloud Storage during daily backups. One of your analysts asks you to join data from Cloud Spanner and Cloud Bigtable for specific users. You want to complete this ad hoc request as efficiently as possible. What should you do?

Options
A. Create a dataflow job that copies data from Cloud Bigtable and Cloud Storage for specific users.
B. Create a dataflow job that copies data from Cloud Bigtable and Cloud Spanner for specific users.
C. Create a Cloud Dataproc cluster that runs a Spark job to extract data from Cloud Bigtable and Cloud Storage for specific users.
D. Create two separate BigQuery external tables on Cloud Storage and Cloud Bigtable. Use the BigQuery console to join these tables through user fields, and apply appropriate filters.

Answer
D

Detailed explanation
A. Incorrect - Dataflow could work, but combining Cloud Storage backups with Bigtable is inefficient for quick ad hoc queries.
B. Incorrect - Similar issue as A, Dataflow is powerful but overkill for quick joins, and querying directly from Spanner is not ideal for analytical joins.
C. Incorrect - Spinning up a Dataproc Spark cluster adds unnecessary overhead and cost for one-time or ad hoc analysis.
D. Correct - Creating BigQuery external tables for both Bigtable and Cloud Storage allows querying directly without data movement. BigQuery SQL joins provide the simplest and most efficient ad hoc nalysis.

Question 267
You are hosting an application from Compute Engine virtual machines (VMs) in us-central1-a. You want to adjust your design to support the failure of a single Compute Engine zone, eliminate downtime, and minimize cost. What should you do?

Options
A. Create Compute Engine resources in us-central1-b. Balance the load across us-central1-a and us-central1-b.
B. Create a Managed Instance Group and specify us-central1-a as the zone. Configure the Health Check with a short Health Interval.
C. Create an HTTP(S) Load Balancer. Create one or more global forwarding rules to direct traffic to your VMs.
D. Perform regular backups of your application. Create a Cloud Monitoring Alert and be notified if your application becomes unavailable. Restore from backups when notified.

Answer
A

Detailed explanation
A. Correct - Deploying resources in multiple zones (e.g., us-central1-a and us-central1-b) ensures high availability and zone failure tolerance. Load balancing across zones minimizes downtime.
B. Incorrect - A Managed Instance Group in a single zone does not protect against zone failures.
C. Incorrect - An HTTP(S) Load Balancer alone doesn’t solve high availability unless you deploy VMs in multiple zones.

D. Incorrect - Relying only on backups does not provide high availability; it only supports disaster recovery after failure.

Question 268
A colleague handed over a Google Cloud Platform project for you to maintain. As part of a security checkup, you want to review who has been granted the Project Owner role. What should you do?

Options
A. In the console, validate which SSH keys have been stored as project-wide keys.
B. Navigate to Identity-Aware Proxy and check the permissions for these resources.
C. Enable Audit Logs on the IAM & admin page for all resources, and validate the results.
D. Use the command gcloud projects get-iam-policy to view the current role assignments.

Answer
D

Detailed explanation
A. Incorrect - Checking SSH keys only shows VM access, not project-wide IAM role assignments.
B. Incorrect - Identity-Aware Proxy manages access to web apps, not project-wide IAM roles.
C. Incorrect - Audit logs track IAM changes but don't directly show current assignments.
D. Correct - Running gcloud projects get-iam-policy directly lists all members and their IAM roles, including Project Owners. This is the most efficient way.

Question 269
You are running multiple VPC-native Google Kubernetes Engine clusters in the same subnet. The IPs available for the nodes are exhausted, and you want to ensure that the clusters can grow in nodes when needed. What should you do?

Options
A. Create a new subnet in the same region as the subnet being used.
B. Add an alias IP range to the subnet used by the GKE clusters.
C. Create a new VPC, and set up VPC peering with the existing VPC.
D. Expand the CIDR range of the relevant subnet for the cluster.

Answer
D

Detailed explanation
A. Incorrect - Creating a new subnet doesn't help unless workloads are migrated; existing clusters won't automatically use it.
B. Incorrect - Alias IP ranges are used for Pod IPs, not for adding capacity for more nodes.
C. Incorrect - A new VPC with peering adds complexity and doesn't solve IP exhaustion in the current subnet.
D. Correct - Expanding the CIDR range of the existing subnet directly increases available IPs, allowing existing GKE clusters to scale nodes without disruption.

Question 270
You have a batch workload that runs every night and uses a large number of virtual machines (VMs). It is fault-tolerant and can tolerate some VMs being terminated. The current cost of VMs is too high. What should you do?

Options
A. Run a test using simulated maintenance events. If the test is successful, use preemptible N1 Standard VMs when running future jobs.
B. Run a test using simulated maintenance events. If the test is successful, use N1 Standard VMs when running future jobs.
C. Run a test using a managed instance group. If the test is successful, use N1 Standard VMs in the managed instance group when running future jobs.
D. Run a test using N1 standard VMs instead of N2. If the test is successful, use N1 Standard VMs when running future jobs.

Answer
A

Detailed explanation
A. Correct - Preemptible VMs are highly cost-effective (up to 80% cheaper) and suitable for fault-tolerant batch workloads. Testing ensures they work with your jobs.
B. Incorrect - Standard VMs (non-preemptible) do not reduce costs compared to the current setup.
C. Incorrect - Managed instance groups provide scalability but won't reduce costs if standard VMs are still used.
D. Incorrect - Switching from N2 to N1 may reduce cost slightly, but not as much as preemptible VMs.

Question 271
Your organization is a financial company that needs to store audit log files for 3 years. Your organization has hundreds of Google Cloud projects. You need to implement a cost-effective approach for log file retention. What should you do?

Options
A. Create an export to the sink that saves logs from Cloud Audit to BigQuery.
B. Create an export to the sink that saves logs from Cloud Audit to a Coldline Storage bucket.
C. Write a custom script that uses logging API to copy the logs from Stackdriver logs to BigQuery.
D. Export these logs to Cloud Pub/Sub and write a Cloud Dataflow pipeline to store logs to Cloud SQL.

Answer
B

Detailed explanation
A. Incorrect - BigQuery is useful for analytics, but it's costly for long-term storage of logs over 3 years.
B. Correct - Coldline Storage is designed for long-term archival with infrequent access. It is the most cost-effective option to store audit logs for compliance purposes.
C. Incorrect - Writing a custom script adds complexity, is error-prone, and not the recommended practice. It also won't optimize storage cost.
D. Incorrect - Exporting to Pub/Sub and Dataflow into Cloud SQL is unnecessarily complex, and Cloud SQL is not designed for storing large volumes of logs.

Question 272
You want to run a single caching HTTP reverse proxy on GCP for a latency-sensitive website. This specific reverse proxy consumes almost no CPU. You want to have a 30-GB in-memory cache, and need an additional 2 GB of memory for the rest of the processes. You want to minimize cost. How should you run this reverse proxy?

Options
A. Create a Cloud Memorystore for Redis instance with 32-GB capacity.
B. Run it on Compute Engine, and choose a custom instance type with 6 vCPUs and 32 GB of memory.
C. Package it in a container image, and run it on Kubernetes Engine, using n1-standard-32 instances as nodes.
D. Run it on Compute Engine, choose the instance type n1-standard-1, and add an SSD persistent disk of 32 GB.

Answer
A

Detailed explanation
A. Correct - Cloud Memorystore for Redis provides an in-memory cache, fully managed, and supports exactly the cache requirement (32 GB). It minimizes cost and complexity.
B. Incorrect - A 6 vCPU machine with 32 GB RAM is costly and unnecessary because CPU requirements are minimal.
C. Incorrect - Using Kubernetes with n1-standard-32 nodes is overkill and highly expensive for this lightweight use case.
D. Incorrect - Adding SSD persistent disk does not solve the in-memory requirement. Disk ≠ memory cache.

Question 273
You are hosting an application on bare-metal servers in your own data center. The application needs access to Cloud Storage. However, security policies prevent the servers hosting the application from having public IP addresses or access to the internet. You want to follow Google-recommended practices to provide the application with access to Cloud Storage. What should you do?

Options
A. 1. Use nslookup to get the IP address for storage.googleapis.com. 2. Negotiate with the security team to be able to give a public IP address to the servers. 3. Only allow egress traffic from those servers to the IP addresses for storage.googleapis.com.
B. 1. Using Cloud VPN, create a VPN tunnel to a Virtual Private Cloud (VPC) in Google Cloud. 2. In this VPC, create a Compute Engine instance and install the Squid proxy server on this instance. 3. Configure your servers to use that instance as a proxy to access Cloud Storage.
C. 1. Use Migrate for Compute Engine (formerly known as Velostrata) to migrate those servers to Compute Engine. 2. Create an internal load balancer (ILB) that uses storage.googleapis.com as backend. 3. Configure your new instances to use this ILB as proxy.
D. 1. Using Cloud VPN or Interconnect, create a tunnel to a VPC in Google Cloud. 2. Use Cloud Router to create a custom route advertisement for 199.36.153.4/30. 3. Announce that network to your on-premises network through the VPN tunnel. 4. In your on-premises network, configure your DNS server to resolve *.googleapis.com as a CNAME to restricted.googleapis.com.

Answer
D

Detailed explanation
A. Incorrect - This workaround is brittle, not scalable, and goes against recommended practices for restricted.googleapis.com.
B. Incorrect - While it works, deploying a Squid proxy adds overhead and is not the recommended or scalable approach.
C. Incorrect - You cannot configure Cloud Storage as a backend for an internal load balancer. This is invalid.
D. Correct - Using Private Google Access via restricted.googleapis.com with Cloud VPN/Interconnect and Cloud Router is the recommended practice for private, secure access to GCP APIs like Cloud Storage.

Question 274
You want to deploy an application on Cloud Run that processes messages from a Cloud Pub/Sub topic. You want to follow Google-recommended practices. What should you do?

Options
A. Create a Cloud Function that uses a Cloud Pub/Sub trigger on that topic. 2. Call your application on Cloud Run from the Cloud Function for every message.
B. 1. Grant the Pub/Sub Subscriber role to the service account used by Cloud Run. 2. Create a Cloud Pub/Sub subscription for that topic. 3. Make your application pull messages from that subscription.
C. 1. Create a service account. 2. Give the Cloud Run Invoker role to that service account for your Cloud Run application. 3. Create a Cloud Pub/Sub subscription that uses that service account and uses your Cloud Run application as the push endpoint.
D. 1. Deploy your application on Cloud Run on GKE with the connectivity set to Internal. 2. Create a Cloud Pub/Sub subscription for that topic. 3. In the same Google Kubernetes Engine cluster as your application, deploy a container that takes the messages and sends them to your application.

Answer
C

Detailed explanation
A. Incorrect - Using Cloud Functions as an intermediary adds unnecessary cost and latency. Not the recommended approach.
B. Incorrect - Cloud Run is serverless and cannot directly “pull” messages like a VM; it expects push-based delivery.
C. Correct - Cloud Pub/Sub can directly push messages to a Cloud Run service endpoint when the proper service account is used. This is the streamlined, recommended approach.
D. Incorrect - Cloud Run on GKE is more complex and unnecessary when standard Cloud Run already integrates directly with Pub/Sub.

Question 275
You need to deploy an application, which is packaged in a container image, in a new project. The application exposes an HTTP endpoint and receives very few requests per day. You want to minimize costs. What should you do?

Options
A. Deploy the container on Cloud Run.
B. Deploy the container on Cloud Run on GKE.
C. Deploy the container on App Engine Flexible.
D. Deploy the container on GKE with cluster autoscaling and horizontal pod autoscaling enabled.

Answer
A

Detailed explanation
A. Correct - Cloud Run is serverless, scales to zero, and only incurs cost when requests are received. Ideal for low-traffic HTTP workloads.
B. Incorrect - Cloud Run on GKE incurs cluster costs even when idle.
C. Incorrect - App Engine Flexible has a minimum instance requirement, meaning costs are higher even with low traffic.
D. Incorrect - GKE incurs cluster costs even if the application is idle.

Question 276
Your company has an existing GCP organization with hundreds of projects and a billing account. Your company recently acquired another company that also has hundreds of projects and its own billing account. You would like to consolidate all GCP costs of both GCP organizations onto a single invoice. You would like to consolidate all costs as of tomorrow. What should you do?

Options
A. Link the acquired company's projects to your company's billing account.
B. Configure the acquired company's billing account and your company's billing account to export the billing data into the same BigQuery dataset.
C. Migrate the acquired company's projects into your company's GCP organization. Link the migrated projects to your company's billing account.
D. Create a new GCP organization and a new billing account. Migrate the acquired company's projects and your company's projects into the new GCP organization and link the projects to the new billing account.

Answer
A

Detailed explanation
A. Correct - Simply relinking the acquired company's projects to your billing account achieves immediate consolidation, starting the next billing cycle.
B. Incorrect - Exporting billing data to BigQuery consolidates reporting, but invoices remain separate.
C. Incorrect - Migrating organizations is possible but not immediate, and it's more complex than relinking projects to a billing account.
D. Incorrect - Creating a new organization and billing account delays the process and adds unnecessary steps.

Question 277
You built an application on Google Cloud that uses Cloud Spanner. Your support team needs to monitor the environment but should not have access to table data. You need a streamlined solution to grant the correct permissions to your support team, and you want to follow Google-recommended practices. What should you do?

Options
A. Add the support team group to the roles/monitoring.viewer role
B. Add the support team group to the roles/spanner.databaseUser role.
C. Add the support team group to the roles/spanner.databaseReader role.
D. Add the support team group to the roles/stackdriver.accounts.viewer role.

Answer
A

Detailed explanation
A. Correct - The monitoring.viewer role grants visibility into monitoring data without access to the actual database contents. This is least privilege.
B. Incorrect - spanner.databaseUser allows modification, which is beyond monitoring needs.
C. Incorrect - spanner.databaseReader allows table reads, exposing sensitive data.
D. Incorrect - stackdriver.accounts.viewer is not the appropriate role for monitoring Spanner resources.

Question 278
For analysis purposes, you need to send all the logs from all of your Compute Engine instances to a BigQuery dataset called platform-logs. You have already installed the Cloud Logging agent on all the instances. You want to minimize cost. What should you do?

Options
A. 1. Give the BigQuery Data Editor role on the platform-logs dataset to the service accounts used by your instances. 2. Update your instances' metadata to add the following value: logs-destination: bq://platform-logs.
B. 1. In Cloud Logging, create a logs export with a Cloud Pub/Sub topic that collects logs as a sink. 2. Create a Cloud Function that is triggered by messages in the logs topic. 3. Configure that Cloud Function to drop logs that are not from Compute Engine and to insert Compute Engine logs in the platform-logs dataset.
C. 1. In Cloud Logging, create a filter to view only Compute Engine logs. 2. Click Create Export. 3. Choose BigQuery as Sink Service, and the platform-logs dataset as Sink Destination.
D. 1. Create a Cloud Function that has the BigQuery User role on the platform-logs dataset. 2. Configure this Cloud Function to create a BigQuery Job that executes this query: INSERT INTO dataset.platform-logs (timestamp, log) SELECT timestamp, log FROM compute.logs WHERE timestamp > DATE_SUB(CURRENT_DATE(), INTERVAL 1 DAY) 3. Use Cloud Scheduler to trigger this Cloud Function once a day.

Answer
C

Detailed explanation
A. Incorrect - You cannot configure instances to directly push logs into BigQuery using metadata.
B. Incorrect - This solution works but adds complexity and extra cost with Cloud Pub/Sub and Cloud Functions.
C. Correct - Cloud Logging export filters allow direct, cost-effective exporting of only the desired logs (Compute Engine) into BigQuery. This is the recommended approach.
D. Incorrect - Using Cloud Functions and scheduled jobs to insert logs is unnecessarily complex and inefficient.

Question 279
You are using Deployment Manager to create a Google Kubernetes Engine cluster. Using the same Deployment Manager deployment, you also want to create a DaemonSet in the kube-system namespace of the cluster. You want a solution that uses the fewest possible services. What should you do?

Options
A. Add the cluster's API as a new Type Provider in Deployment Manager, and use the new type to create the DaemonSet.
B. Use the Deployment Manager Runtime Configurator to create a new Config resource that contains the DaemonSet definition.
C. With Deployment Manager, create a Compute Engine instance with a startup script that uses kubectl to create the DaemonSet.
D. In the cluster's definition in Deployment Manager, add metadata that has kube-system as key and the DaemonSet manifest as value.

Answer
A

Detailed explanation
A. Correct - Adding the Kubernetes API as a Type Provider lets Deployment Manager manage Kubernetes objects like DaemonSets directly. It uses fewer services and is Google-recommended.
B. Incorrect - Runtime Configurator is for dynamic variables, not deploying Kubernetes resources.
C. Incorrect - Creating a VM with startup scripts is a workaround and adds unnecessary complexity.
D. Incorrect - Metadata cannot be used to directly create DaemonSets in GKE.

Question 280
You are building an application that will run in your data center. The application will use Google Cloud Platform (GCP) services like AutoML. You created a service account that has appropriate access to AutoML. You need to enable authentication to the APIs from your on-premises environment. What should you do?

Options
A. Use service account credentials in your on-premises application.
B. Use gcloud to create a key file for the service account that has appropriate permissions.
C. Set up direct interconnect between your data center and Google Cloud Platform to enable authentication for your on-premises applications.
D. Go to the IAM & admin console, grant a user account permissions similar to the service account permissions, and use this user account for authentication from your data center.

Answer
B

Detailed explanation
A. Incorrect - This is vague. You need a key file to actually authenticate.
B. Correct - Creating a key file for the service account and using it in the on-premises app is the standard, recommended way to authenticate with Google APIs.
C. Incorrect - Interconnect only provides network connectivity, not authentication.
D. Incorrect - User accounts should not be used for service-to-service authentication; service accounts are designed for this.

Question 281
You are using Container Registry to centrally store your company's container images in a separate project. In another project, you want to create a Google Kubernetes Engine (GKE) cluster. You want to ensure that Kubernetes can download images from Container Registry. What should you do?

Options
A. In the project where the images are stored, grant the Storage Object Viewer IAM role to the service account used by the Kubernetes nodes.
B. When you create the GKE cluster, choose the Allow full access to all Cloud APIs option under 'Access scopes'.
C. Create a service account, and give it access to Cloud Storage. Create a P12 key for this service account and use it as an imagePullSecrets in Kubernetes.
D. Configure the ACLs on each image in Cloud Storage to give read-only access to the default Compute Engine service account.

Answer
A

Detailed explanation
A. Correct - Granting Storage Object Viewer to the GKE node service account gives nodes secure access to pull images from Container Registry.
B. Incorrect - Allowing full access to all Cloud APIs violates least-privilege principles.
C. Incorrect - P12 keys are deprecated, and imagePullSecrets are unnecessary when IAM roles can be used.
D. Incorrect - Managing ACLs at image level is tedious and not scalable compared to IAM.

Question 282
You deployed a new application inside your Google Kubernetes Engine cluster using the YAML file specified below.

```
apiVersion: apps/v1

kind: Deployment

metadata:

  name: myapp-deployment

spec:

  selector:

    matchLabels:

      app: myapp

  replicas: 2

  template:

    metadata:

      labels:

        app: myapp

    spec:

      containers:

      - name: myapp
```

```
    image: myapp:1.1
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
  name: myapp-service
spec:
  ports:
  - port: 8000
    targetPort: 80
    protocol: TCP
  selector:
    app: myapp
```

You check the status of the deployed pods and notice that one of them is still in PENDING status:

```
kubectl get pods -l app=myapp
NAME                                  READY   STATUS    RESTARTS   AGE
myapp-deployment-58ddbbb995-lp86m     0/1     Pending   0          9m
myapp-deployment-58ddbbb995-djpkg     1/1     Running   0          9m
```

You want to find out why the pod is stuck in pending status. What should you do?

Options

A. Review details of the myapp-service Service object and check for error messages.
B. Review details of the myapp-deployment Deployment object and check for error messages.
C. Review details of myapp-deployment-58ddbbb995-lp86m Pod and check for warning messages.
D. View logs of the container in myapp-deployment-58ddbbb995-lp86m Pod and check for warning messages.

Answer

C

Detailed explanation

A. Incorrect - The Service object is for networking and won't show why a pod is Pending.
B. Incorrect - The Deployment shows high-level status, not specific scheduling issues.
C. Correct - kubectl describe pod <pod-name> reveals scheduling problems such as insufficient resources, taints, or node selector mismatches.
D. Incorrect - Logs are unavailable because the container never started (pod stuck in Pending).

Question 283
You are setting up a Windows VM on Compute Engine and want to make sure you can log in to the VM via RDP. What should you do?

Options
A. After the VM has been created, use your Google Account credentials to log into the VM.
B. After the VM has been created, use gcloud compute reset-windows-password to retrieve the login credentials for the VM.
C. When creating the VM, add metadata to the instance using 'windows-password' as the key and a password as the value.
D. After the VM has been created, download the JSON private key for the default Compute Engine service account. Use the credentials in the JSON file to log in to the VM.

Answer
B

Detailed explanation
A. Incorrect - You cannot use Google account credentials directly for RDP.
B. Correct - The gcloud compute reset-windows-password command generates a Windows username and password that can be used with RDP.
C. Incorrect - Adding Windows password metadata is not supported.
D. Incorrect - Service account JSON keys are not used for VM logins.

Question 284
You want to configure an SSH connection to a single Compute Engine instance for users in the dev1 group. This instance is the only resource in this particular Google Cloud Platform project that the dev1 users should be able to connect to. What should you do?

Options
A. Set metadata to enable-oslogin=true for the instance. Grant the dev1 group the compute.osLogin role. Direct them to use the Cloud Shell to ssh to that instance.
B. Set metadata to enable-oslogin=true for the instance. Set the service account to no service account for that instance. Direct them to use the Cloud Shell to ssh to that instance.
C. Enable block project-wide keys for the instance. Generate an SSH key for each user in the dev1 group. Distribute the keys to the users.
D. Add all members of the dev1 group to the project. Grant them the compute.instanceAdmin role. Direct them to use the Cloud Console to connect to the instance.

Answer
A

Detailed explanation
A. Correct - Enabling OS Login on the instance and granting the group compute.osLogin allows fine-grained SSH access. This is the recommended, secure way to manage SSH access.
B. Incorrect - Removing the service account is unrelated to SSH access control.
C. Incorrect - Manually creating and distributing SSH keys is harder to manage and less secure than OS Login.
D. Incorrect - Granting instanceAdmin is overly permissive and gives more access than just SSH.

Question 285
You need to produce a list of the enabled Google Cloud Platform APIs for a GCP project using the gcloud command line in the Cloud Shell. The project name is my-project. What should you do?

Options
A. Run gcloud projects list to get the project ID, and then run gcloud services list --project <project ID>.
B. Run gcloud init to set the current project to my-project, and then run gcloud services list --available.
C. Run gcloud info to view the account value, and then run gcloud services list --account <Account>.
D. Run gcloud projects describe <project ID> to verify the project value, and then run gcloud services list --available.

Answer
A

Detailed explanation
A. Correct - Listing the project's enabled APIs requires using gcloud services list --project. First get the project ID, then list the services.
B. Incorrect - --available lists all possible APIs, not just enabled ones.
C. Incorrect - gcloud info shows account and environment info, not API services.
D. Incorrect - describe shows metadata of the project, not APIs enabled.

Question 286
You are building a new version of an application hosted in an App Engine environment. You want to test the new version with 1% of users before completely switching. What should you do?

Options
A. Deploy a new version of your application in Google Kubernetes Engine instead of App Engine and then use GCP Console to split traffic.
B. Deploy a new version in Compute Engine instead of App Engine and use GCP Console to split traffic.
C. Deploy a new version as a separate app in App Engine. Configure App Engine using GCP Console to split traffic between two apps.
D. Deploy a new version in App Engine. Then go to App Engine settings in GCP Console and split traffic between current and new versions.

Answer
D

Detailed explanation
A. Incorrect - Moving to GKE is unnecessary for traffic splitting.
B. Incorrect - Same issue, Compute Engine doesn't provide built-in traffic splitting for App Engine apps.
C. Incorrect - Creating separate apps is not required. Traffic splitting is supported within App Engine services.
D. Correct - App Engine has built-in support for gradual traffic splitting.

Question 287
You need to provide a cost estimate for a Kubernetes cluster using the GCP pricing calculator. Workload requires high IOPS and snapshots. You start by entering number of nodes, average hours, and average days. What should you do next?

Options
A. Fill in local SSD. Fill in persistent disk storage and snapshot storage.
B. Fill in local SSD. Add estimated cost for cluster management.
C. Select Add GPUs. Fill in persistent disk storage and snapshot storage.
D. Select Add GPUs. Add estimated cost for cluster management.

Answer
A

Detailed explanation
A. Correct - Local SSD is required for high IOPS, persistent disk for regular storage, and snapshot storage for backups.
B. Incorrect - This ignores snapshot cost, which is essential here.
C. Incorrect - GPUs are unrelated to IOPS and snapshot requirement.
D. Incorrect - Again, GPUs don't fit the workload need.

Question 288
You are using GKE with autoscaling to host a new application. You want to expose the app publicly using HTTPS. What should you do?

Options
A. Create a Kubernetes Service of type NodePort and a Kubernetes Ingress via Cloud Load Balancer.
B. Create a Kubernetes Service of type ClusterIP. Configure DNS name with the Service's IP.
C. Create a Service of type NodePort exposing port 443 on each node. Configure DNS name with all node IPs.
D. Create an HAProxy pod and configure DNS name with the node's public IP.

Answer
A

Detailed explanation
A. Correct - Best practice is NodePort + Ingress backed by HTTPS Load Balancer.
B. Incorrect - ClusterIP only works within the cluster, not public.
C. Incorrect - NodePort works but exposing all nodes directly is poor practice and not scalable.
D. Incorrect - HAProxy with manual DNS mapping is inefficient and not cloud-native.

Question 289
You need to enable traffic between multiple groups of Compute Engine instances running in two GCP projects. Each group runs in its own VPC. What should you do?

Options
A. Verify both projects are in a GCP Organization. Create a new VPC and add all instances.
B. Verify both projects are in a GCP Organization. Share the VPC from one project and request the other to use it.
C. Verify you are the Project Administrator. Create two new VPCs and add instances.
D. Verify you are the Project Administrator. Create a new VPC and add instances.

Answer
B

Detailed explanation
A. Incorrect - You cannot move existing instances into a new VPC.
B. Correct - Shared VPC is the right solution for enabling communication across projects.
C. Incorrect - Creating new VPCs doesn't solve inter-project communication.
D. Incorrect - Same issue as C.

Question 290
You want to add a new auditor to a Google Cloud Platform project. The auditor should be allowed to read, but not modify, all project items. How should you configure the auditor's permissions?

Options
A. Create a custom role with view-only project permissions. Add the user's account to the custom role.
B. Create a custom role with view-only service permissions. Add the user's account to the custom role.
C. Select the built-in IAM project Viewer role. Add the user's account to this role.
D. Select the built-in IAM service Viewer role. Add the user's account to this role.

Answer
C

Detailed explanation
A. Incorrect - Custom roles could achieve this, but they add unnecessary complexity when a built-in role exists.
B. Incorrect - Service-level roles don't provide full project-wide read-only access.
C. Correct - The built-in IAM project Viewer role grants read-only access across all resources in the project.
D. Incorrect - Service Viewer is resource-specific, not project-wide.

Question 291
You are operating a GKE cluster for multiple teams. Your ML team needs access to Nvidia Tesla P100 GPUs. You want to minimize effort and cost. What should you do?

Options
A. Ask your ML team to add the accelerator: gpu annotation to their pod specification.
B. Recreate all the nodes of the GKE cluster to enable GPUs on all of them.
C. Create your own Kubernetes cluster on top of Compute Engine nodes with GPUs and dedicate it to the ML team.
D. Add a new GPU-enabled node pool to the GKE cluster. Ask the ML team to add the nodeSelector for GPUs to their pod specification.

Answer
D

Detailed explanation
A. Incorrect - Simply annotating pods with GPU requests will not work unless GPU-enabled nodes exist.
B. Incorrect - Recreating the entire cluster with GPUs on all nodes is expensive and unnecessary.
C. Incorrect - Building a separate Kubernetes cluster increases cost and management complexity.
D. Correct - The recommended way is to add a GPU node pool to the cluster and let workloads that request GPUs be scheduled there.

Question 292
Your VMs are in a subnet with a mask of 255.255.255.240. The subnet has no free IP addresses, and you require 10 additional IPs. All VMs (new and existing) must still reach each other without extra routes. What should you do?

Options
A. Use gcloud to expand the IP range of the current subnet.
B. Delete the subnet, and recreate it using a wider range of IP addresses.
C. Create a new project. Use Shared VPC to share the current network with the new project.
D. Create a new subnet with the same starting IP but a wider range to overwrite the current subnet.

Answer
A

Detailed explanation
A. Correct - GCP allows subnet range expansion without disrupting existing workloads. This is the simplest and correct fix.
B. Incorrect - Deleting and recreating the subnet would disrupt all running workloads.
C. Incorrect - Shared VPC does not solve the issue of subnet IP exhaustion.
D. Incorrect - You cannot overwrite an existing subnet with a new one starting from the same base IP.

Question 293
Your organization uses G Suite for communication. You want to grant some G Suite users access to your Cloud Platform project. What should you do?

Options
A. Enable Cloud Identity in the GCP Console for your domain.
B. Grant them the required IAM roles using their G Suite email address.
C. Create a CSV sheet with all users' email addresses. Use gcloud to convert them into Google accounts.
D. In the G Suite console, add the users to a group called cloud-group@domain.com. Rely on default behavior to grant Cloud Platform access.

Answer
B

Detailed explanation
A. Incorrect - Cloud Identity is for organizations without G Suite. G Suite already integrates with IAM.
B. Correct - G Suite users are treated as valid Google identities, so you can directly assign IAM roles using their G Suite email.
C. Incorrect - No conversion to Google accounts is needed; G Suite accounts are already valid identities.
D. Incorrect - Adding users to a G Suite group doesn't automatically grant project permissions. IAM role assignment is required.

Question 294
You have a GCP account with production and development projects. You want an automated process to list all compute instances in both projects daily. What should you do?

Options
A. Create two configurations using gcloud config. Write a script that sets each config as active. For each, run gcloud compute instances list to get the list of instances.
B. Create two configurations using gsutil config. Write a script to use gsutil compute instances list to get the list of instances.
C. Go to Cloud Shell and export this information to Cloud Storage on a daily basis.
D. Go to GCP Console and export this information to Cloud SQL on a daily basis.

Answer
A

Detailed explanation
A. Correct - Multiple configurations in gcloud allow switching between projects. A script can run daily to collect instance lists for both.
B. Incorrect - gsutil is for Cloud Storage, not Compute Engine.
C. Incorrect - Cloud Shell is interactive, not meant for daily automation.
D. Incorrect - Console export to Cloud SQL is not a feature and cannot be automated this way.

Question 295
You have a large 5-TB AVRO file stored in a Cloud Storage bucket. Your analysts are proficient only in SQL and need access to the data stored in this file. You want to find a cost-effective way to complete their request as soon as possible. What should you do?

Options
A. Load data in Cloud Datastore and run a SQL query against it.
B. Create a BigQuery table and load data in BigQuery. Run a SQL query on this table and drop this table after you complete your request.
C. Create external tables in BigQuery that point to Cloud Storage buckets and run a SQL query on these external tables to complete your request.
D. Create a Hadoop cluster and copy the AVRO file to HDFS by compressing it. Load the file in a Hive table and provide access to your analysts so that they can run SQL queries.

Answer
C

Detailed explanation
A. Incorrect – Cloud Datastore is a NoSQL database, not suitable for large analytical queries with SQL.
B. Incorrect – Loading 5 TB into BigQuery incurs ingestion cost and time, not efficient if queries are temporary.
C. Correct – BigQuery external tables allow querying Cloud Storage files directly with SQL, cost-effective and fast since no full load is required.
D. Incorrect – Creating a Hadoop cluster adds unnecessary complexity and cost compared to BigQuery's external table feature.

Question 296
You need to verify that a Google Cloud Platform service account was created at a particular time. What should you do?

Options
A. Filter the Activity log to view the Configuration category. Filter the Resource type to Service Account.
B. Filter the Activity log to view the Configuration category. Filter the Resource type to Google Project.
C. Filter the Activity log to view the Data Access category. Filter the Resource type to Service Account.
D. Filter the Activity log to view the Data Access category. Filter the Resource type to Google Project.

Answer
A

Detailed explanation
A. Correct – Creation of service accounts is a configuration event, which appears in the Activity log under the Configuration category for Service Account resources.
B. Incorrect – Filtering by Project would show project-level events, not service account creation.
C. Incorrect – Data Access logs record read/write operations, not creation/configuration events.
D. Incorrect – Same as above, Data Access category for Project will not show service account creation.

Question 297
You deployed an LDAP server on Compute Engine that is reachable via TLS through port 636 using UDP. You want to make sure it is reachable by clients over that port. What should you do?

Options
A. Add the network tag allow-udp-636 to the VM instance running the LDAP server.
B. Create a route called allow-udp-636 and set the next hop to be the VM instance running the LDAP server.
C. Add a network tag of your choice to the instance. Create a firewall rule to allow ingress on UDP port 636 for that network tag.
D. Add a network tag of your choice to the instance. Create a firewall rule to allow egress on UDP port 636 for that network tag.

Answer
C

Detailed explanation
A. Incorrect – Simply adding a network tag doesn't create a firewall rule; no traffic would be allowed.
B. Incorrect – Routes don't control access at the port level; firewall rules do.
C. Correct – Attaching a network tag and creating a firewall ingress rule allows inbound traffic on UDP 636 to the LDAP server.
D. Incorrect – Egress rule would control outbound traffic, but we need inbound access for clients.

Question 298
You need to set a budget alert for Compute Engine services on one of the three Google Cloud Platform projects that you manage. All three projects are linked to a single billing account. What should you do?

Options
A. Verify that you are the project billing administrator. Select the associated billing account and create a budget and alert for the appropriate project.
B. Verify that you are the project billing administrator. Select the associated billing account and create a budget and alert for the appropriate billing account.
C. Verify that you are the project administrator. Select the associated billing account and create a budget and alert for the appropriate project.
D. Verify that you are the project administrator. Select the associated billing account and create a budget and custom alert.

Answer
A

Detailed explanation
A. Correct – To set project-specific budget alerts, you must be a billing administrator and configure the budget/alert at the project level tied to the billing account.
B. Incorrect – Setting a budget at the billing account level would apply across all projects, not one specific project.
C. Incorrect – Project administrator role does not have billing permissions.
D. Incorrect – Same as above, project administrator cannot configure billing budgets.

Question 299
You are migrating a production-critical on-premises application that requires 96 vCPUs. You want to make sure the application runs in a similar environment on GCP. What should you do?

Options
A. When creating the VM, use machine type n1-standard-96.
B. When creating the VM, use Intel Skylake as the CPU platform.
C. Create the VM using Compute Engine default settings. Use gcloud to modify the instance to have 96 vCPUs.
D. Start the VM using Compute Engine default settings, and adjust as you go based on Rightsizing Recommendations.

Answer
A

Detailed explanation
A. Correct – GCP provides predefined machine types such as n1-standard-96 that offer 96 vCPUs. This matches the requirement exactly.
B. Incorrect – CPU platform choice does not guarantee the required vCPUs.
C. Incorrect – You cannot arbitrarily modify default VM to scale to 96 vCPUs after creation unless you pick the correct machine type.
D. Incorrect – Rightsizing is advisory, not for meeting strict resource requirements upfront.

Question 300
You want to configure a solution for archiving data in a Cloud Storage bucket. The solution must be cost-effective. Data with multiple versions should be archived after 30 days. Previous versions are accessed once a month for reporting. This archive data is also occasionally updated at month-end. What should you do?

Options
A. Add a bucket lifecycle rule that archives data with newer versions after 30 days to Coldline Storage.
B. Add a bucket lifecycle rule that archives data with newer versions after 30 days to Nearline Storage.
C. Add a bucket lifecycle rule that archives data from regional storage after 30 days to Coldline Storage.
D. Add a bucket lifecycle rule that archives data from regional storage after 30 days to Nearline Storage.

Answer
B

Detailed explanation
A. Incorrect – Coldline is best for data accessed less than once a quarter; here, older versions are accessed monthly.
B. Correct – Nearline is optimized for infrequent but monthly access, making it the most cost-effective choice.
C. Incorrect – Regional to Coldline would make monthly access too expensive and slow.
D. Incorrect – Regional to Nearline doesn't address versioning; the requirement is to archive *older versions*.